Geothermal Drilling Operation

JDS – 24 October 2026



Objectives

- Typical drilling operation discussion
- Capture drilling operation needs and well-prepared during drilling design phase







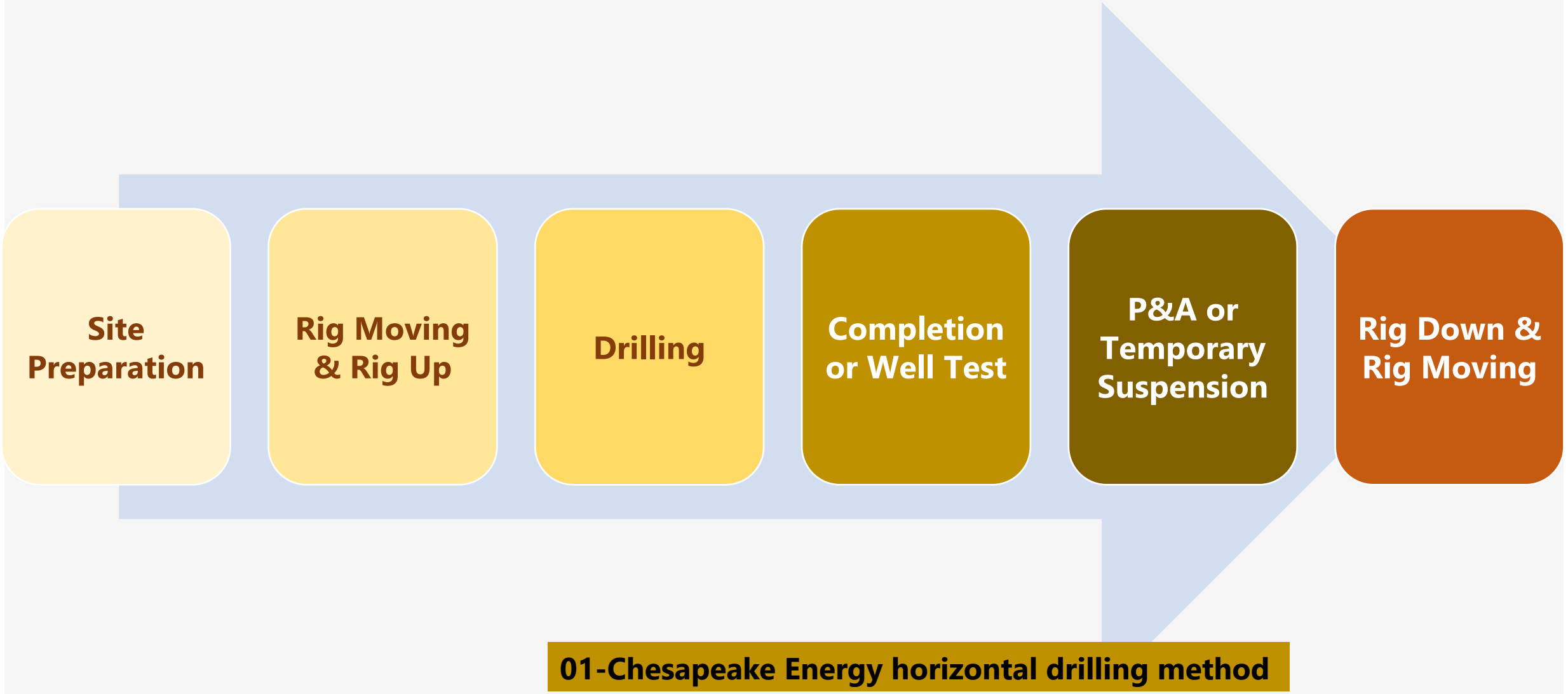




Drilling Operation Nomenclature

- RM → Rig Move
- RU → Rig Up
- RD → Rig Down
- MU → Make Up
- BO → Break Out
- POOH → Pull Out Of Hole
- RIH → Run In Hole
- DRLG → Drilling
- CIRC → Circulation
- PT → Pressure and Temperature
- ST → Side Track
- BU → Bottom Up
- Lat. → Latitude
- Long. → Longitude
- KOP → Kick Off Point
- BUR → Build Up Rate
- EOB → End Of Build
- TVD → True Vertical Depth
- MD → Measure Depth
- EW → East West
- NS → North South
- AZ → Azimuth

Drilling Operation General Sequences



Oil and Gas Drilling Work Location

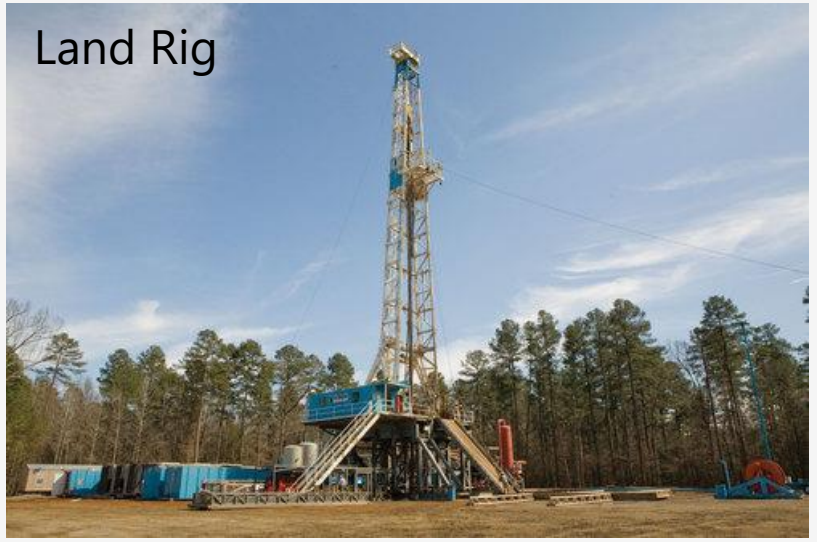












Geothermal Drilling Work Location











Geothermal Wellsite view



Karaha Bodas



Wayang Windu







KS Orka – Sorik Marapi







KS-Orka - Sokoria







Geodipa: Patuha and Dieng





Typical Drilling Rig Operation

Preparation	Drilling	Completion
• Site Construction • Water Supply • Pre-installed 30" Conductor • Rig Layout	• Drill 26" Hole to 450 m-MD • Run and Cement 20" Casing • Nipple Up 21-1/4" Wellhead & BOP	• Well Test • Jet wash to clean Perforated Liner • Rig Released • Rig Down
• Rig Mobilization • Rigging Up • Nipple up 30" Diverter • Make up Drill String	• Drill 17-1/2" Hole to 1000 m-MD • Run and Cement 13-3/8" Liner • Run and Cement 13-3/8" Tie Back • Run CBL • Nipple Up Master Valve & BOP	• Hand Over to Production • Extended Well Test
• Rig Inspection • Permit to spud • Spud ceremonial	• Drill 12-1/4" Hole to 2000 m-MD • Run E-Log • Run 10-3/4" Perforated Liner	• EPC Construction • Pipe Installation • Tie in well to Production Facility
• Local Hire	• Drill 9-7/8" Hole to 2600 m-MD • Run E-Log • Run 8-5/8" Perforated Liner	
	• Drill 7-7/8" Hole to 3200 m-MD • Run 7" Perforated Liner	

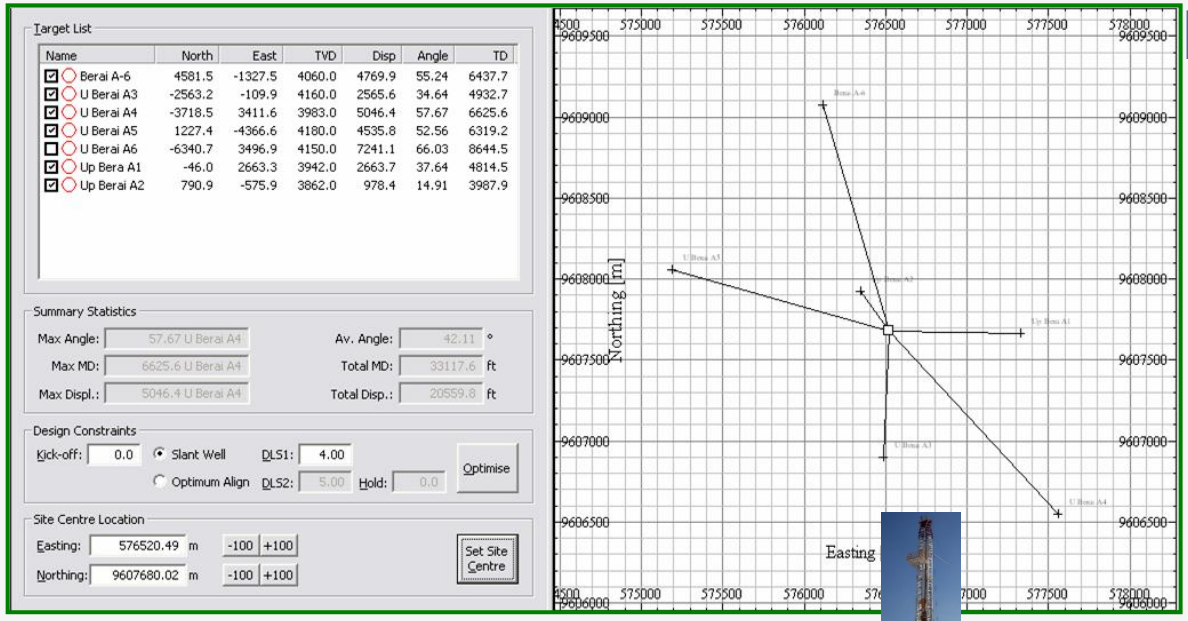
PRE – DRILLING: Site Construction

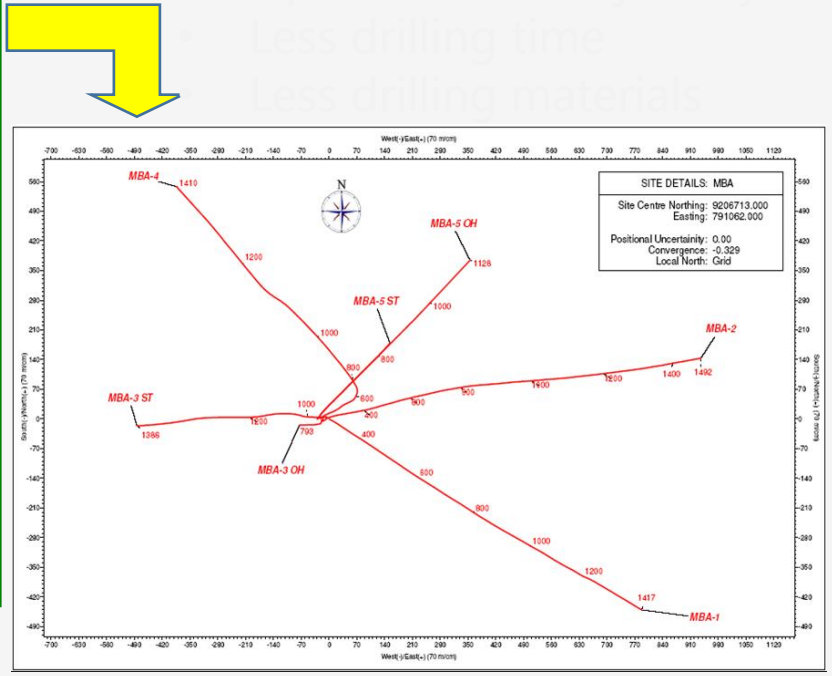
Site Construction – SNI 13-6910-2002- Oil and Gas

- Wells should be located at least **100 meters away from Public Roads**, Railways, Public Works, Houses or other places win which a source of ignition may arise.
- Vent, flare pits and extremities of flare lines should be located at least **90 meters from railways, roads, public works, processing units or tanks**. They should be at least **30 meters from a well**, gas/oil separator, site drainage or other possible source of ignitable vapours.
- Combustible material where possible all tress, undergrowth, and other such combustible material should be cleared from drilling sites and for a distance of **50 meters around the site perimeter**.
- Full earthing shall be provided, verified and documented. The maximum permissible resistance to earth shall be **4 ohms**. Where necessary boreholes to provide a ground to earth will have to be provided.

Optimum Surface Location

- Less Overall Well TD
- Optimum Well trajectory
- Less drilling time
- Less drilling materials



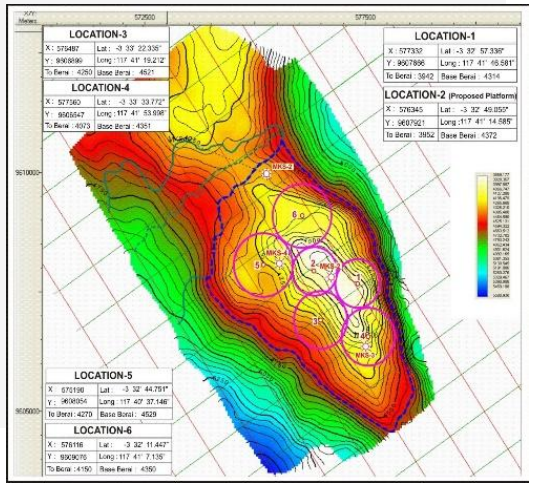






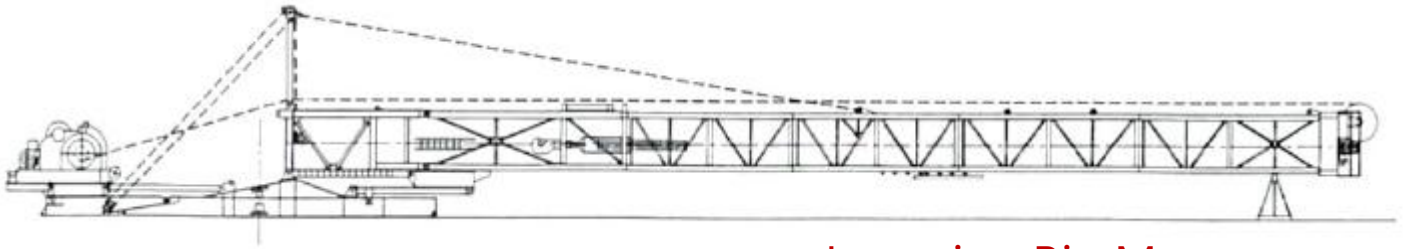
High
Drilling Risk

Medium-Low
Drilling Risk



Site Construction and Rig Layout





Lowering Rig Mast

Behind:

- Space for Rig Engine.

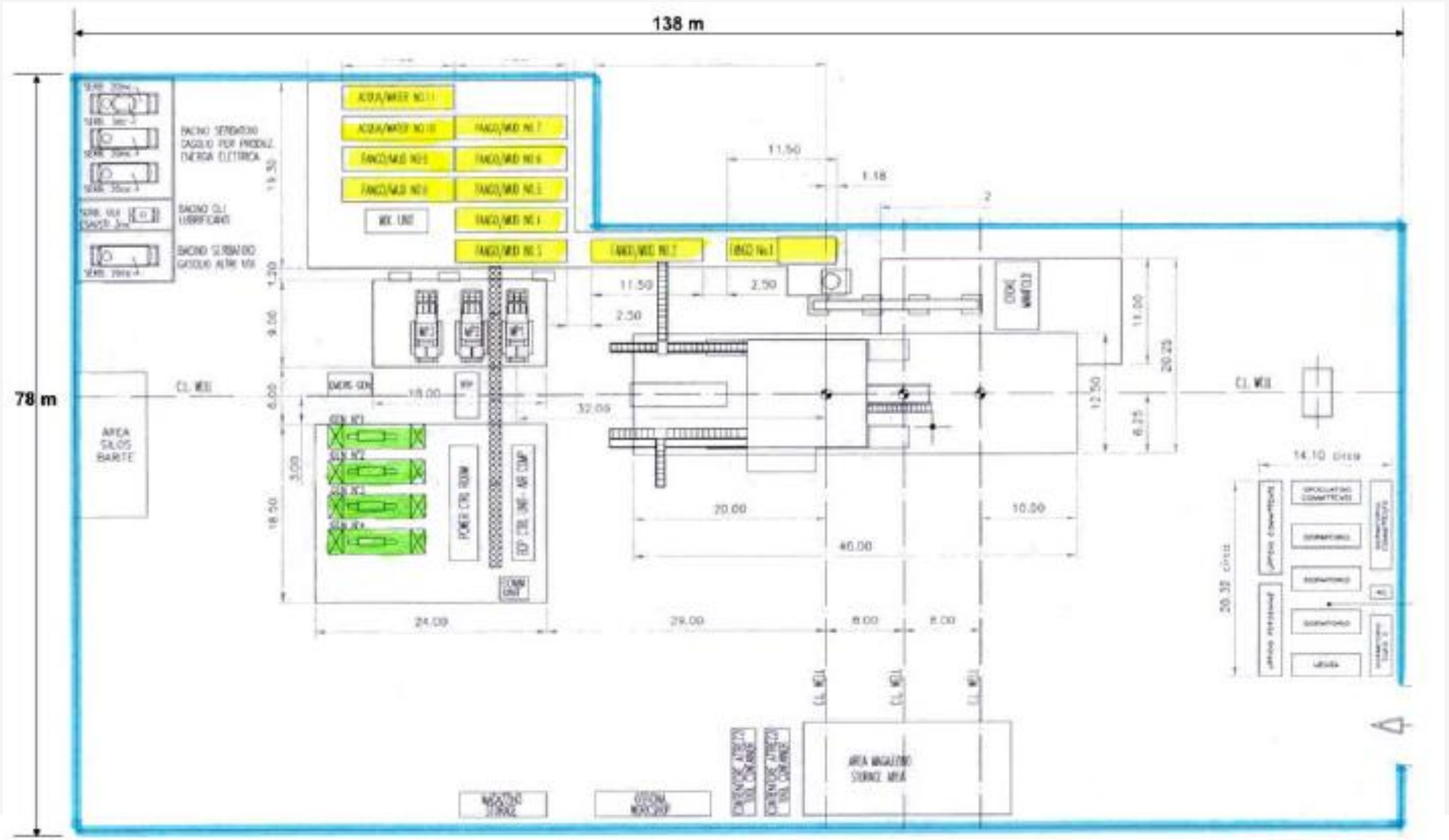
Front:

- Space for Lowering Rig Mast

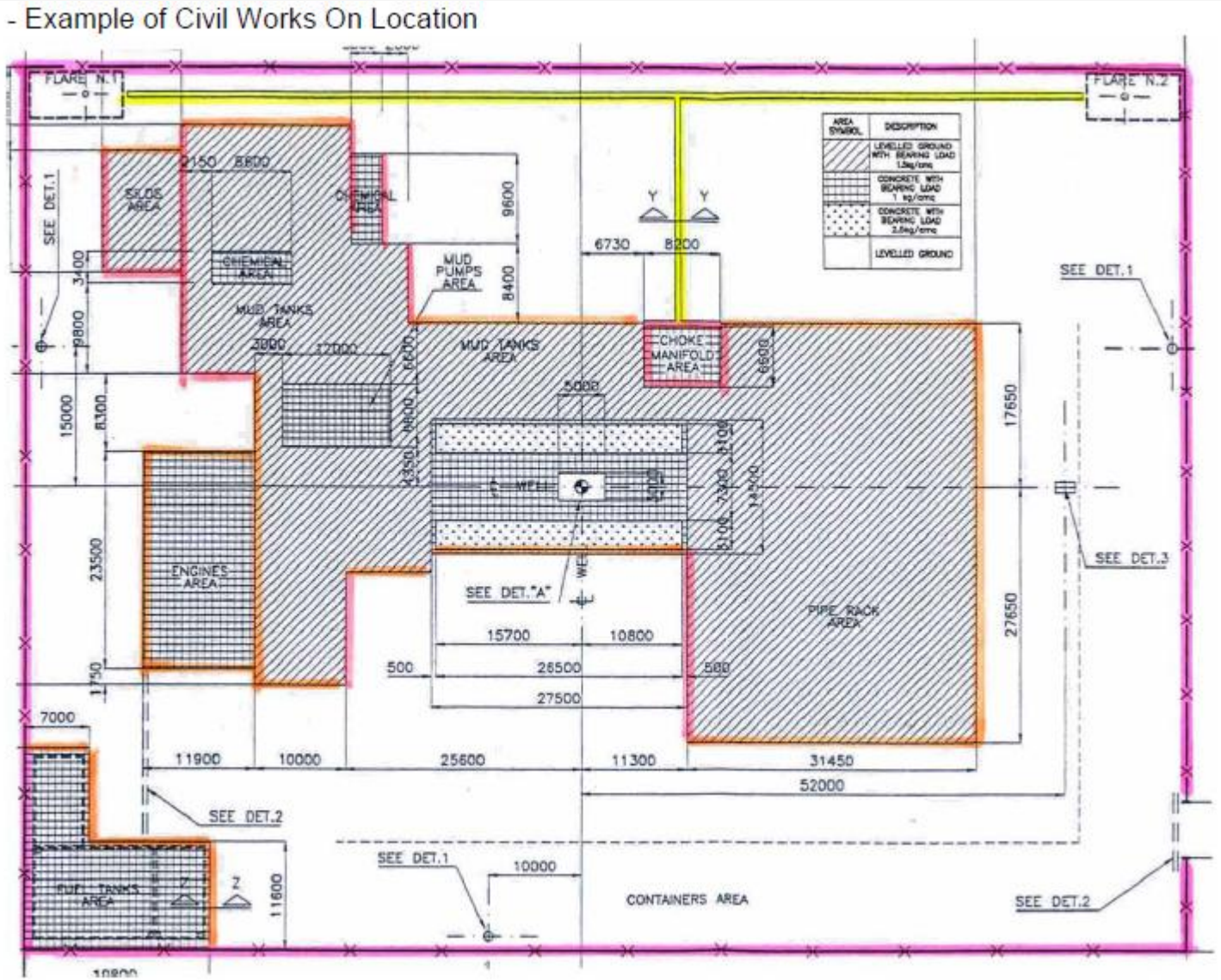
Left and right side

- Shale shakers and Mud Tanks
- Mini Camps
- Water Ponds
- Service Company
- Access for Heavy Equipment

Example of 3 Wells Cluster for 2000 HP Rig

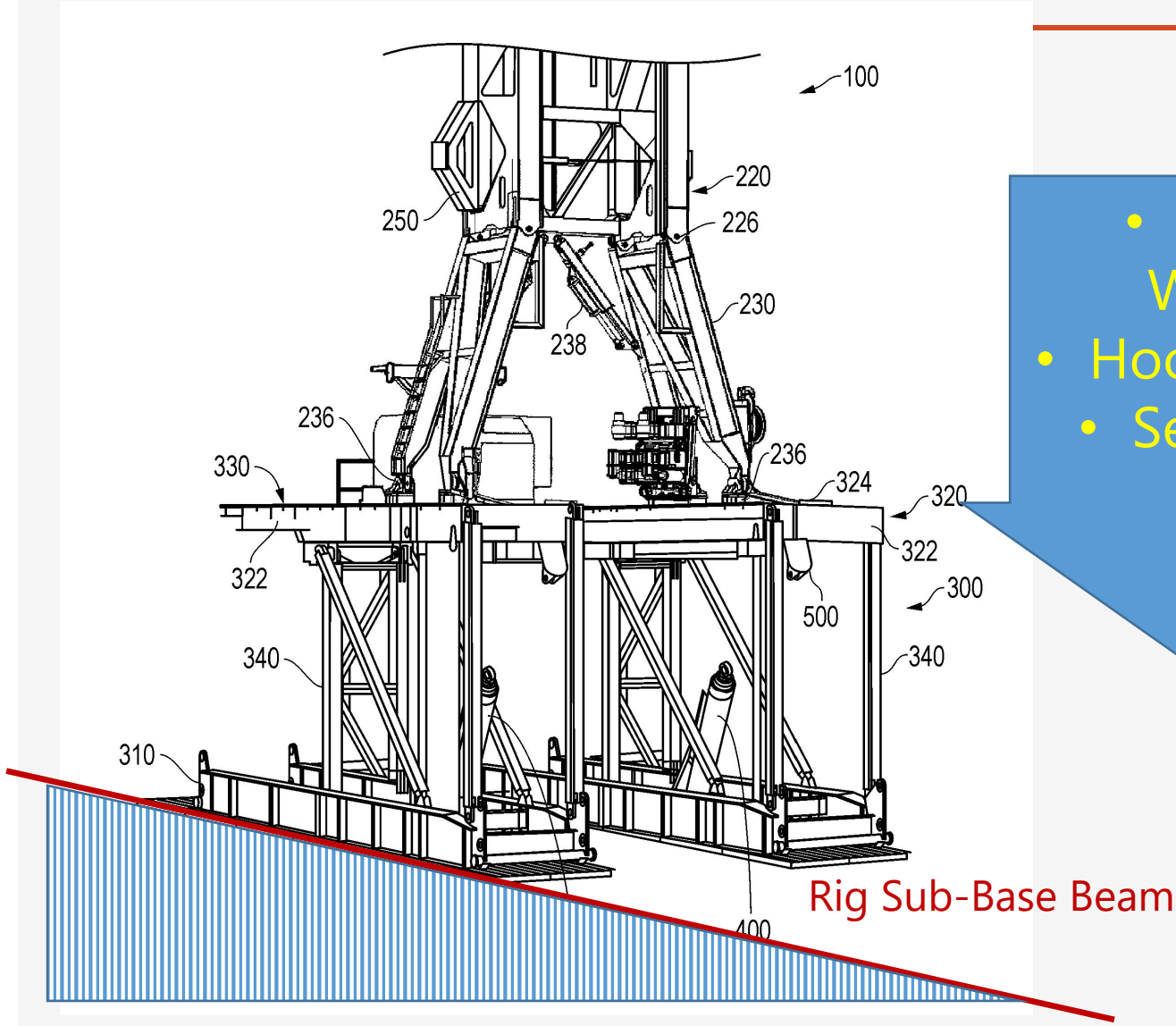


Example of Civil Works On Location



- Soil Bearing Load for Rig Foot Print.
- Area for leveling and compaction:
  - Mud tanks
  - Mud Pumps
  - Rig engine
  - Cementing
- Drainage system
- **Cutting bunker**
- **Cellar design**
- Water and cutting pond Capacity and elevation
- ~~Flare Pit area~~
- Mini camps area and septic tank
- Master Point Area
- Distance between well
- Etc……..

Soil Bearing Load for Rig Sub-base



- Total Weight
- Hook Loads
- Set Back Load

- Rig Foot-print elevation tolerance
- Soil Bearing Load Requirement

D2P2 Well Pad → Minimum Soil Bearing Pressure Requirement for Rig Foot Print :

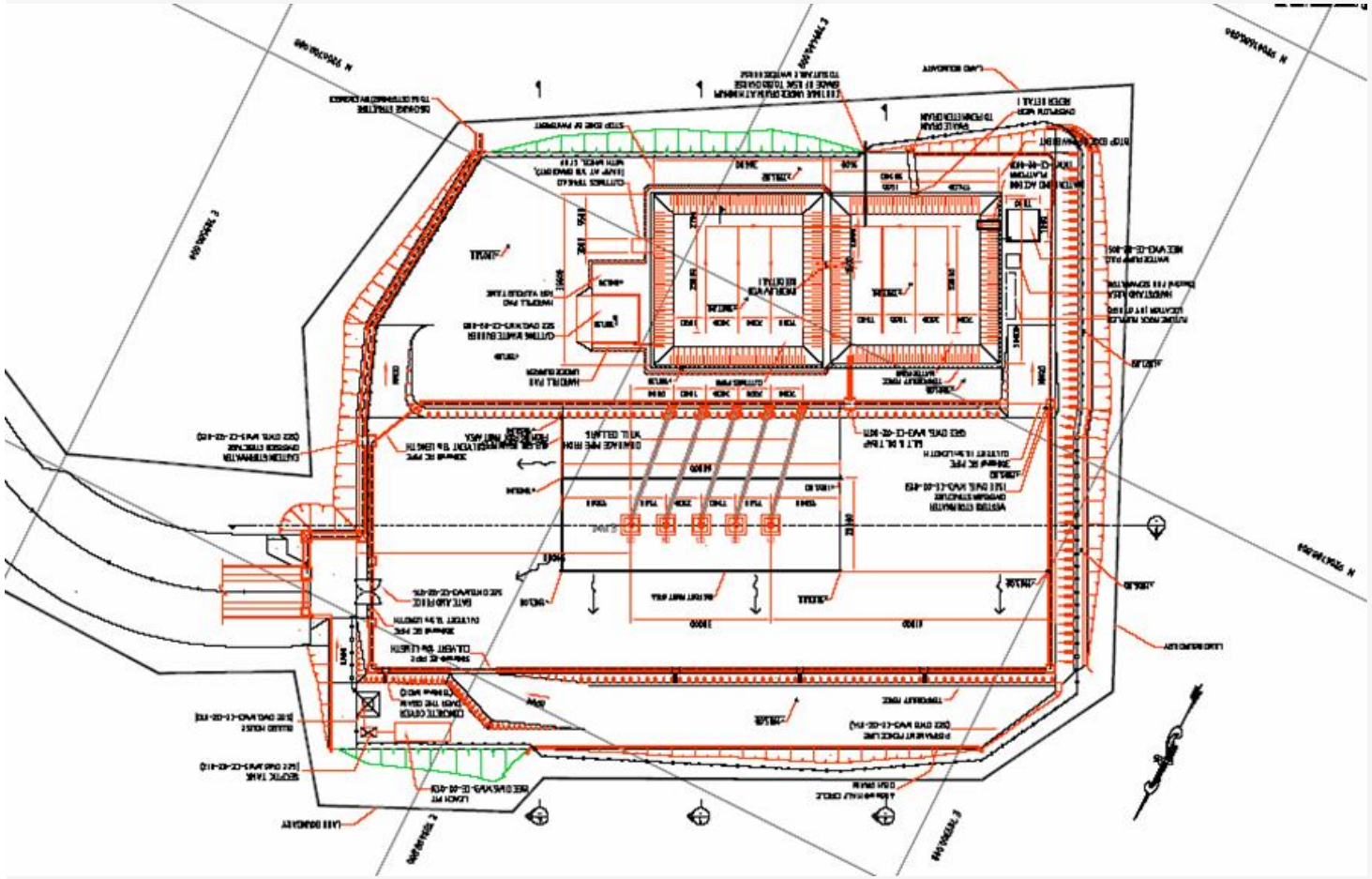
- 4500 PSF

Site Construction – Geothermal Well



- Smaller well pad area due to in mountain condition.
- Surface hard rock
- Additional Space for water ponds: **5000 - 8000 m3**
- Safety Reason: Minimum number of Mini Camps, less rig personnel stay at rig, i.e H2S hazard
- Space for simultaneous operation with Well Test
  - Pipeline
  - Separator
- Rig with skidding system is a must.
- Ensure Rig has additional power cable → adjust rig equipment lay out.

Optimum Well Pad Design



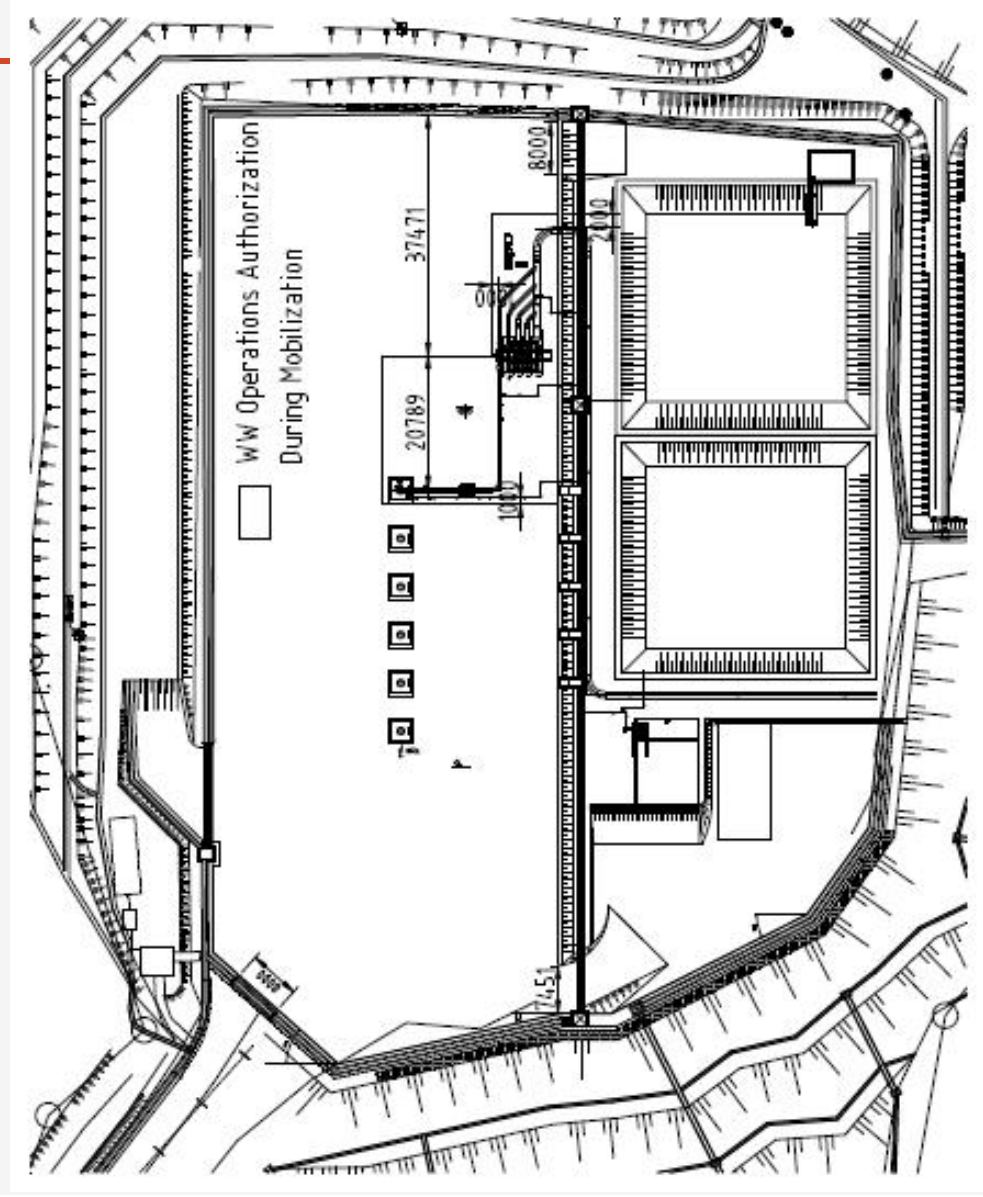
Typical Rig Site Layout

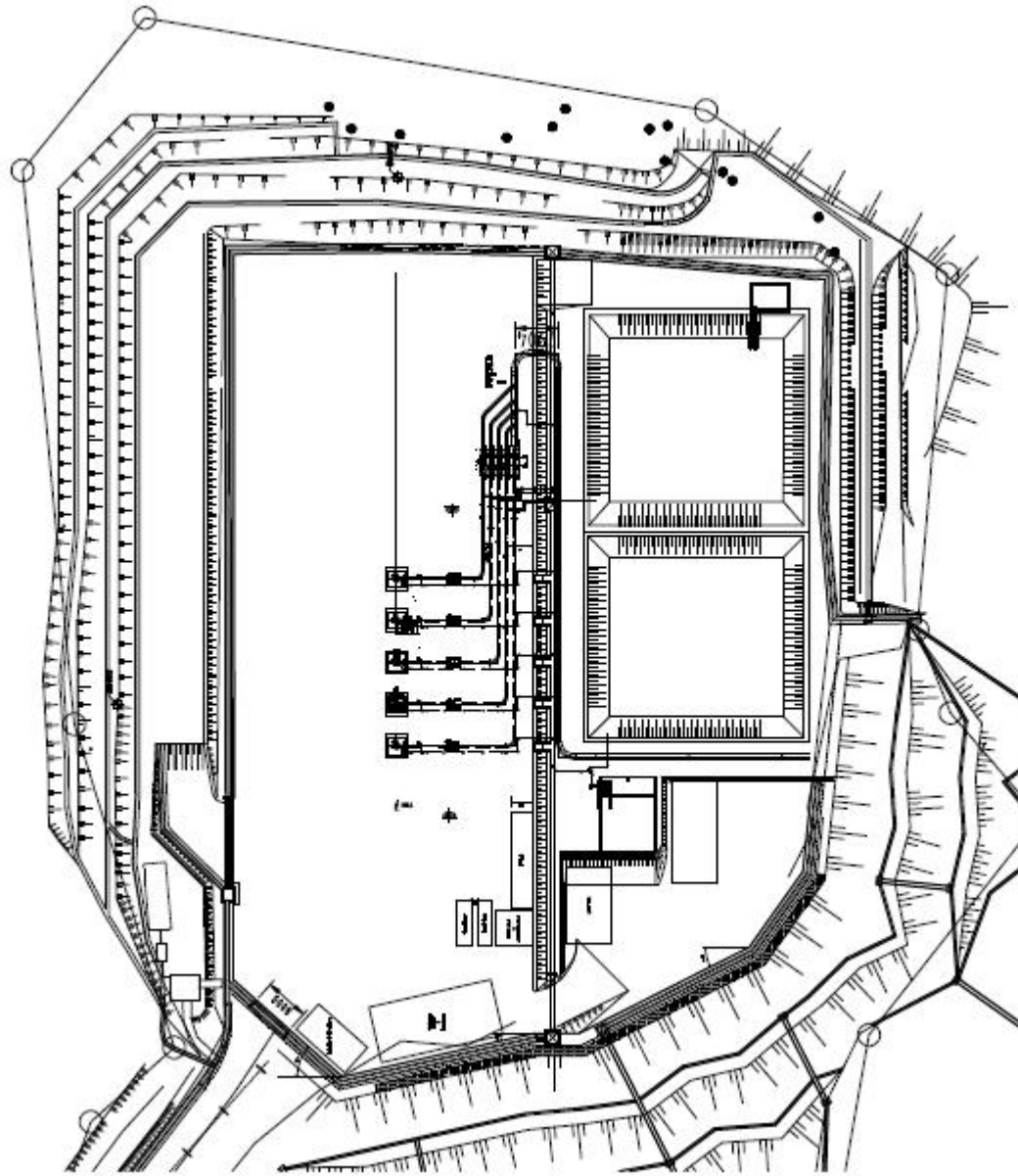
- 5000-8000 m3 water pond
- Lower elevation for cutting and water pond
- Pre design for well test pipe line layout
- Proper drainage and control box
- Direct Under ground drainage from Cellar to cutting Pond



Ideal Geothermal Well Pad – Star Energy

- Mini camp area could be higher elevation than rig elevation.
- Distance between Cellar → Well accessibility after being drilled
- Water flow by gravity to water pond.
- SIMOP: Pipeline layout not interfere with Rig Equipment layout.

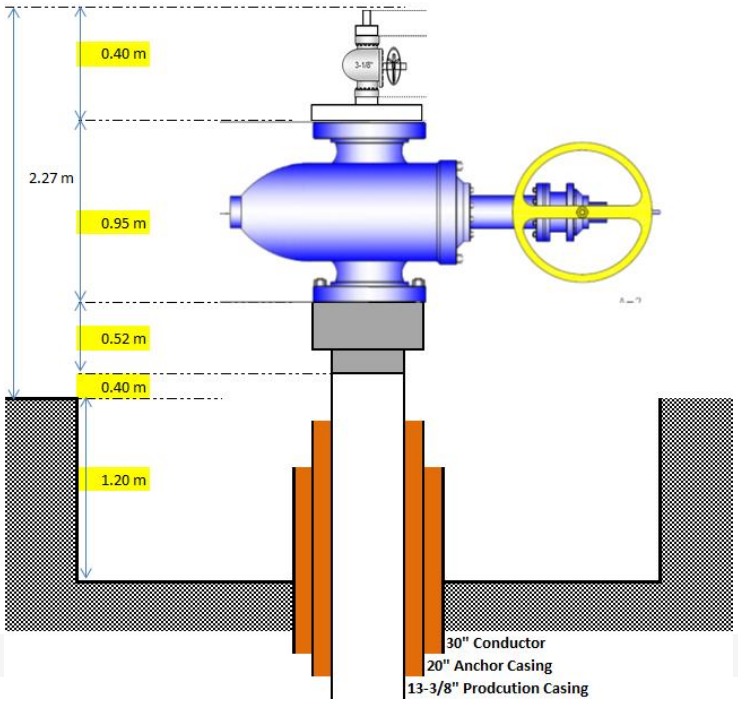


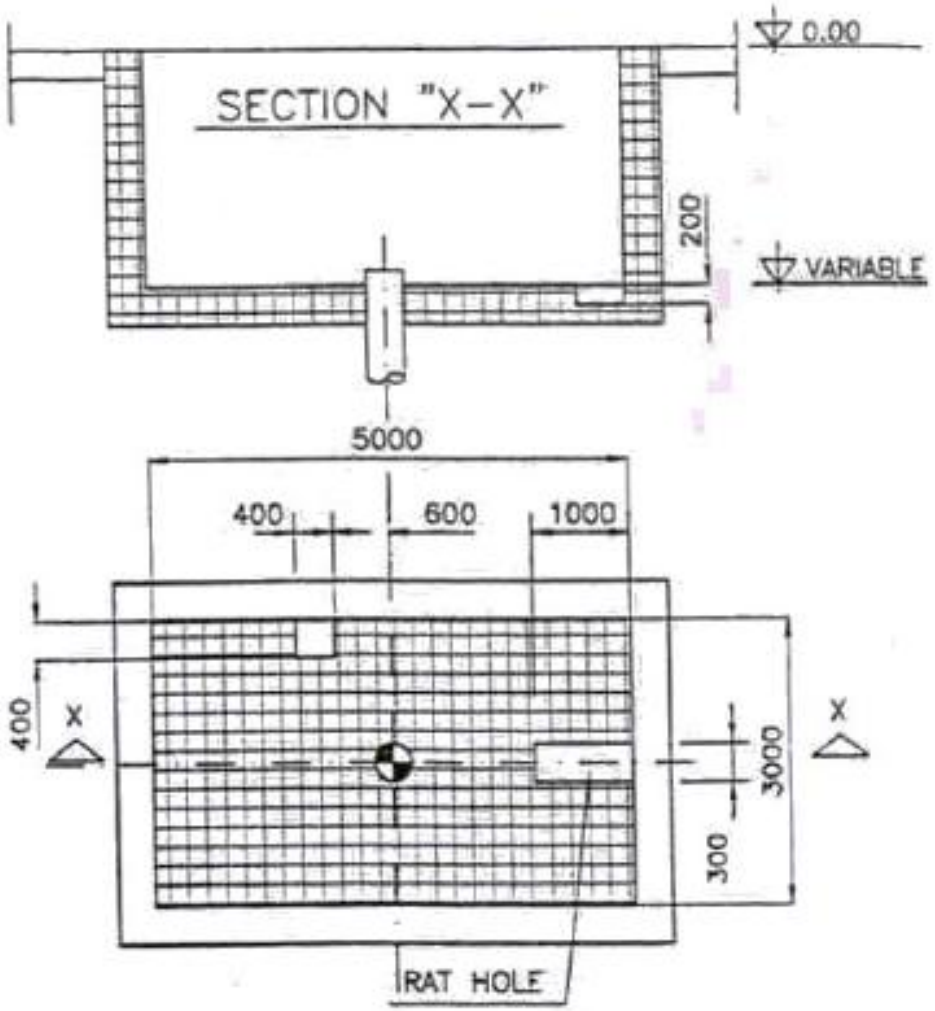


Cellars Design

- Cellar breadth

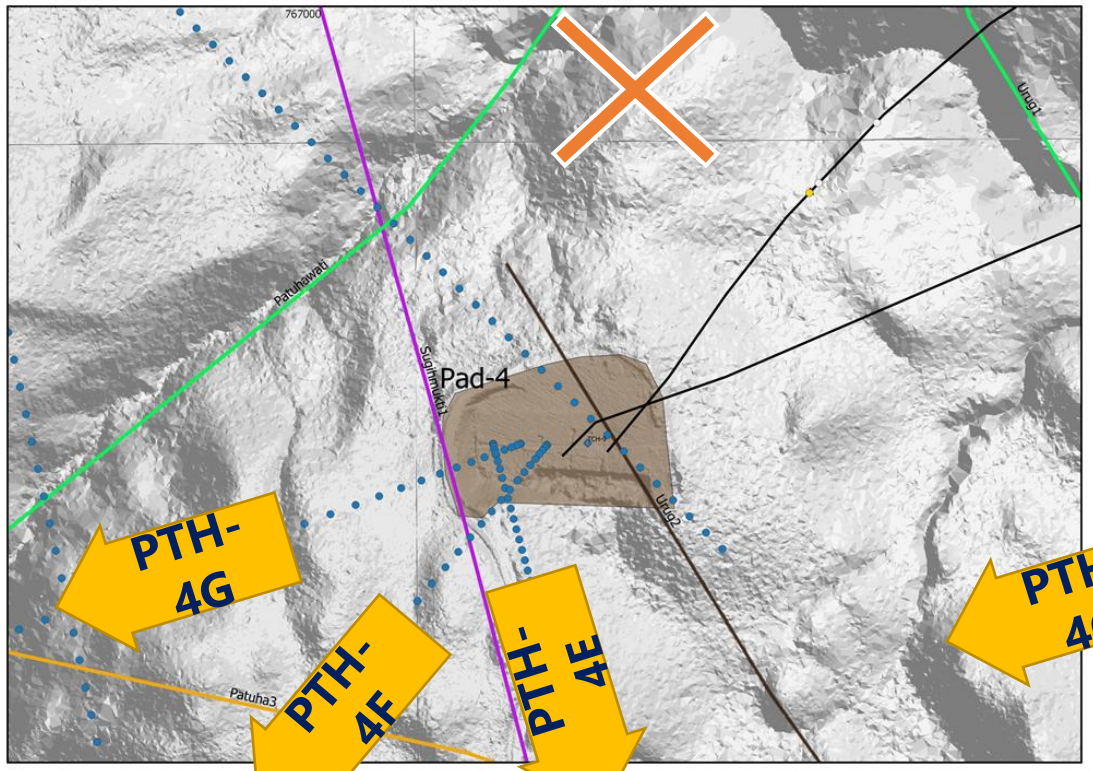
Cellar breadth is usually decided with the Rig Contractor, considering well head, BOP and substructure. The cellar is usually cased in concrete to avoid collapse with the weight of the rig.

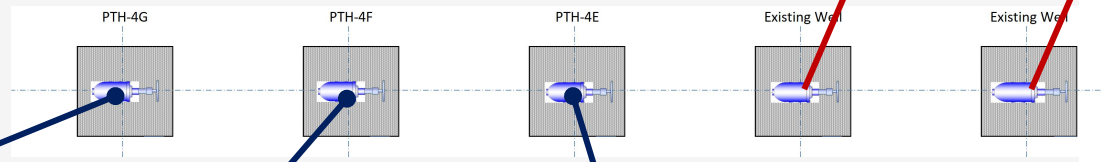




- Common size 3x3 m
- Depth depend on wellhead and BOP configuration

Well Pad – G: Cellar Assignment to minimize Surface collision

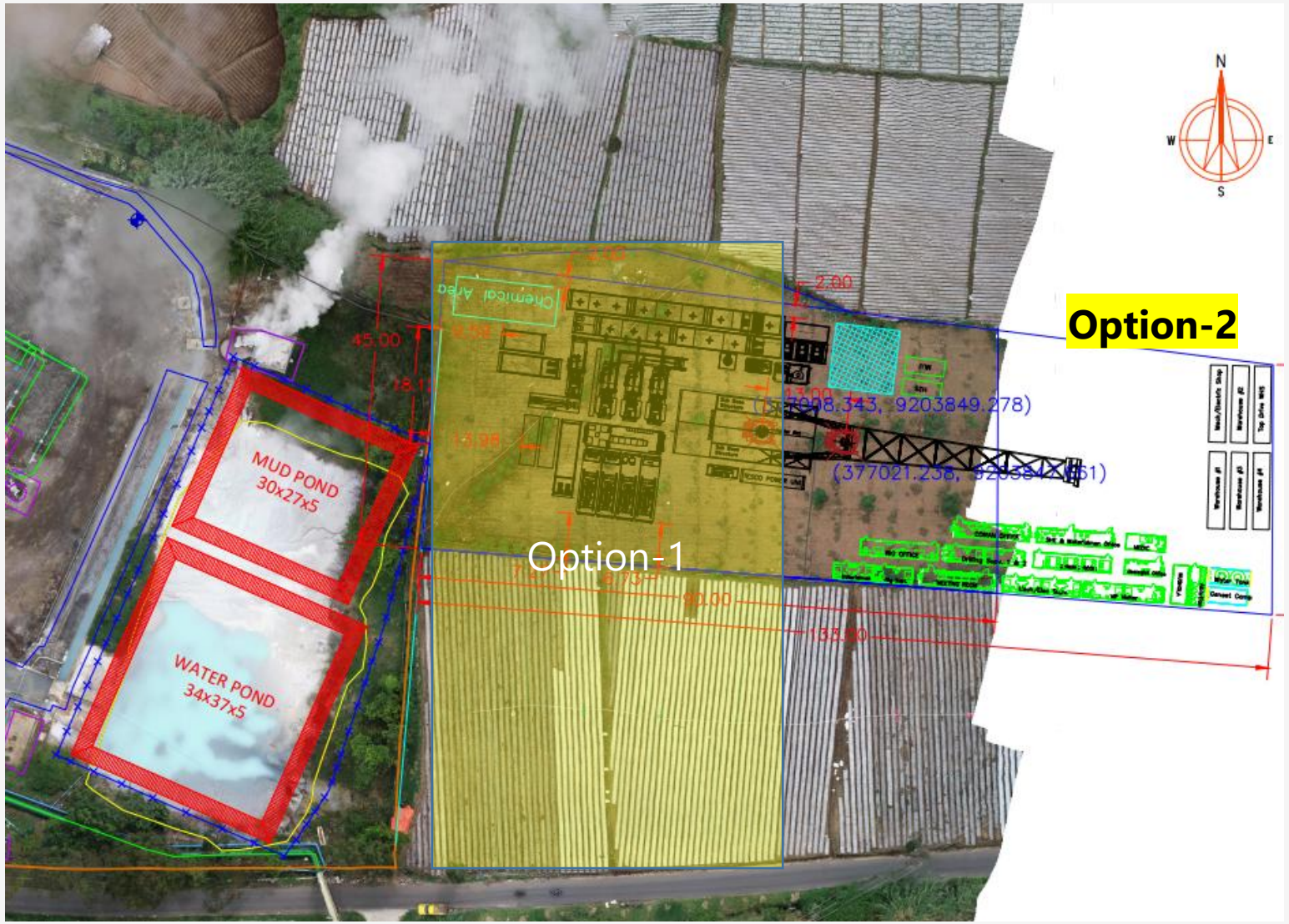






Pad No	New Wells		Existing Wells	
	Well Name	Azimuth	Well Name	Azimuth
Pad-G Pad-4	PTH-4F	220		
	PTH-4G	250		
	PTH-4E	165		

Detail Rig Lay out Design



Case-1:

Best Case?

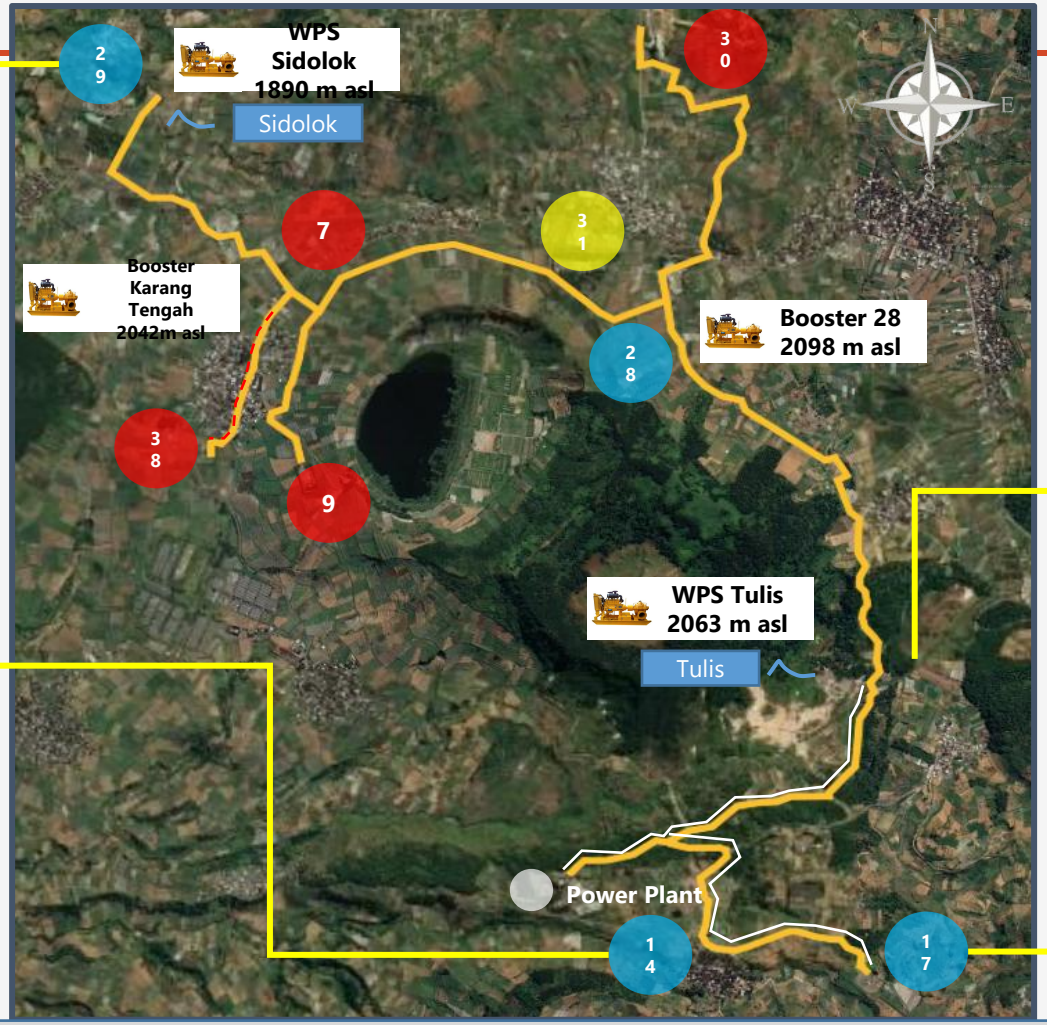

Case-2:
Water Pond?
ENTRY
379005, 9203260

Water Supply

Water Supplies-Dieng











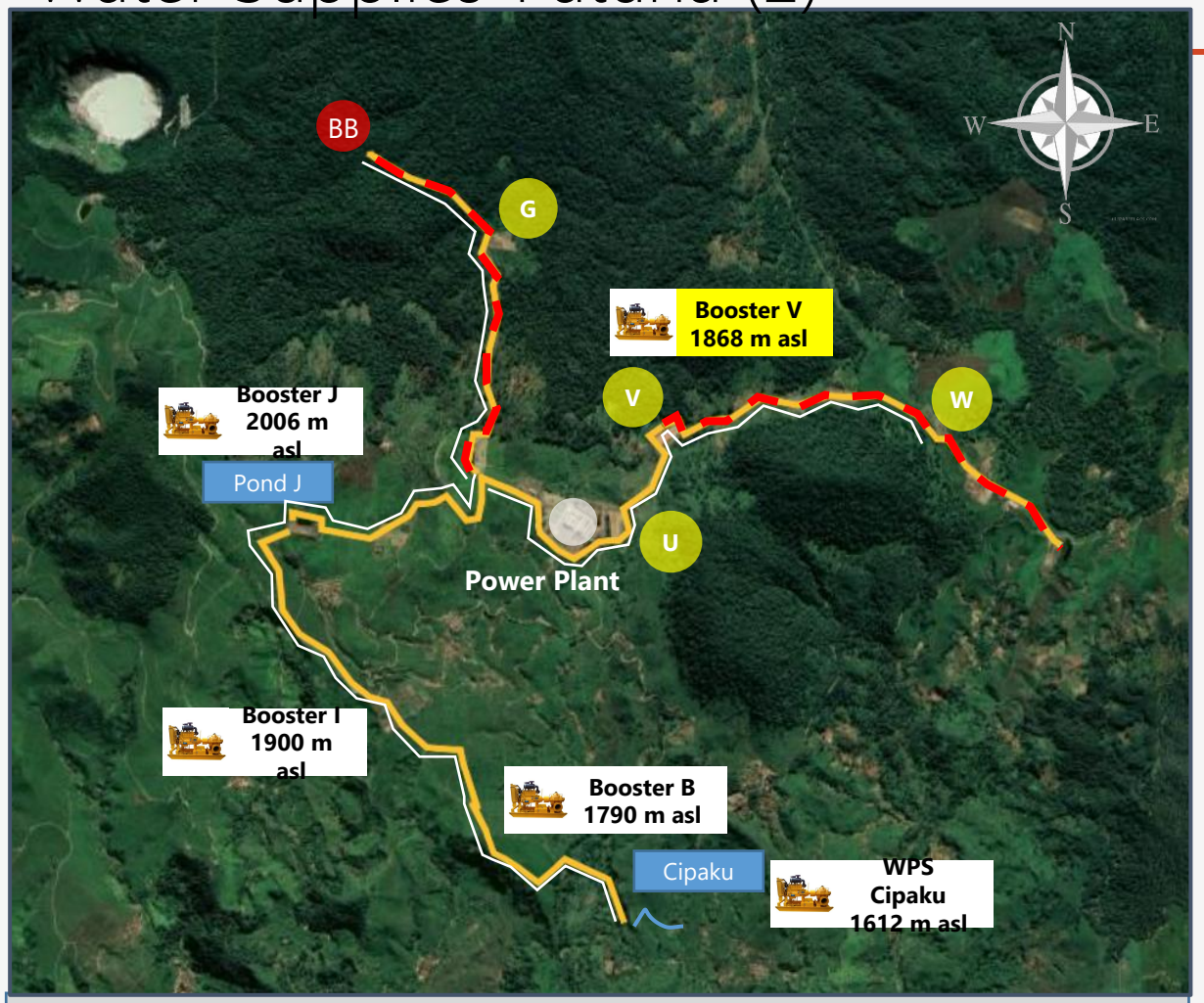
Building up *"Master Pond"* to utilize water for dry season usage

Table 3.1. Estimated ponds capacity

NO	LOCATION	AREA (m^2)	DEPTH (m)	VOLUME (m^3)
1	Pad 14	16,250	5	81,250
2	Pad 17	3,900	5	19,500
3	WPS Siterus	1,000	5	5,000
4	Pad 29	5,500	5	27,500
			TOTAL	133,250

Wellpad
Drilling Wellpad
New Wellpad
WPS/Booster
Water Sources
8" existing water line
8" temporary water line (Victaulic)
8" condensate line

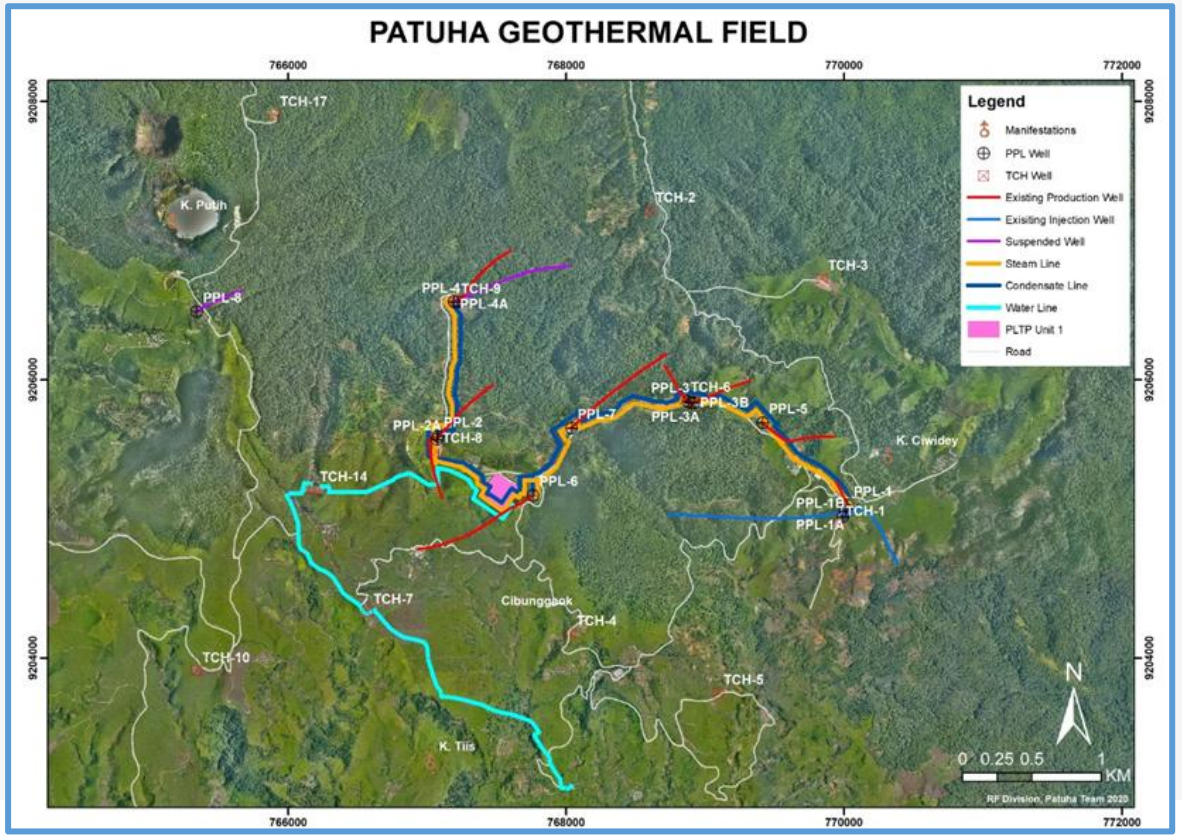
Water Supplies-Patuha (2)



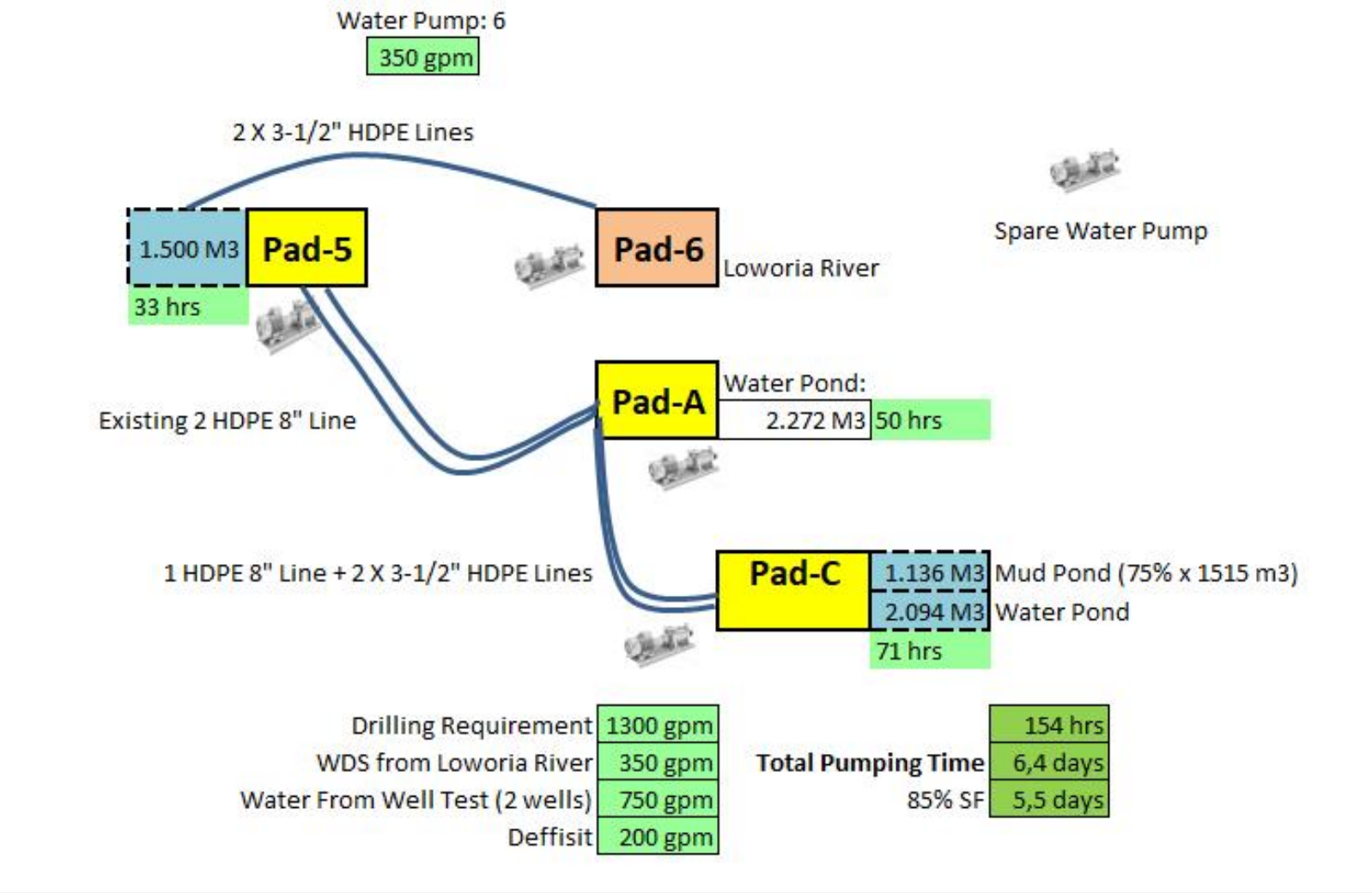
Legend

- Water Sources
- Drilling Wellpad
- New Wellpad
- WPS/Booster
- Backup Booster
- 6" existing water line
- 6" existing temporary water line (Victaulic)
- 8" victaulic new water line

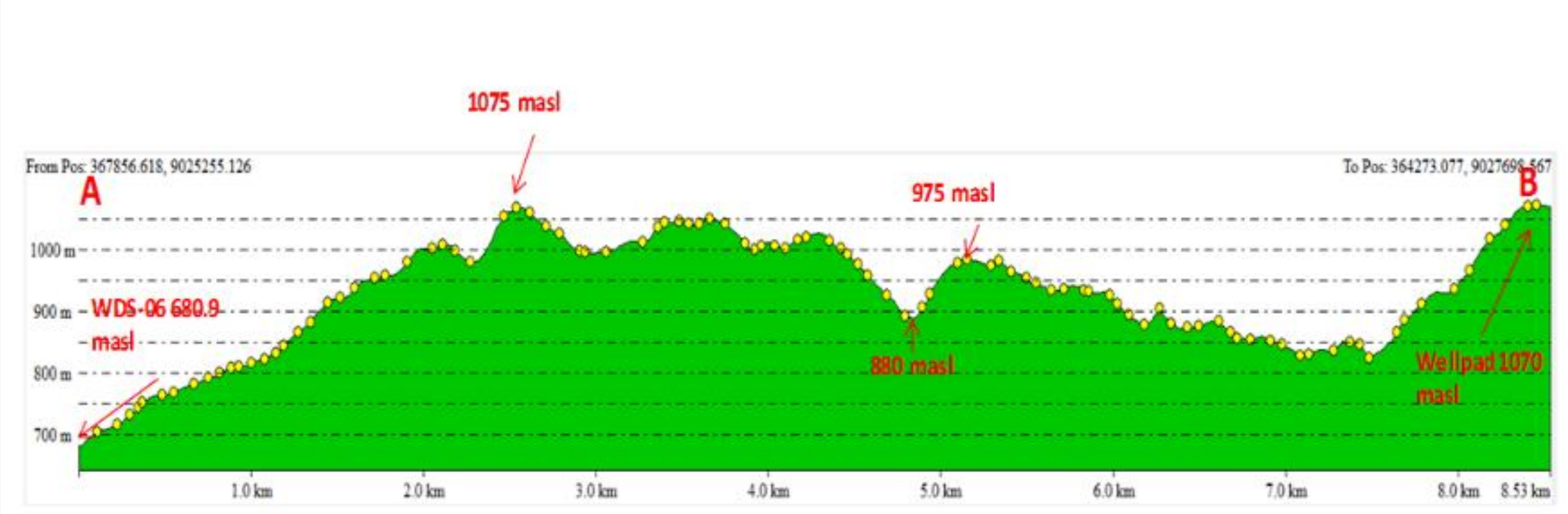
Based from observation, Water Debit from Cipaku River sufficient for dry and rainy season (pH ± 7)



WATER SUPPLY – 1300 GPM

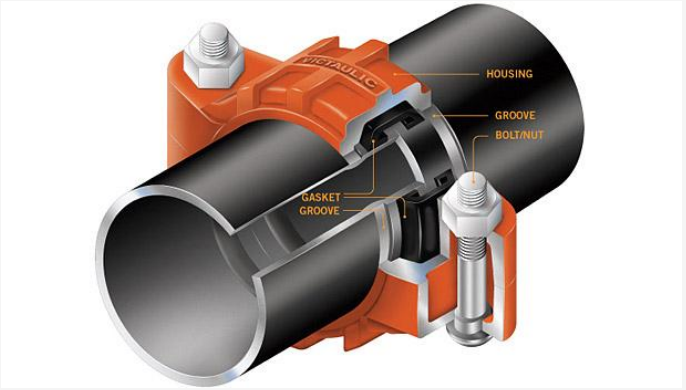


Water Supply – Pipe Route Elevation



WATER SUPPLY – 1300 GPM

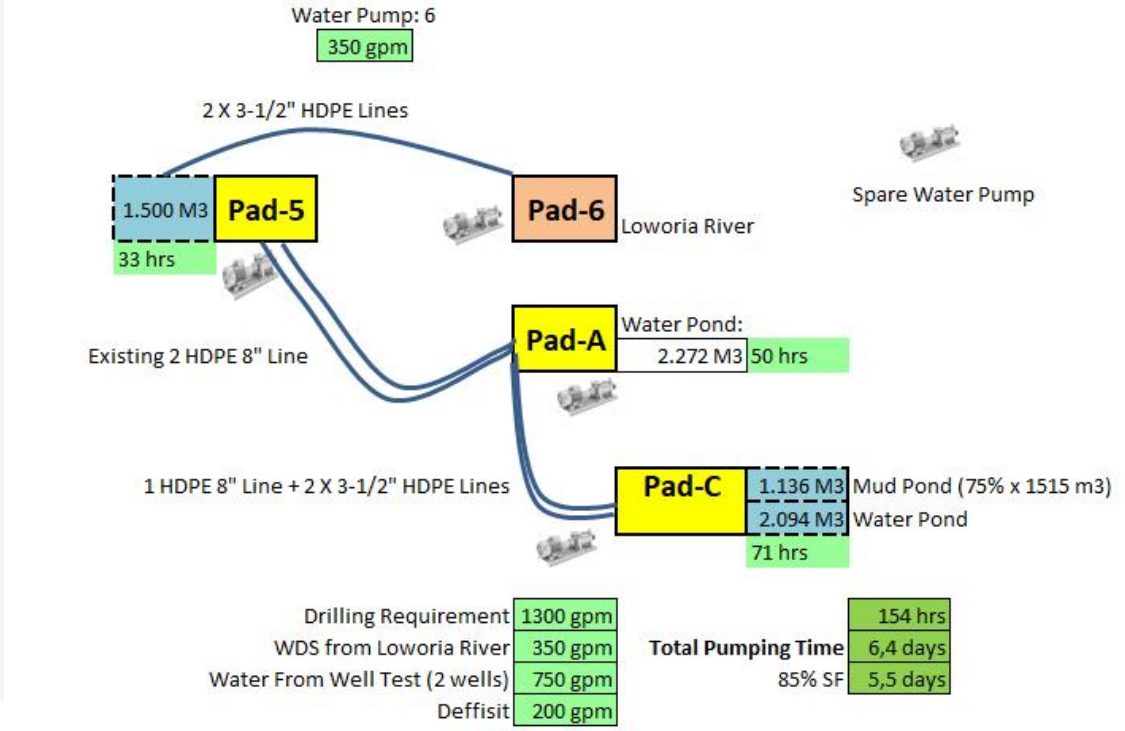
- **Water Sources:**
  - **Water pH?**
  - **Water Flow rate.**
  - **Distance and elevation**
  - **Community issue**





- **Pipe Line Sizes:**
  - **2 lines of 8”ODx6”ID HDPE pipe OR**
  - **2 line of victaulic or steel pipe**

- **Water Pond Storage**
  - **8000 – 10000 m3**
  - **If no water source → bigger water pons → up 100.000 m3**



WATER SUPPLY – 1400 GPM

WATER PIPE LINE: Use Galvanize or Steel Pipe Line instead of HDPE:

- Less pressure drop due to friction
- Higher Burst Pressure capacity → allow to utilize maximum water pump FLOW RATE and PRESSURE capacity.
  - **Water Pump: 600 psi**
  - **Steel Pipe: ±1000 psi**
  - **HDPE Pipe: 300 psi.**
- Less water pump requirement → Less Personnel requirement → Less Community issue
- Easy to repair if any pipe leak











Other Drilling Support Area

a. Open Yard for: Casing storage, excess rig equipment, etc
b. Warehouse
c. Base camp
d. Drill Cutting Disposal
e. Temporary Staging Area
f. Access Road







Question

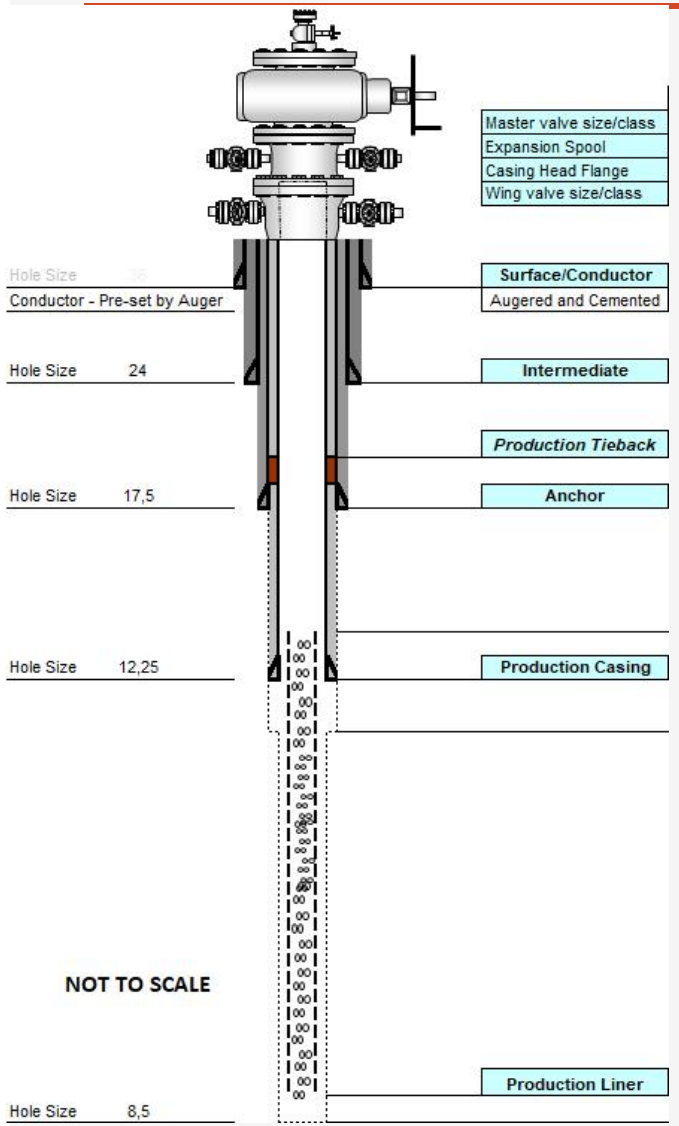


1. Minimum distance Flare Pit from center of the well ?
2. Common Cellar Size ?
3. Water Pond Capacity for Geothermal drilling well?

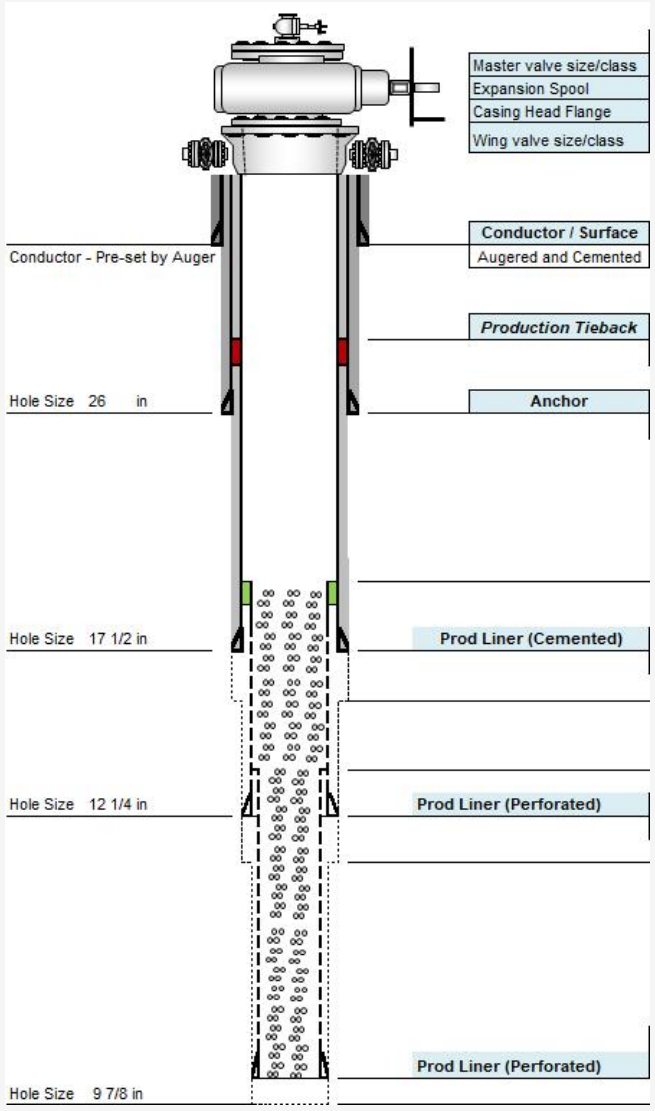
DRILLING OPERATION

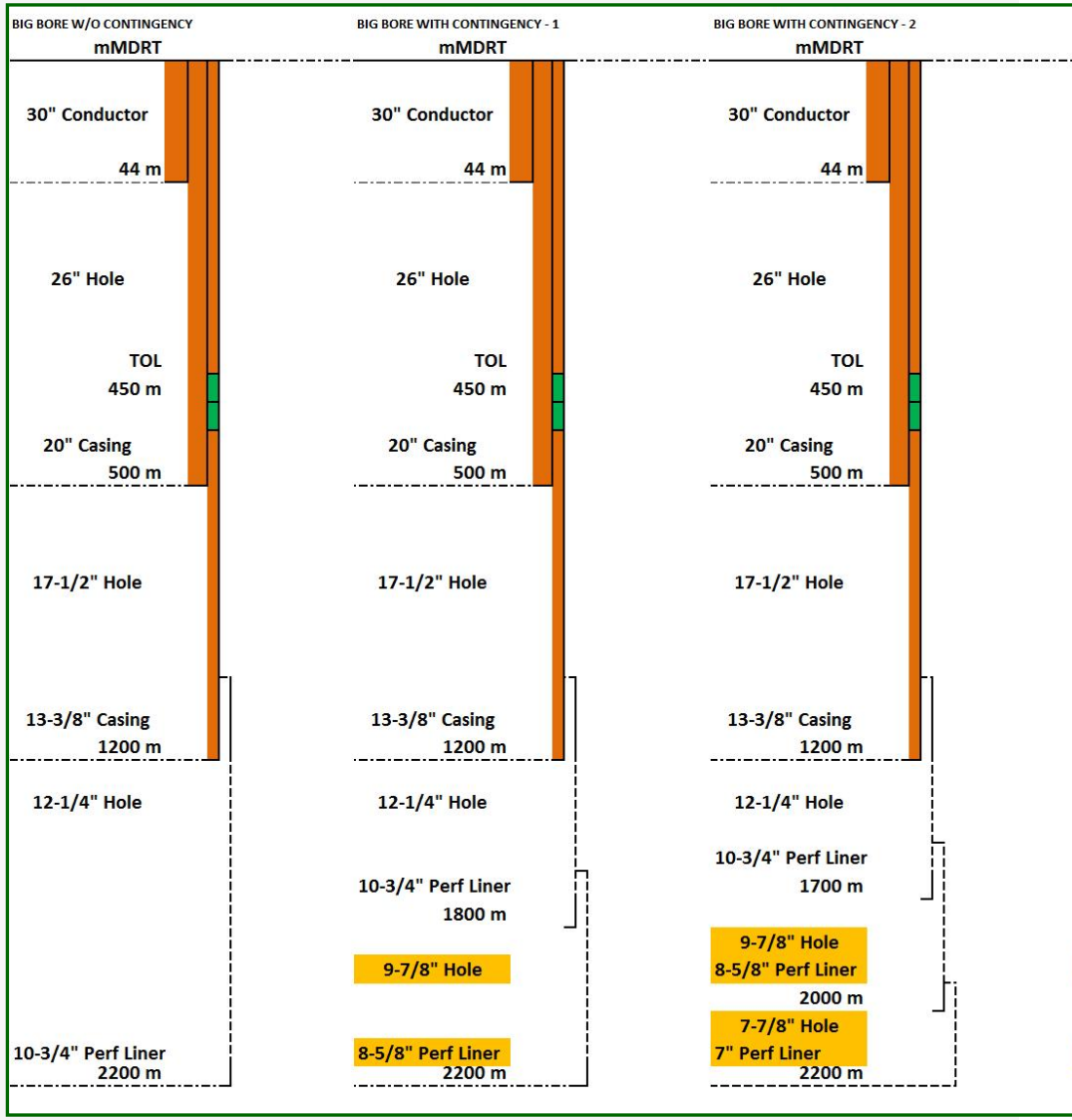
01_Bentec_Onshore Rig Building

Typical Well Schematic - Geothermal



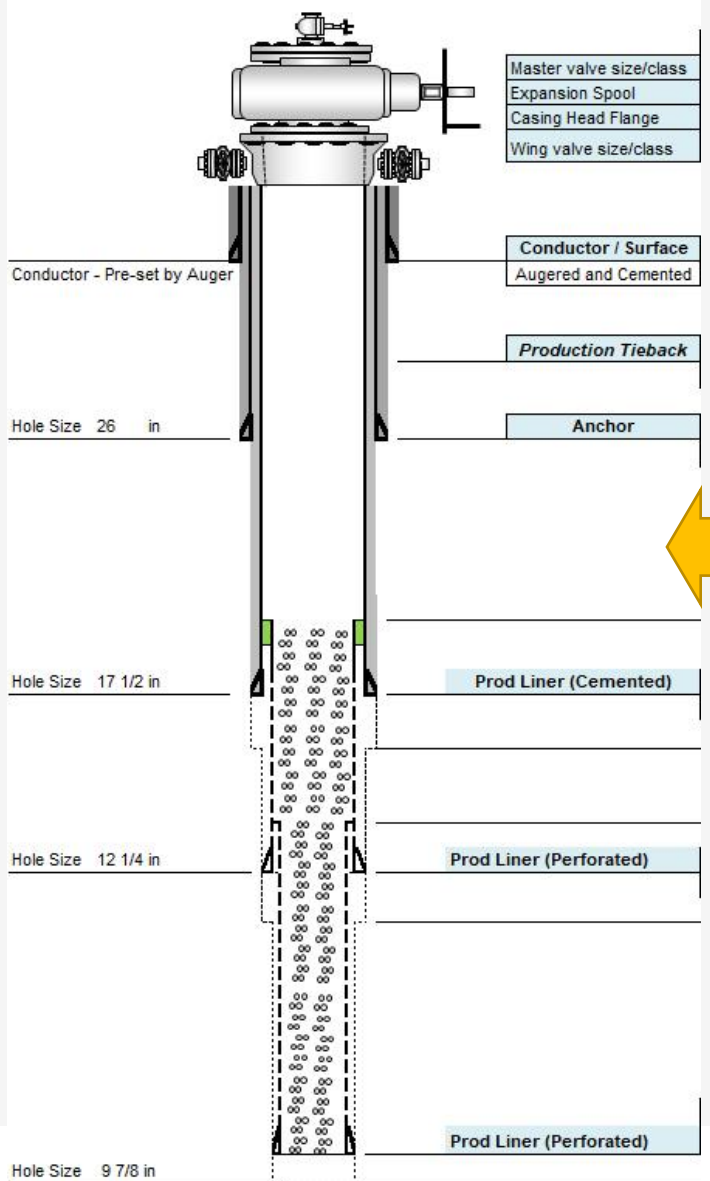
Standard Hole





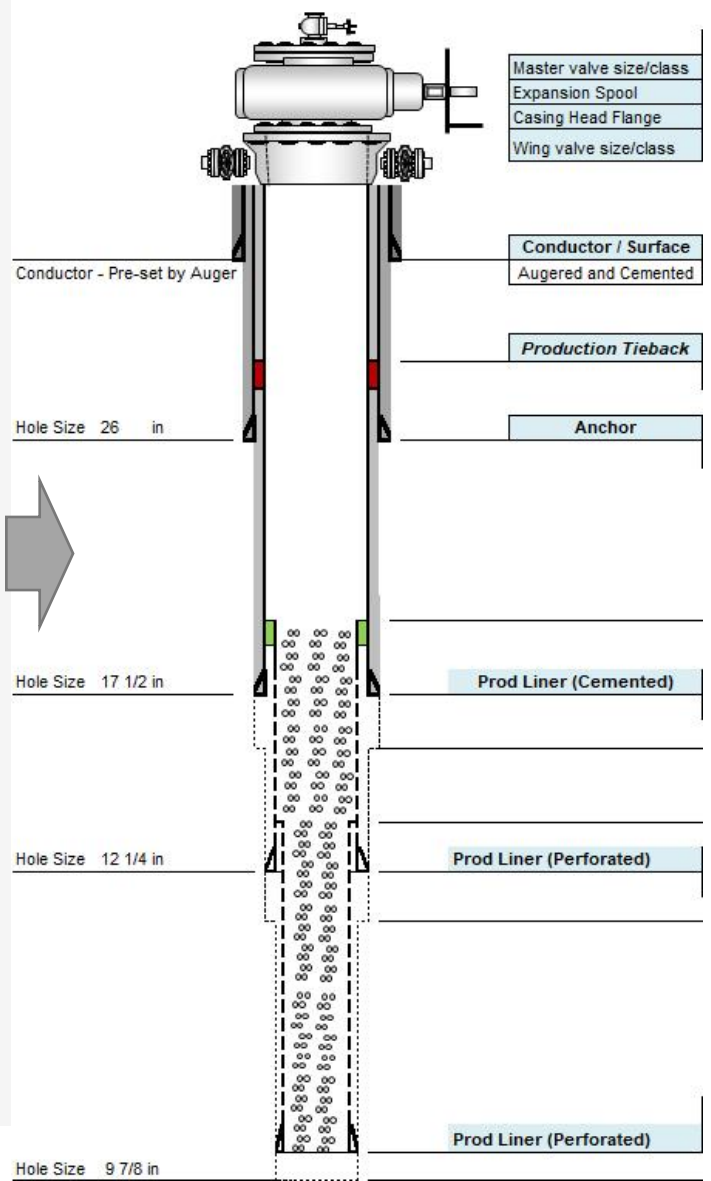
Big Bore

Typical Well Schematic - Geothermal

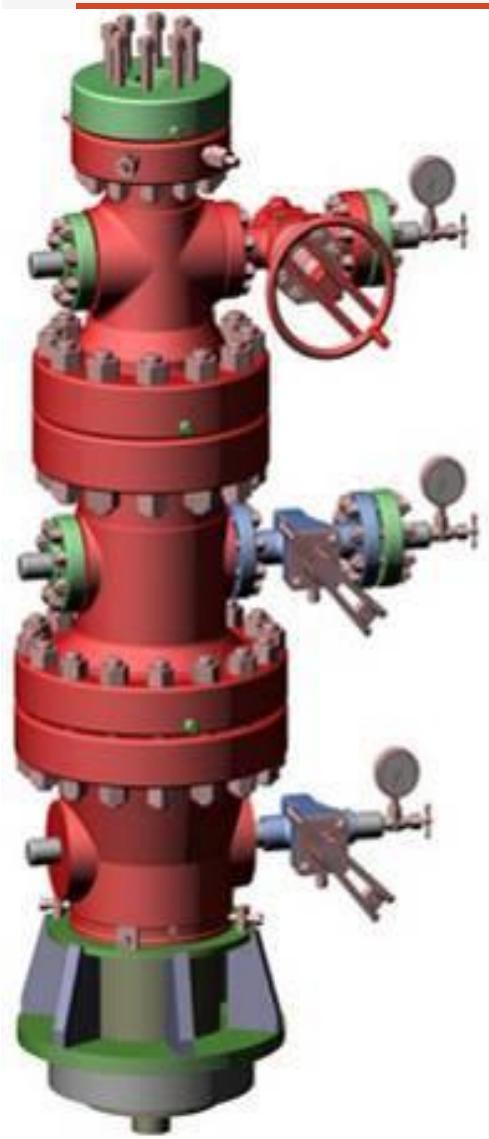


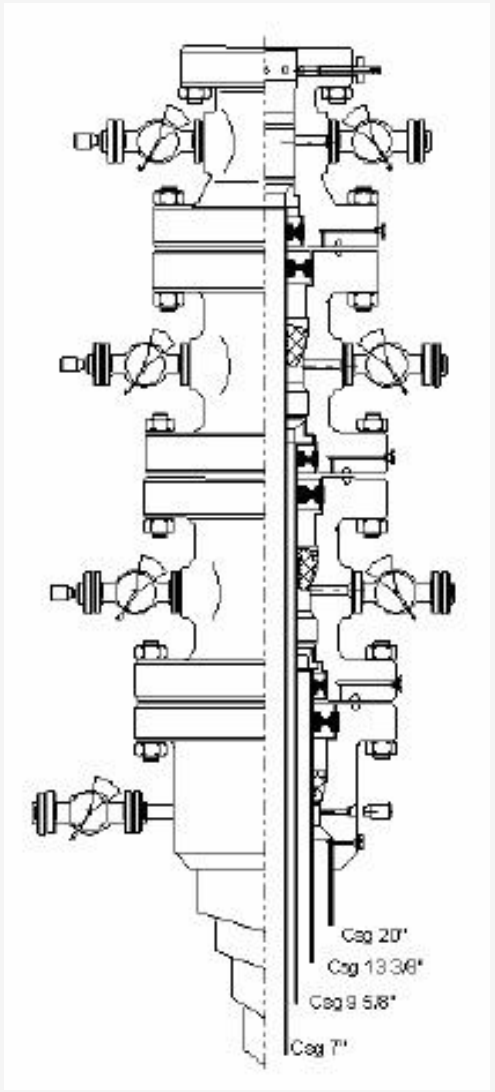
Big Bore with
13-3/8” Long string

Big Bore with
13-3/8” Dual string
OR
Tie Back String

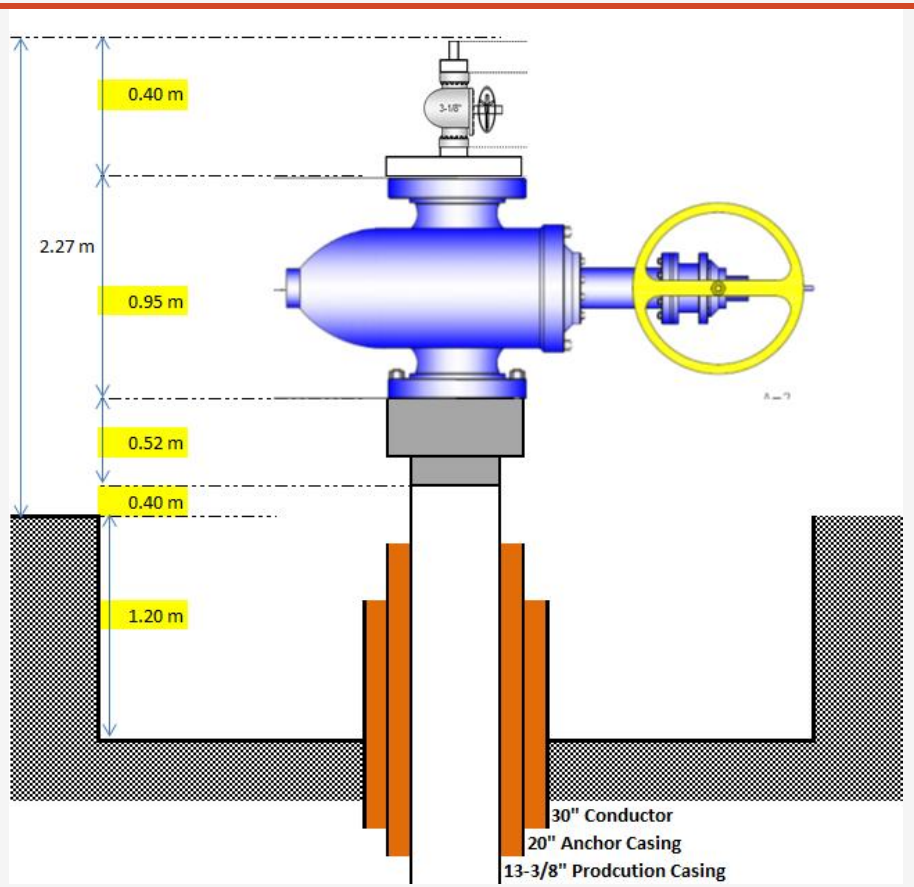


Typical Wellhead Schematic





Oil and Gas Well

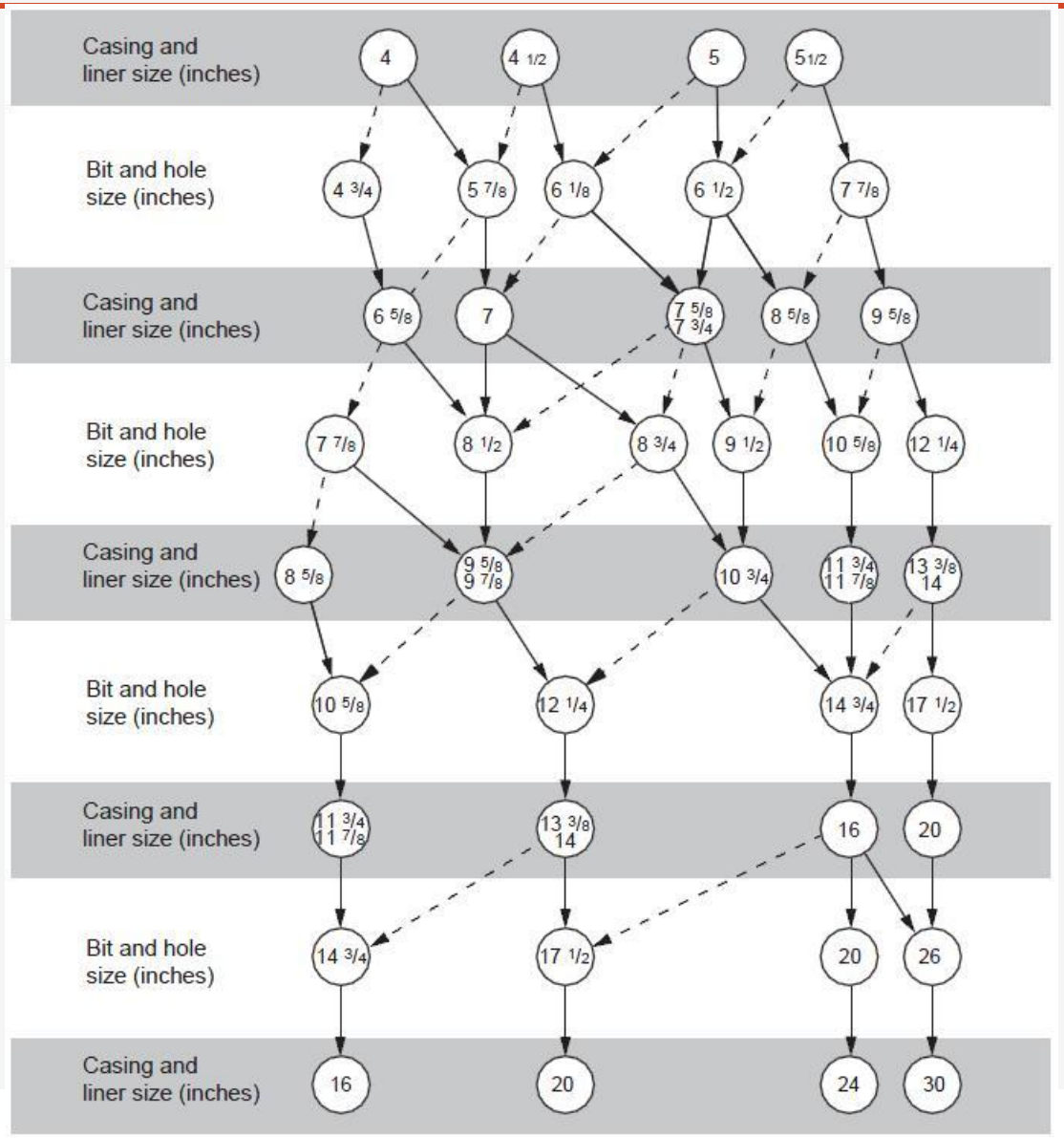


Geothermal Well

- **29-1/2” 500 psi Starting Head**
- **21-1/4”, 2M, Starting Head**
- **13-5/8” 3M Section A Wellhead**
- **12” ANSI 900, Master Valve**
- **Companion Flange**

TYPICAL LAND RIG DRILLING OPERATION

1. Pre-installed 30″ Conductor
2. **Drill 26″ Hole**
3. **Run and cement 20″ casing**
4. **Wellhead and BOP**
5. Drill 17-1/2″ Hole
6. Run and cement 13-3/8″ casing
7. Wellhead and BOP
8. **Drill 12-1/4″ Hole**
9. **E-Log**
10. **Run 10-3/4″ Perforated casing**
11. **Contingency Plan:**
    a. **Drill 9-7/8″ Hole**
    b. **Run 8-5/8″ Perforated casing**
    c. Drill 7-7/8″ Hole
    d. Run 7″ Perforated casing
12. **Well Test**
13. **Rig Release**
14. **Rig Down – Skid or Demob**



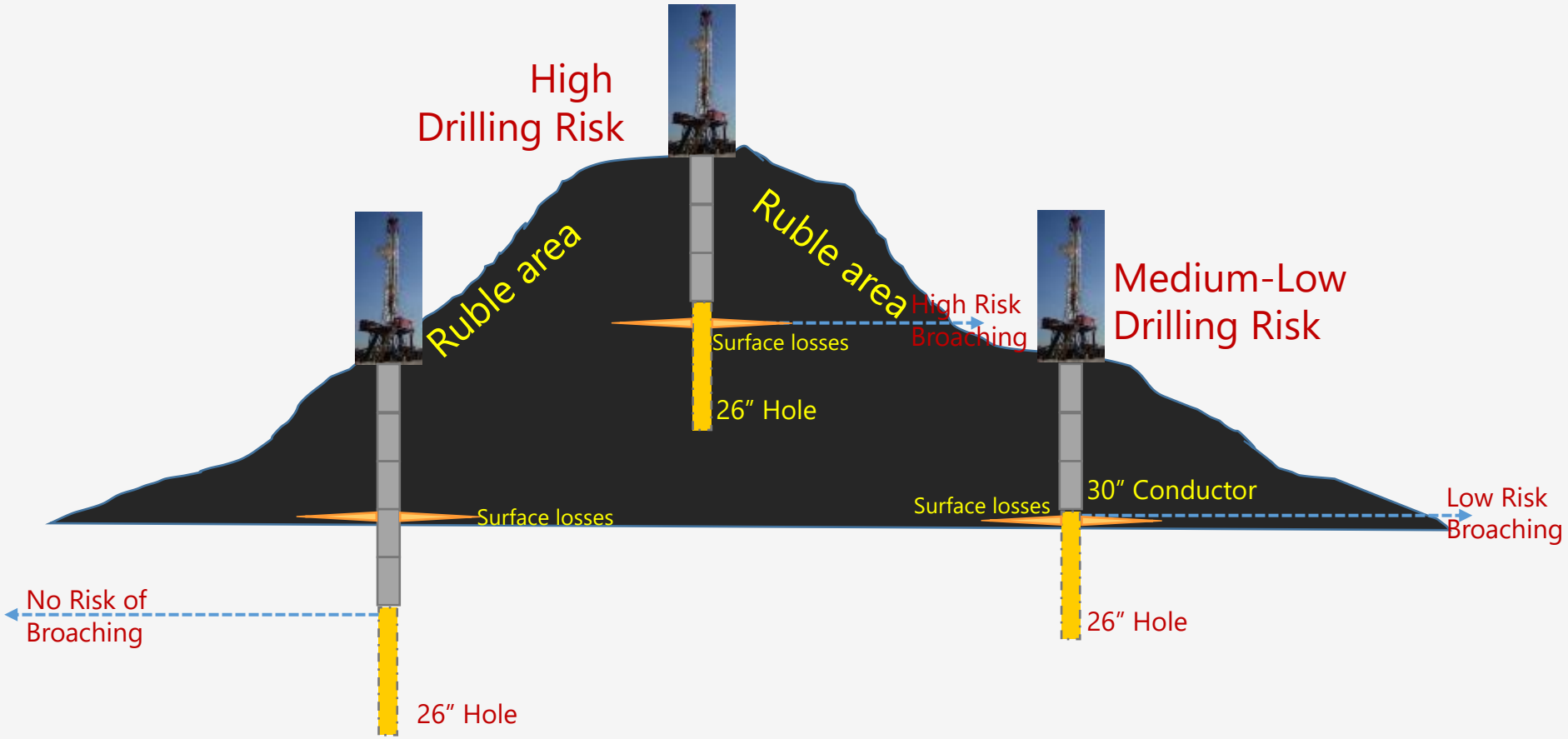
Pre-Drilling Preparation

- Service Company Equipment Inspection
- Drill On Paper
- Pre-Rig Mobilization Meeting
- Pre-Spud meeting
- 3rd Party Rig Inspection
- Dress Mud Pump Liner
- Drill String Inspection
- Drilling Equipment Checklist
- EBTKE Spud Checklist and Spud Approval
- Praying
- Pick Up Drill string and Rack Back
- Discuss Drilling shallow hazard: mud losses, shallow gas, etc.

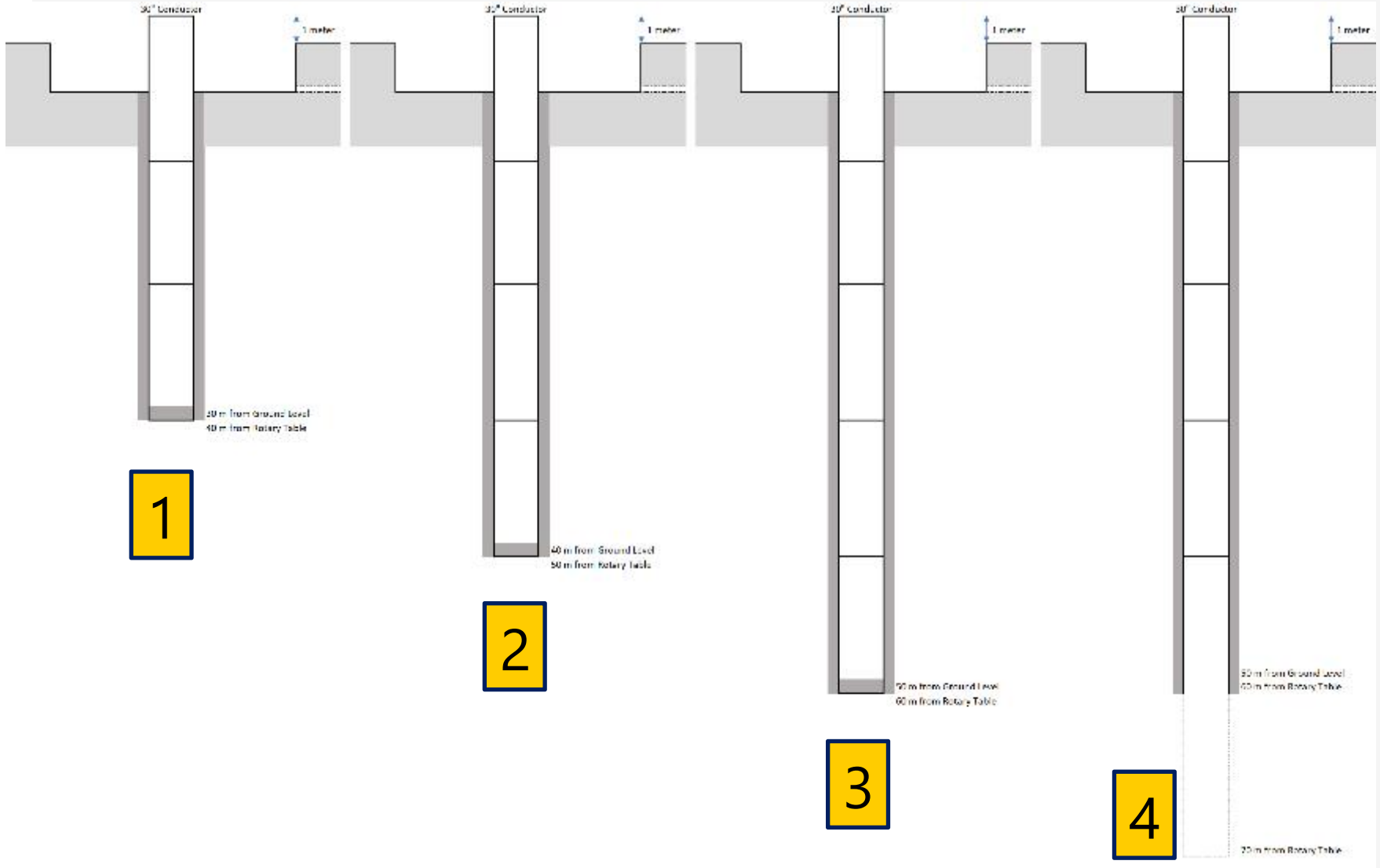
DRILLING OPERATION

30″ CONDUCTOR INSTALLATION

30″ Conductor Depth



Conductor Depth Illustration



26" Directional BHA:

9-1/2" MM + 9-1/2" EM MWD Tool

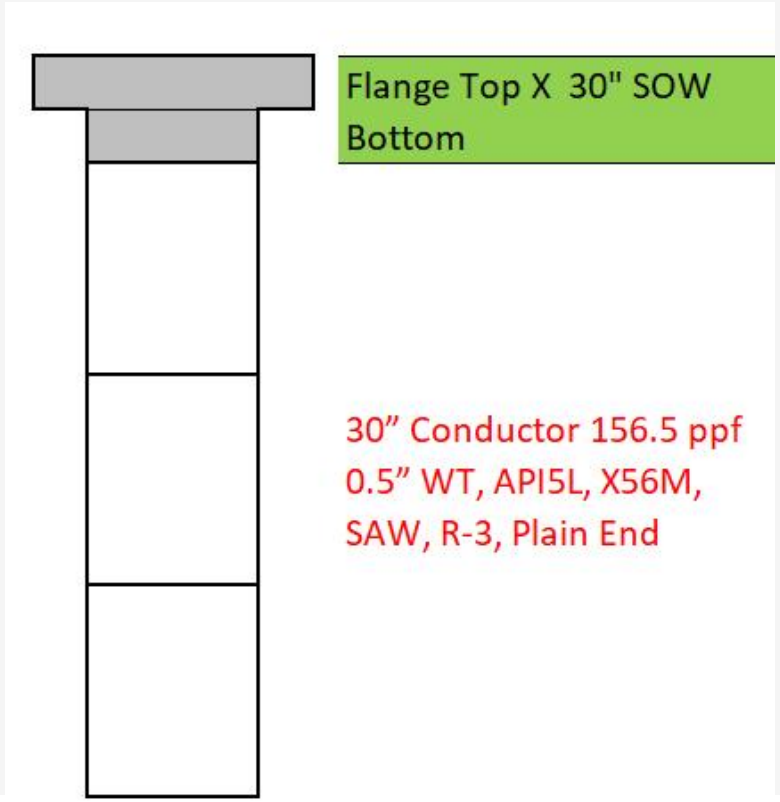
No	BHA Component	OD in	ID in	Connection Up	Connection Down	Conn.	LENGTH m	Cumm Length
1	26" Bit	26	TFA	7⅝ Reg		P	0.5	0.5
2	9-5/8" Mud Motor	9⅝	4.75	7⅝ Reg	7⅝ Reg	B/B	10.7	11.2
3	9 1/2" Catcher Sub w/ Float Sub	9½	3	7⅝ Reg	7⅝ Reg	B/P	1	12.2
4	24" IBS Stabilizer	9½	3	7⅝ Reg	7⅝ Reg	B/P	1.5	13.7
5	Shock Sub 9 1/2"	9½	3	7⅝ Reg	7⅝ Reg	B/P	2	15.7
6	9 1/2" NMDC	9½	3	7⅝ Reg	7⅝ Reg	B/P	9.1	24.8
7	9-1/2" MWD XEM 800	9½	3	7⅝ Reg	7⅝ Reg	B/P	9.1	33.9
8	9-1/2" MWD Gap Sub	9½	3	7⅝ Reg	7⅝ Reg	B/P	9.1	43
9	9-1/2" NMDC (Optional)	9½	3	7⅝ Reg	7⅝ Reg	B/P	9.1	52.1
10	9-1/2" Pony NMDC (Optional)	9½	3	7⅝ Reg	7⅝ Reg	B/P	3.0	55.1
11	X/O (7 5/8" REG Pin x 6 5/8" REG)	9½	3	6⅝ Reg	7⅝ Reg	B/P	3.0	58.1
12	7 x 8" DC	8	2.812	6⅝ Reg	6⅝ Reg	B/P	63.7	121.8
13	8" Hydraulic Jars	8	2.812	6⅝ Reg	6⅝ Reg	B/P	10.7	132.5
14	8" DC	8	2.812	6⅝ Reg	6⅝ Reg	B/P	9.1	141.6
15	X/O 6 5/8" RP x 4 1/2" IF9	8	2.812	4½ IF	6⅝ Reg	B/P	3.0	144.6
16	6 3/4" DC	6¾	2.813	4½ IF	4½ IF	B/P	9.1	153.7
17	15 x 5" HWDP	5	3	4½ IF	4½ IF	B/P	136.5	290.2
18								

- 26" Directional BHA length = > 58 m
- Option-3 and 4, we could run 26" Directional BHA without any drill out rotary BHA
- Rig time saving = 20 hours

Conductor Installation

- Pre-Driven by Pile Driver
- Using Auger System
- Installed by Rig: drill with 36” bit or 36” hole opener, run 30” conductor and cemented

30” conductor configuration
- 2 – 5 joints
- 11 m / joint



Conductor Installation - Pre-Driven



- Diesel Pilling Hammer, Delmag D-44 or equivalent
- 30” casing with 1” thickness
- 45 Ton Crawler Crane
- Bevel machine and cutting equipment
- Welding Equipment

- Shoe Joint: 30° internal bevel.
- The Driving Operation will be stopped:
  - Reached desired Depth
  - Penetration Rate slow to 200 blows per foot.

Pile Hammer – Blow/ft

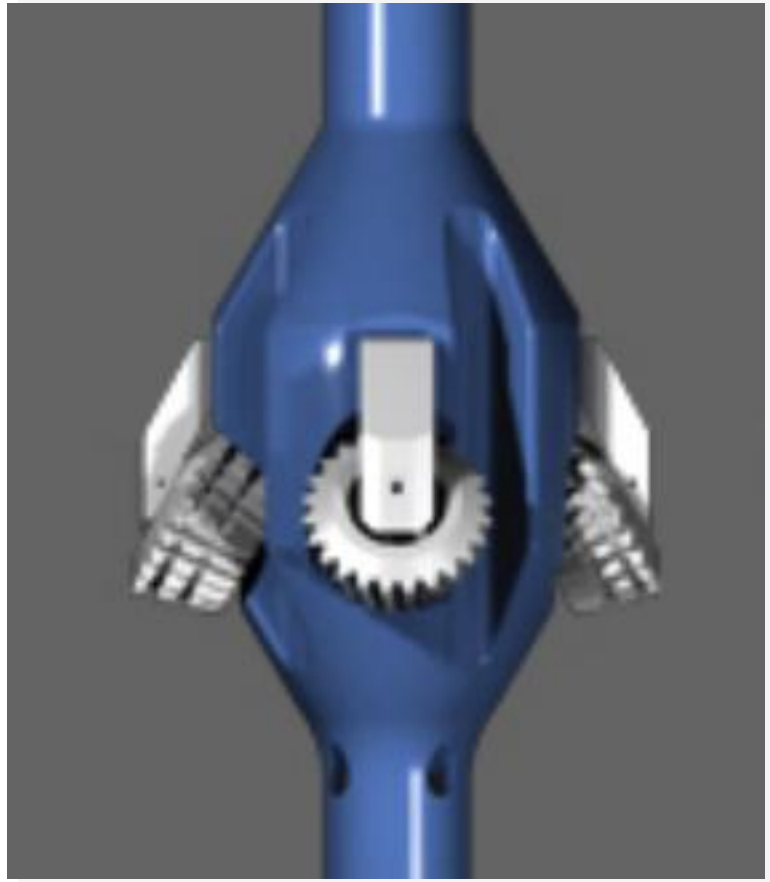
Pipe Size And Wall Thickness	Blows Per ft:	Hammer Size		
		D 22	D 30	D 44
20 x .312	Normal			
	Maximum			
20 x .375	Normal	65 - 70		
	Maximum	90		
20 x .500	Normal	65 - 90	55 - 80	
	Maximum	120	110	
20 x .750	Normal	100 - 150	100 - 120	
	Maximum	160	140	
20 x 1.00	Normal	140 - 180	120 - 150	
	Maximum	200	170	
24 x .500	Normal	90 - 110	80 - 100	
	Maximum	150	140	
24 x .625	Normal	100 - 120	90 - 110	
	Maximum	170	160	
24 x .750	Normal	120 - 150	110 - 140	
	Maximum	200	180	
24 x 1.00	Normal	150 - 200	150 - 180	
	Maximum	250	200	
26 x .500	Normal	100 - 150	90 - 100	
	Maximum	200	170	
26 x .750	Normal	150 - 180	110 - 150	
	Maximum	250	200	
26 x 1.00	Normal	200 - 220	175 - 200	
	Maximum	300	250	
30 x .500	Normal	150 - 200	100 - 150	
	Maximum	250	200	
30 x .625	Normal	200 - 225	140 - 175	100 - 130
	Maximum	275	250	150
30 x .750	Normal	250 - 300	150 - 200	130 - 160
	Maximum	350	300	180
30 x 1.00	Normal	300 - 350	200 - 300	150 - 200
	Maximum	400	350	250
36 x .500	Normal	160 - 210	120 - 170	
	Maximum	260	220	
36 x .625	Normal	210 - 235		120 - 140
	Maximum	280		160
36 x .750	Normal	260 - 310	200 - 250	150 - 170
	Maximum	360	350	190
36 x 1.00	Normal	320 - 360	250 - 350	180 - 210
	Maximum	425	400	260
*48 x .750	Normal			170 - 180
	Maximum			200
*48 x 1.00	Normal			180 - 200
	Maximum			300
* With adapter				

Conductor Installation - Auger System



- 9-11 loads - Expensive Mobilization and Demobilization Cost
- Cost effective if we do many conductor in once time.
- Pre-installed 30" conductor (cemented) prior to rig arrival
- Activity:
  1. Rig Up
  2. Drill 36" Hole
  3. Run 30" Conductor (2-4 joint)
  4. Pump Cement

Conductor Installation – By Rig



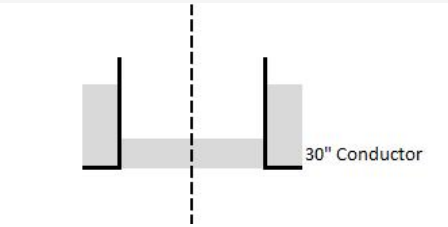


Hole Opener

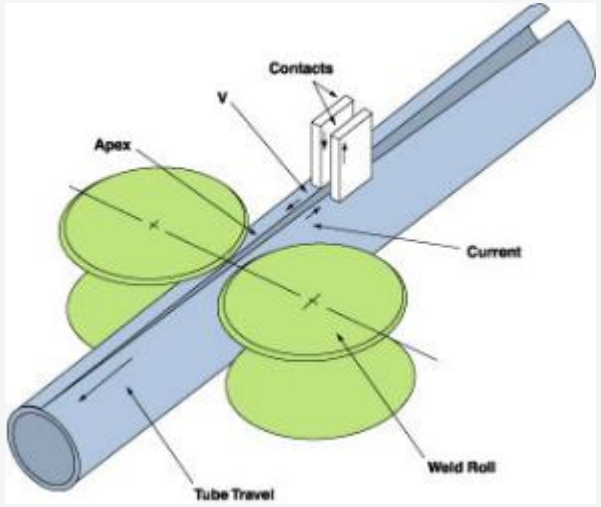


36″ Mill Tooth Bit

- Use 26″ Bit tandem with 36″ Hole Opener
- OR
- Drill with 36″ Mill Tooth Bit.
- Run 30″ Pipe and Cemented



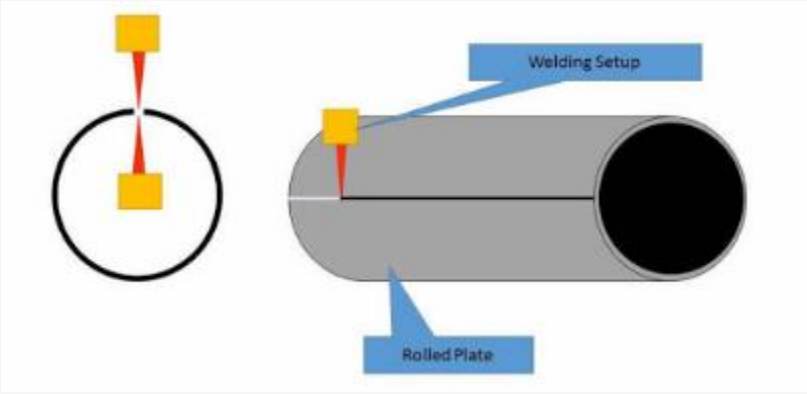
30″ CASING CONDUCTOR MANUFACTURE PROCESS



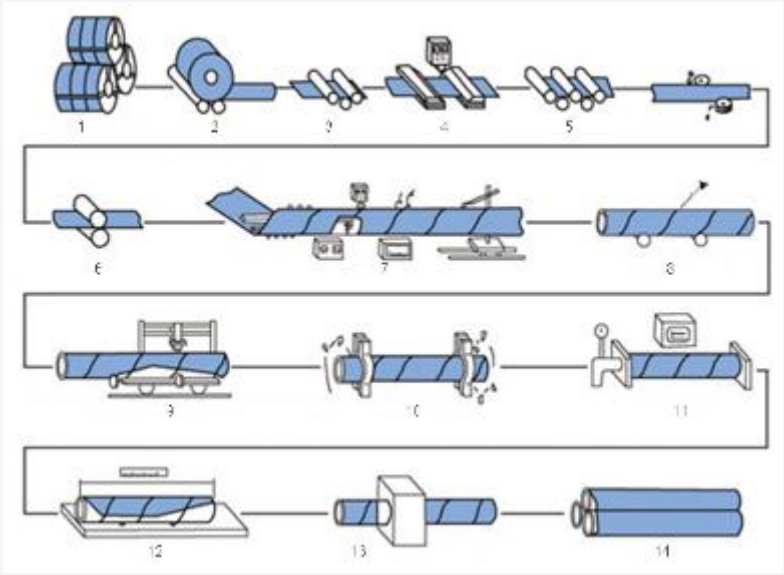
ERW: Electric Resistance Welded



SAW: Submerge Arc Welding



LSAW: Longitudinal Submerge Arc Welding
DSAW: LSAW with double seam weld in inside and outside



Spiral SAW

CASING MANUFACTURE PROCESS

ERW: Electric Resistance Welded

- Manufactured by rolling metal and then welding it longitudinally across its length

SAW: Submerge Arc Welding

LSAW: Longitudinal Submerge Arc Welding,

- Single seam weld on the outer surface

DSAW: LSAW with double seam weld in inside and outside

SPIRAL SAW



SEAMLESS:

- Manufactured by extruding the metal to the desired length. 20-30% higher price than ERW casing

30” Conductor Price Comparison

30” Casing	Welded	USD/Jt
1” Wall Thickness (For Piling)	ERW	
	LSAW	
	DSAW	
	SPIRAL	
	Seamless	

Stronger pipe, for piling installation



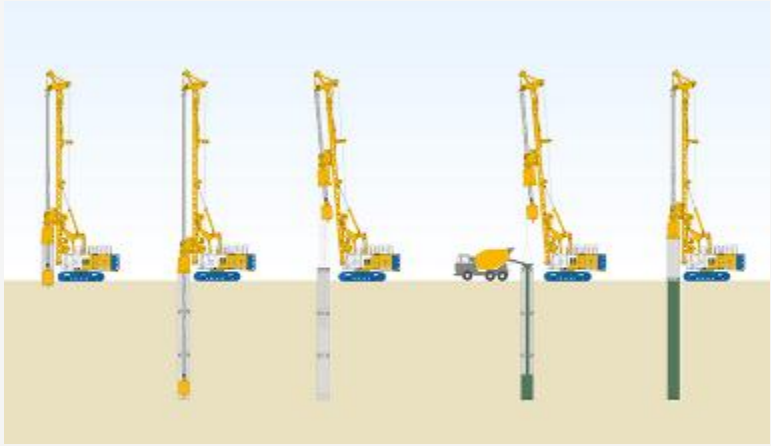
30” Casing	Welded	USD/Jt
0.75” Wall Thickness (Recommended for Auger)	ERW	
	LSAW	
	DSAW	
	SPIRAL	
	Seamless	

Thin pipe, for Auger installation



30” Casing	Welded	USD/Jt
0.5” Wall Thickness (Low Price)	ERW	
	LSAW	
	DSAW	
	SPIRAL	
	Seamless	

Thin pipe, for Auger installation



DRILLING OPERATION

RIG MOVE AND RIG UP

RIG MOVE AND RIG UP

Rig Mobilization

Rig move from rig sources to the first well location:

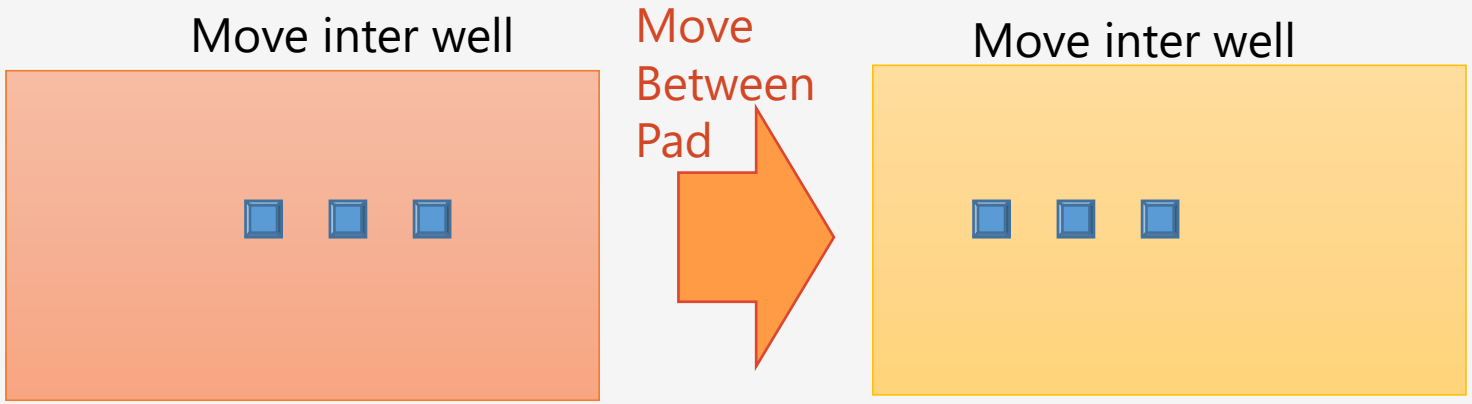
- Pre Rig Move Meeting
- Access Road Survey
- Journey Management Plan (JMP)
- Rig Mobilization Socialization to Community.
- Permit and fee
- Rig and Mini Camps Lay Out
- Base Camp Layout
- Rig Equipment:
  - Number of Loads
  - Heaviest Loads
  - Sizes (W,L,T)
  - Loads sequences
- Transporter Company
- List of Heavy Equipment
- Landing Port

Rig Skid

- Skidding or Walking
- With or Without DP stand in derrick
- Skid system specification (stroke length, speed, Hydraulic Power System)
- Leveling and compaction
- Maximum Wellhead Height

Rig Move Between Pad

- Rig Down
- Rig Move and Rig Up → Similar with Rig Mobilization without Camps.



Heavy Equipment for Access Road















Rig Moving - Truck & Trailer


Low Boy


Hi Boy


Winch Truck


Dump Truck


Vacuum Truck


Foco Truck

Heavy Equipment for Rig Moving



Roughter Crane



Crawler Crane



Forklift

Rig Mobilization Plan – Things to do

Pre-Rig Mob

- JMP and Rig Mob Plan
- Load list
- Heavy Equipment Inspection
- Rig Mob Socialization
- Permits
- Rig Mob Risk Assessment
- Local Hire

Rig Mob

- JMP and Rig Mob implementation
- Daily Coordination Meeting
- JSA and Safety meeting
- Daily Rig Mob Report
- Back up Dozer and Prime Mover
- Always check any crossing cable.
- Road maintenance
- CBR test for Rig Foot print
- MCU for all personnel
- SIMOP with Service Company

Rig Up

- 3rd Party Rig Inspection
- EBTKE Spud approval
- Rig Acceptance
- Function test and Pressure test
- Berita Acara for Service Company

Heavy Equipment and Light Vehicles

During Rig Down, Mob and Rig Up: • **Origin to Well Pad** • **Between Well Pad**	**During Drilling Operation**
• 3 cranes (Roughter 50T and 2 X Crawler 80T) • 2 Forklift, 8-10 Ton • 2 Low Bed, 6X6 • 5-10 Hi-Bed, 6x6 or 4x6 • 3-5 Tronton • 1 Foco Truck	• 2 cranes (Roughter 50T and Crawler 80T) • 1 Forklift, 8-10 Ton • 1 Prime Mover + Hi-Bed and Low Bed trailer • 1 Foco Truck • 1 Back Hoe (Call Out)
• 1 Crew Bus, 25 seats • 2 Double Cabin • 1 Minibus • 1 Ambulance	• 1 Crew Bus, 25 seats • 2 Double Cabin • 1 Minibus • 1 Ambulance

TRUCK MOUNTED - RIGING UP



Raised Derrick Proses of TRUCK MOUNTED Rig

SKID RIG SYSTEM





RIG RMI NO 98 – New Generation Rig



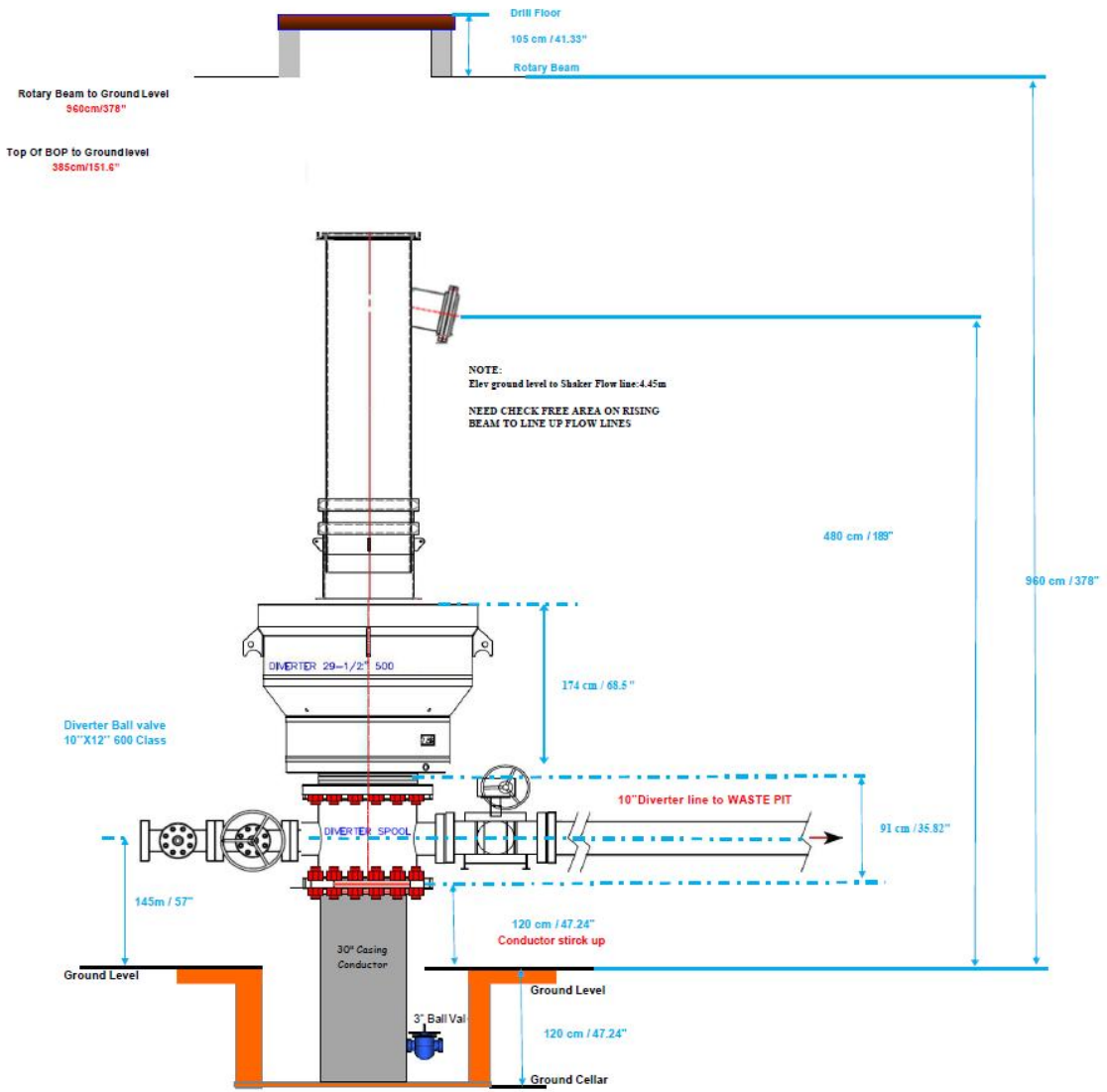




- Walking Rig

DRILLING SECTION

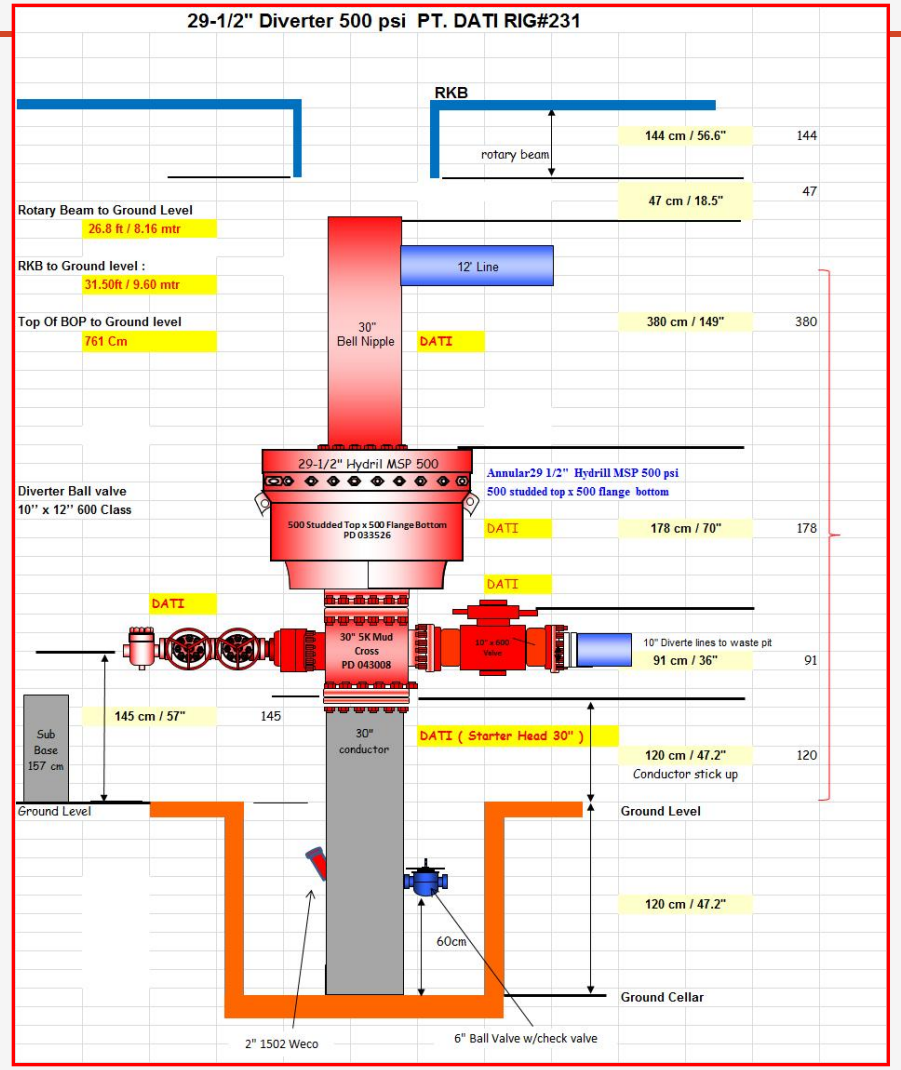
Drilling Activity - Install 29-1/2" Diverter



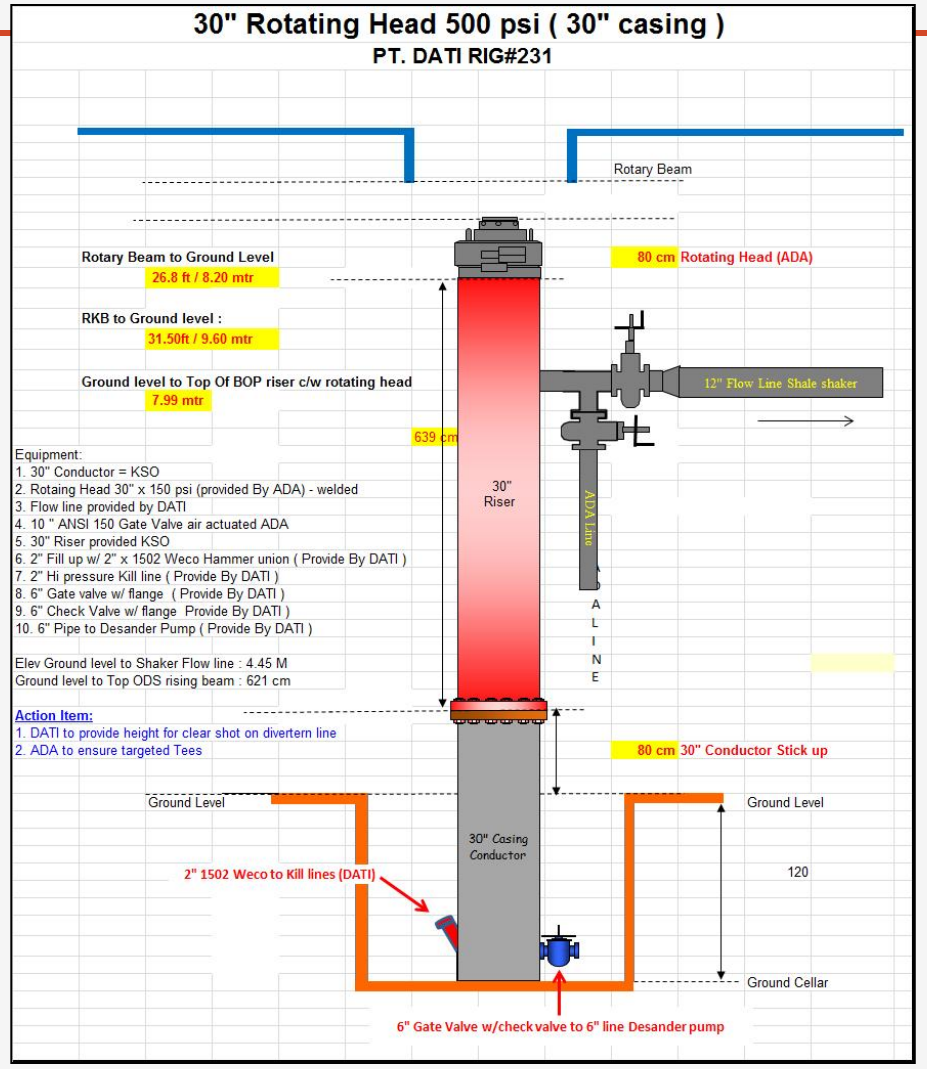
- Cut and bevel 30" Conductor
- Pick up 29-1/2" Flange
- Welded on 30" Conductor
- Nipple Up 29-1/2" Spool
- Nipple Up 29-1/2" Diverter with Riser
- Function Test
- Install Diverter Line – Straight line, no bend.
- Install Flow Line
- Fill with Water for Leak test

- Ready to Spud

BOP CONFIGURATION - Geothermal



Diverter + Diverting Lines
For First Well



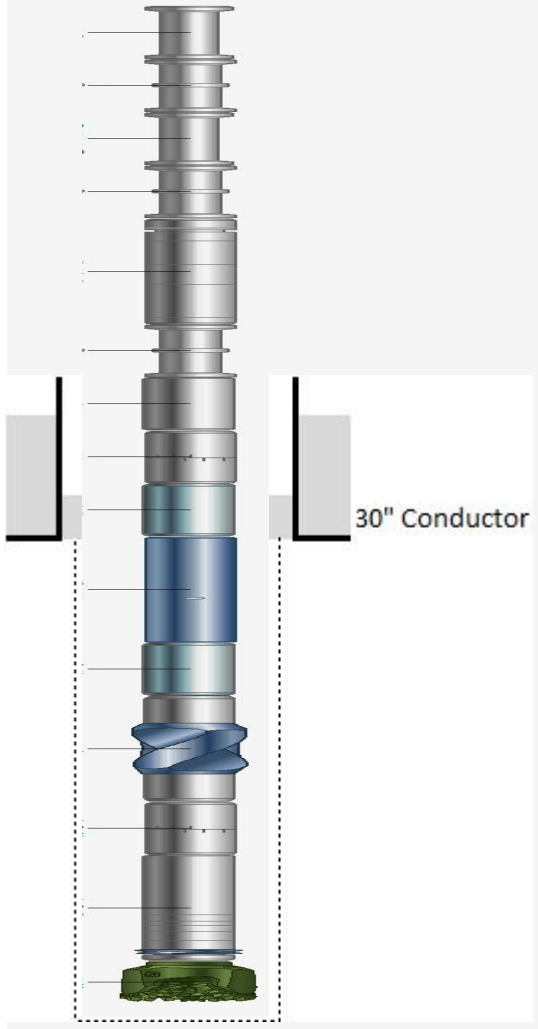
30" Rotating Head + Flow Lines
Next sequence well – if no shallow hazard

DRILLING OPERATION

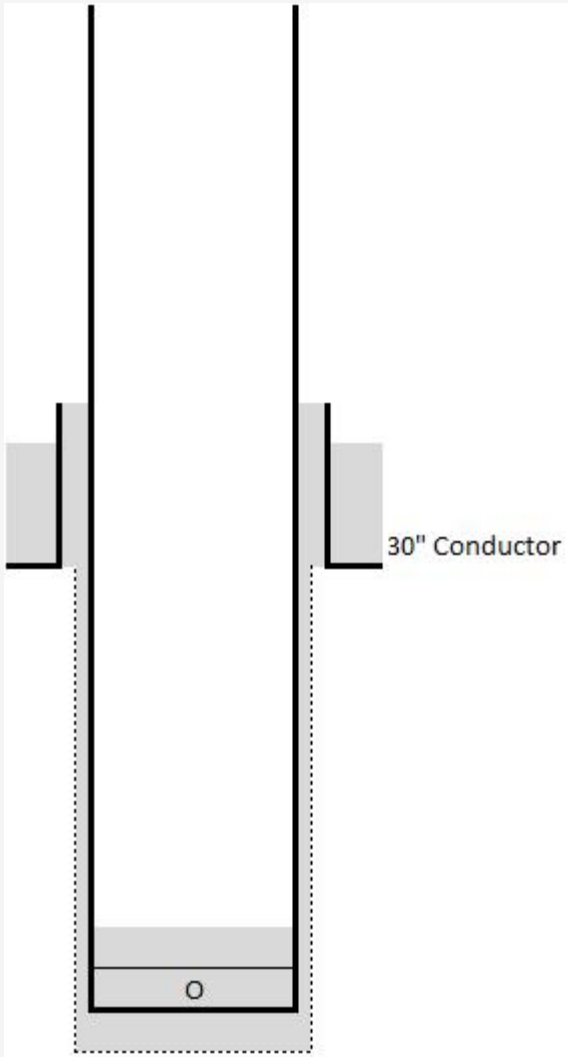
26″ HOLE SECTION

Drilling Operation - 26" Hole Section

Drilling



Run Casing & Cemented

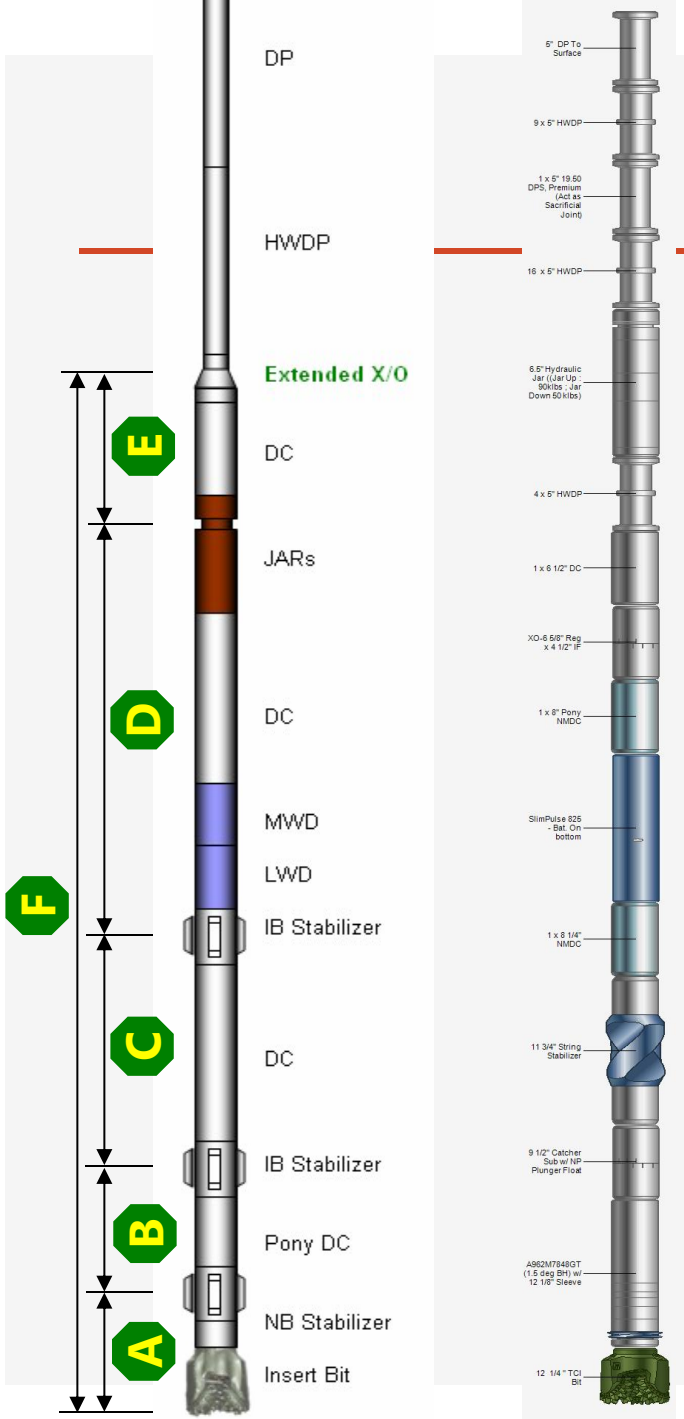


- Make Up 26" Bottom Hole Assembly (BHA)
- Run in hole and tag
- Spud the well
- Drill 26" hole
- Circulation hole clean
- Wiper trip
- Pull Out Of Hole (POOH)
- Run 20" Casing and Cement
- Wait On Cement
- Pick up 29-1/2" Diverter
- Rough Cut 20" casing and Laid Down
- Nipple Down 29-1/2" Diverter
- Final Cut 20" casing

DRILLING OPERATION – Drill 26” Hole Section

BHA COMPONENT

BHA Component



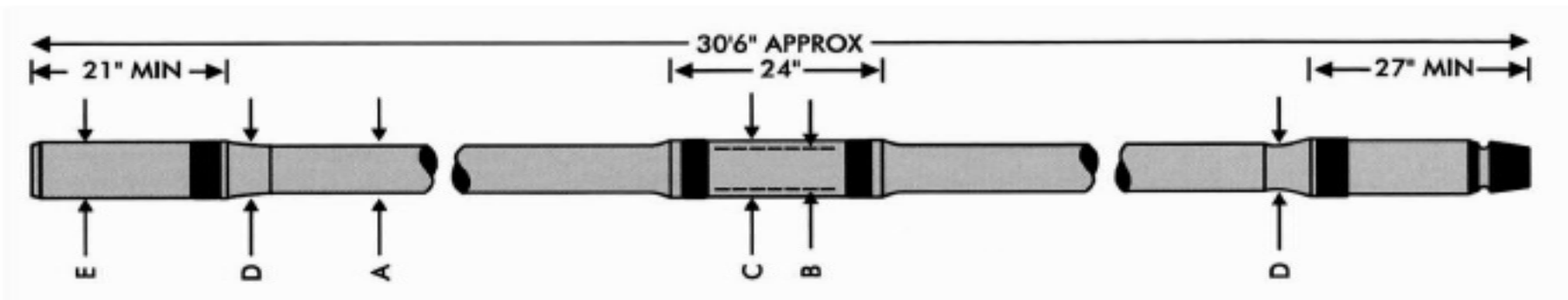
BHA Component
1. DP
2. HWDP
3. DC
4. NMDC
5. Jars
6. Stabilizer
7. MWD
8. LWD
9. Motor
10. Bit

Special Application
1. Reamer
2. Under Reamer
3. Hole Opener
4. Circ. Sub
5. Csg Cutters
6. Bumper Sub
7. Shock Sub
8. Thruster
9. Milling
10. Fishing Tools
11. Running Tools





Heavy Weight Drill Pipe



Nom. Size (A)	Tool Joint OD (E)	Nom. ID (B)	Wall Thickness	Centre Upset OD (C)	Elevator Upset OD (D)	Connection Size and Type	Approx. Weight LBS		Make Up Torque Ft.-Lbs.
							Per Foot	Per Joint 30 ft.	
3½	4¾	2 ¼	.718	4	3⅝	NC 38 (3½ IF)	25.3	760	9,900
4½	6¼	2 13/16	.844	5	4⅝	NC 46 (4 IF)	41.0	1,230	21,800
5	6½	3	1.000	5½	5⅛	NC 50 (4½ IF)	49.3	1,480	29,400
5½	7	3 ¼	1.125	6	5 11/16	5½ FH	57.0	1,710	33,200
6⅝	8	4	1.312	7⅛	6¾	6⅝ FH	71.5	2,145	46,900

DRILL COLLAR TYPE

SLICK Type, SPIRAL Type







DRILL COLLAR SPECIFICATIONS

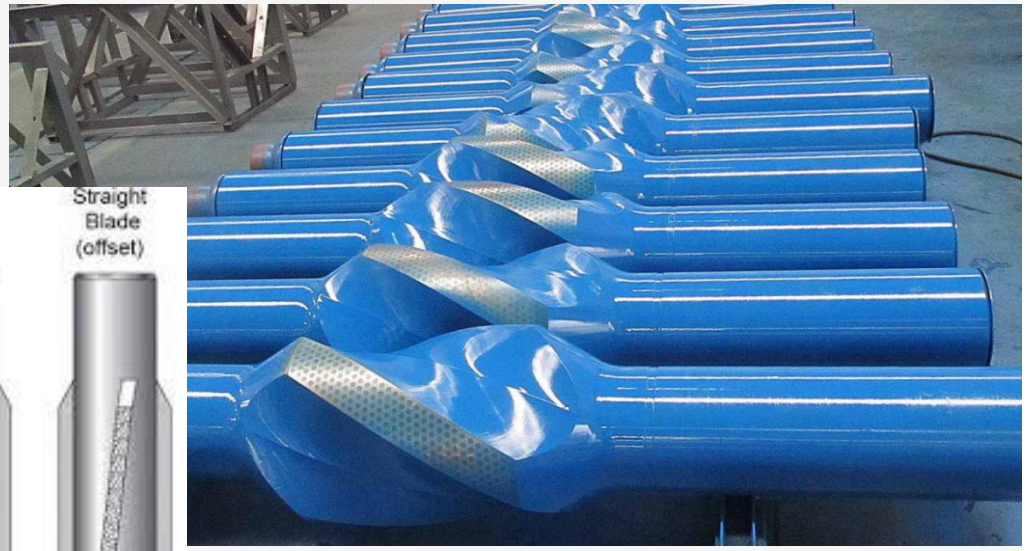
SIZE O.D INS.	BORE		CONNECTION STYLE AND SIZE		APPROX WEIGHT (30FT LONG) LB.	APPROX WEIGHT (3FT LONG) LB.
	Standard Ins.	Optional Ins.	Standard Bore A.P.I. No. Other	Optional Bore A.P.I. No. Other		
3½	1½		NC26 2⅜ IF	NC26 2⅜ IF	1012	1045
4⅛	2	1¾	NC31 2⅞ IF	NC31 2⅞ IF	1045	1078
4¼	2	1¾	NC31 2⅞ IF	NC31 2⅞ IF	1122	1155
4½	2	1¾	NC31 2⅞ IF	NC31 2⅞ IF	1298	1342
4¾	2¼	2¼	NC35	NC38 3½ IF	1412	1463
5	2¼	2	NC38 3½ IF	NC38 3½ IF	1606	1659
5¼	2¼	2	NC38 3½ IF	NC38 3½ IF	1793	1848
5½	2¼	2	NC38 3½ IF	NC38 3½ IF	2046	2112
5¾	2¼	$2^{13}/_{16}$	NC40 4 FH	4½ FH	2253	2328
6	2¼	$2^{13}/_{16}$	NC44	NC40 4½ FH	2486	2570
6¼	2¼	$2^{13}/_{16}$	NC46 4 IF	NC46 4 IF	2728	2816
6½	2¼	$2^{13}/_{16}$	NC46 4 IF	NC50 4½ IF	2761	2851
6¾	2¼	$2^{13}/_{16}$	NC46 4 IF	NC50 4½ IF	3025	3124
7	$2^{13}/_{16}$	2¼	NC50 4½ IF	NC50 4½ IF	3304	3412
7¼	$2^{13}/_{16}$	3	NC50 4½ IF	5½ FH	3586	3705
7½	$2^{13}/_{16}$	2¾	NC50 4½ IF	5½ REG	3885	4013
7¾	$2^{13}/_{16}$	3	NC56 6⅝ REG	NC56 6⅝ REG	4191	4327
8	$2^{13}/_{16}$	3	NC56 6⅝ REG	NC56 6⅝ REG	4514	4653
8¼	$2^{13}/_{16}$	3	6⅝ REG	6⅝ REG	4840	4990
8½	$2^{13}/_{16}$	3	6⅝ REG	6⅝ REG	5181	5337
8¾	$2^{13}/_{16}$	3	6⅝ REG	6⅝ REG	5416	5577
9	3	$2^{13}/_{16}$	NC61 7⅝ REG	NC61 7⅝ REG	5786	5973
9¼	3	$2^{13}/_{16}$	NC61 7⅝ REG	NC61 7⅝ REG	6182	6369
9½	3	$2^{13}/_{16}$	7⅝ REG	7⅝ REG	6538	6750
9¾	3	$2^{13}/_{16}$	NC70 7⅝ REG	NC70 7⅝ REG	6930	7154
10	3	$2^{13}/_{16}$	NC70 8⅝ REG	NC70 8⅝ REG	7260	7480
11	3	$2^{13}/_{16}$	NC77 8⅝ REG	NC77 8⅝ REG	9020	9295
11¼	3	$2^{13}/_{16}$	NC77 8⅝ REG	NC77 8⅝ REG	9460	9768
12	3	$2^{13}/_{16}$	NC77 8⅝ REG	NC77 8⅝ REG	10868	11187
14	3	$2^{13}/_{16}$	NC77 8⅝ REG	NC77 8⅝ REG	14949	15400

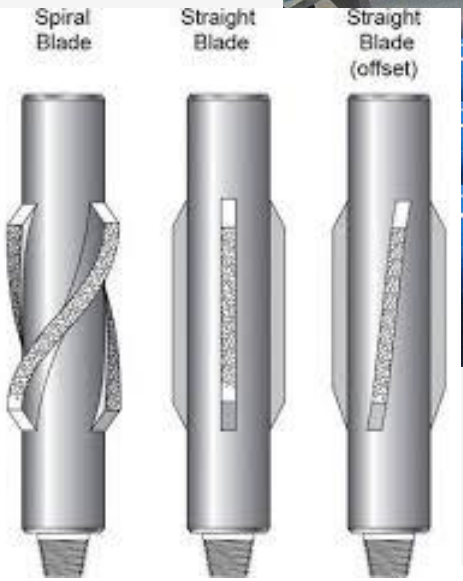
With 2 ends 11¼ O.D.

NOTE: H90 and X hole connections are supplied on request.

STABILIZER

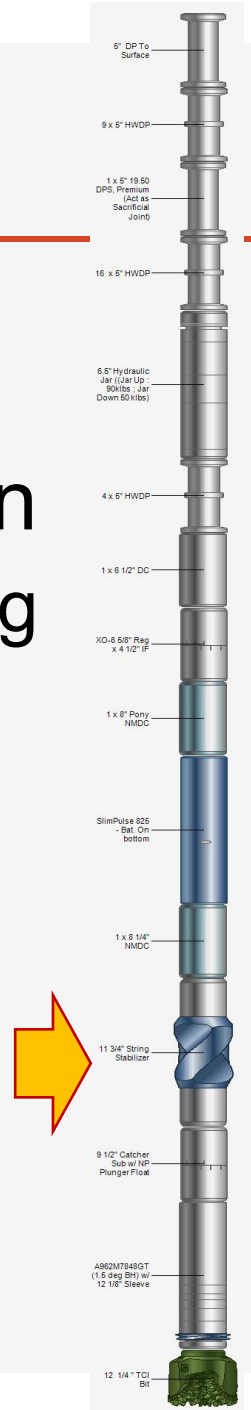




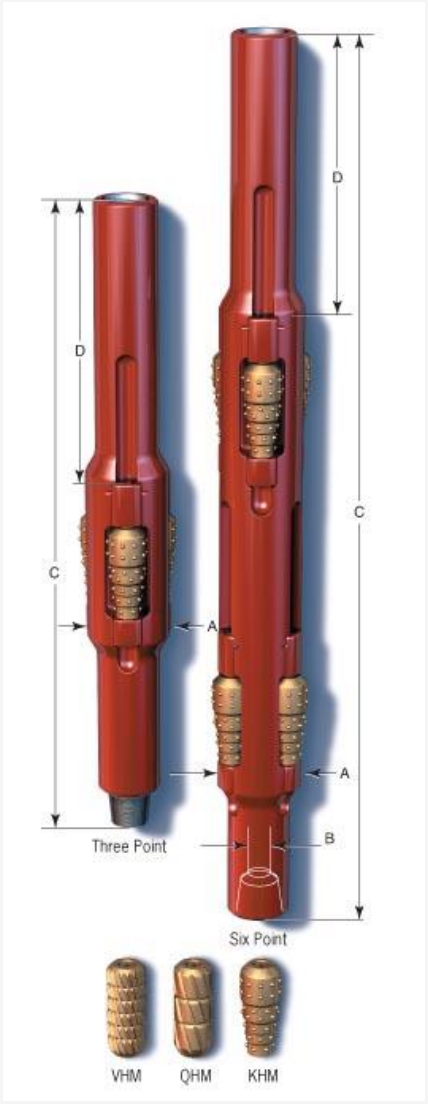


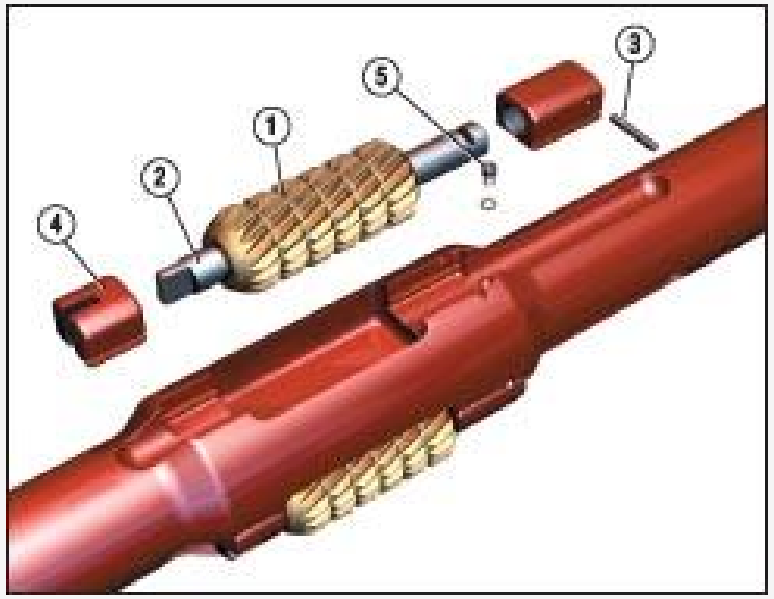


- Type and Size
- Connection
- Hard Facing



ROLLER REAMER TYPES





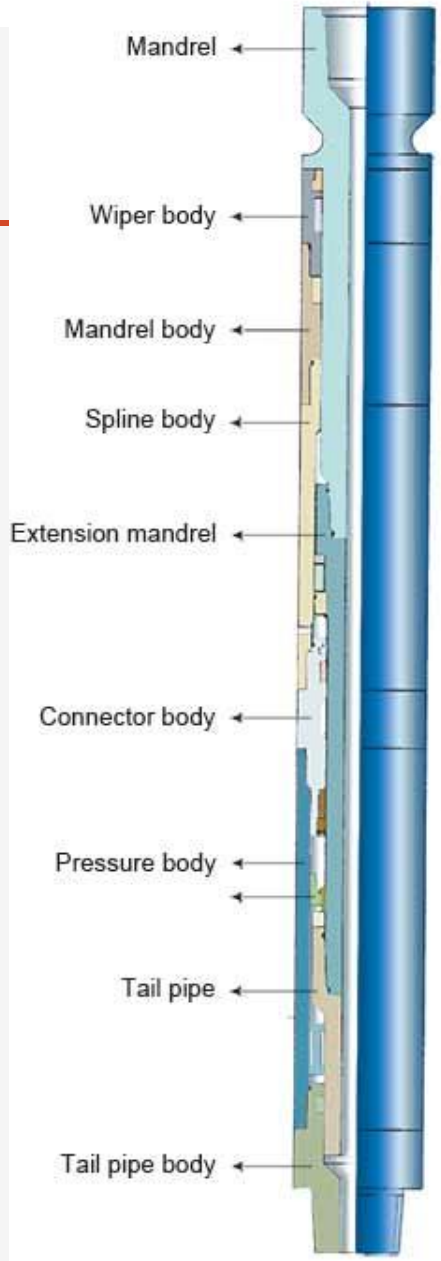
(1) Cutters
(2) Reamer Pin
(3) Cross Pin
(4) Drive-Fit
(5) Socket Head Cap Screws and Lock Washers

SHOCK-SUBS

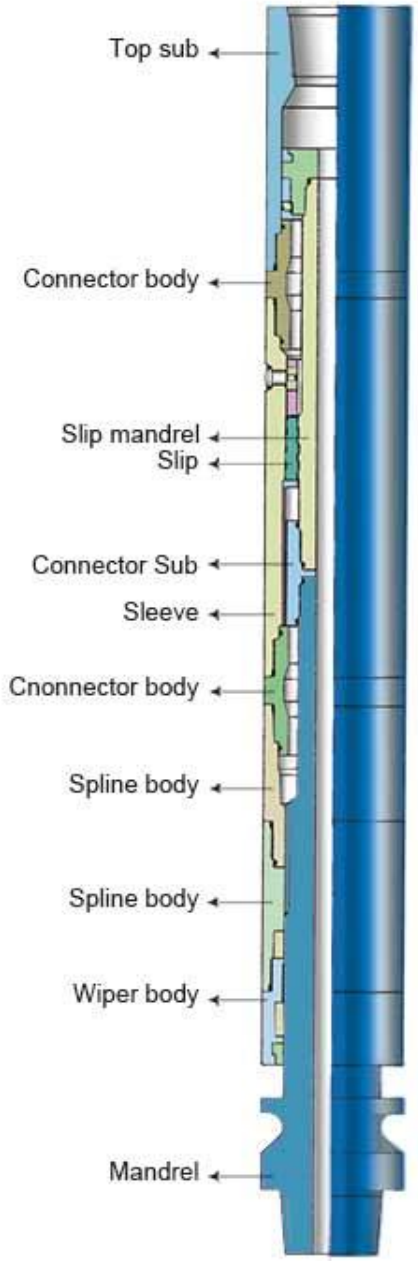



5" DP To Surface
9 x 5" HWDP
1 x 5" 19.50 DPS, Premium (Act as Sacrificial Joint)
16 x 5" HWDP
6.5" Hydraulic Jar ((Jar Up : 90klbs ; Jar Down 50 klbs)
4 x 5" HWDP
1 x 6 1/2" DC
XO-6 5/8" Reg x 4 1/2" IF
1 x 8" Pony NMDC
SlimPulse 825 - Bat. On bottom
1 x 8 1/4" NMDC
11 3/4" String Stabilizer
9 1/2" Catcher Sub w/ NP Plunger Float
A962M7848GT (1.5 deg BH) w/ 12 1/8" Sleeve
12 1/4 " TCI Bit

Drilling Jars

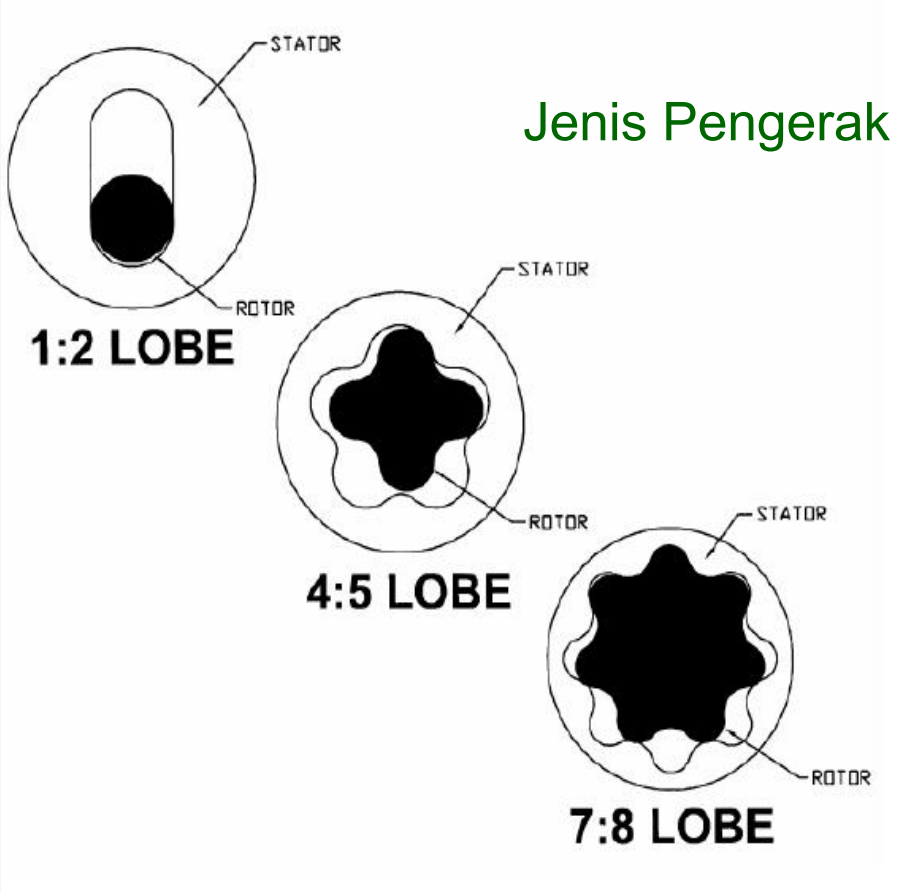


The drilling (up) jar is as shown in block drawing enclosed

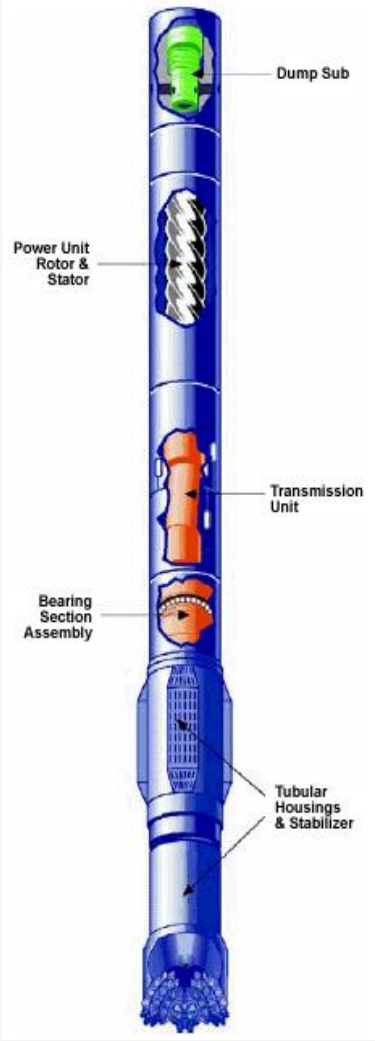


drilling (down) jar is shown in block drawing enclosed

MUD MOTOR

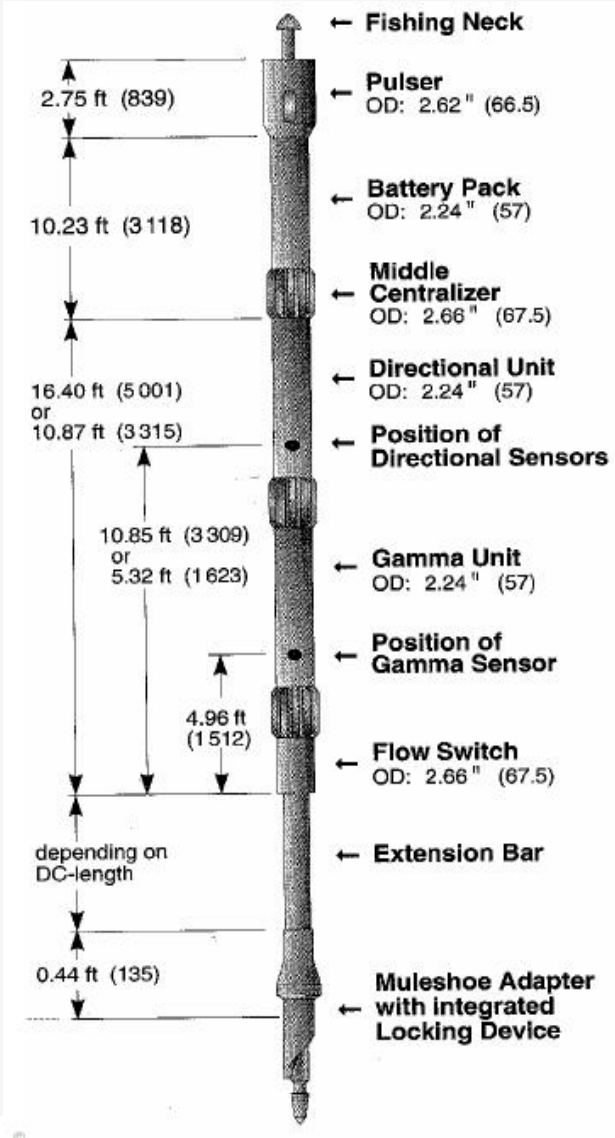


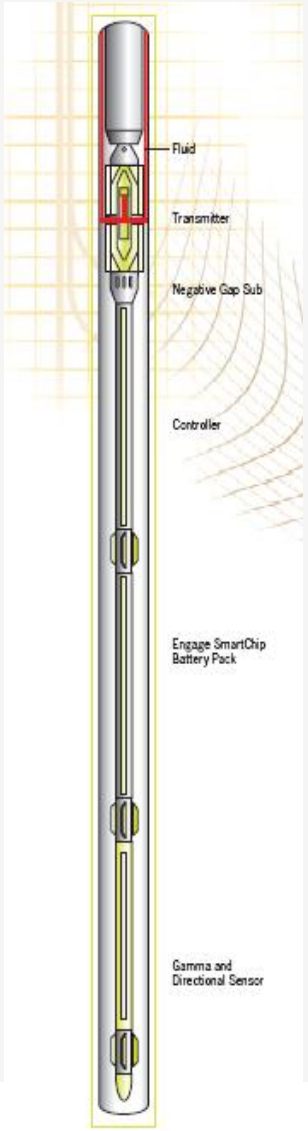
Jenis Pengerak Mud Motor



MWD (Measure While Drilling)

MWD w/ Pulse + MWD w/ Pulse -



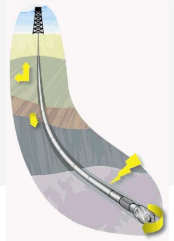




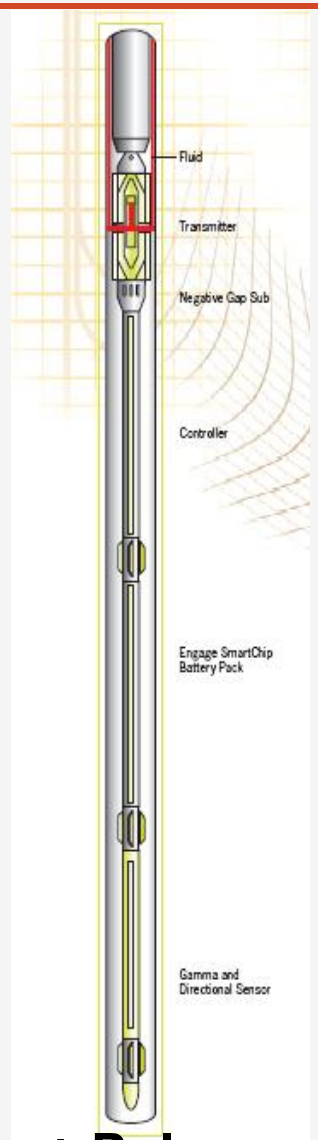
Pulsa-pulsa ini akan dihantarkan melalui media lumpur dalam bentuk bilangan Biner

Dengan program komputer, pulsa2 ini dikonvesikan dari bilangan biner menjadi decimal

Hasil survey berupa: sudut lubang, azimuth lubang, temperatur.



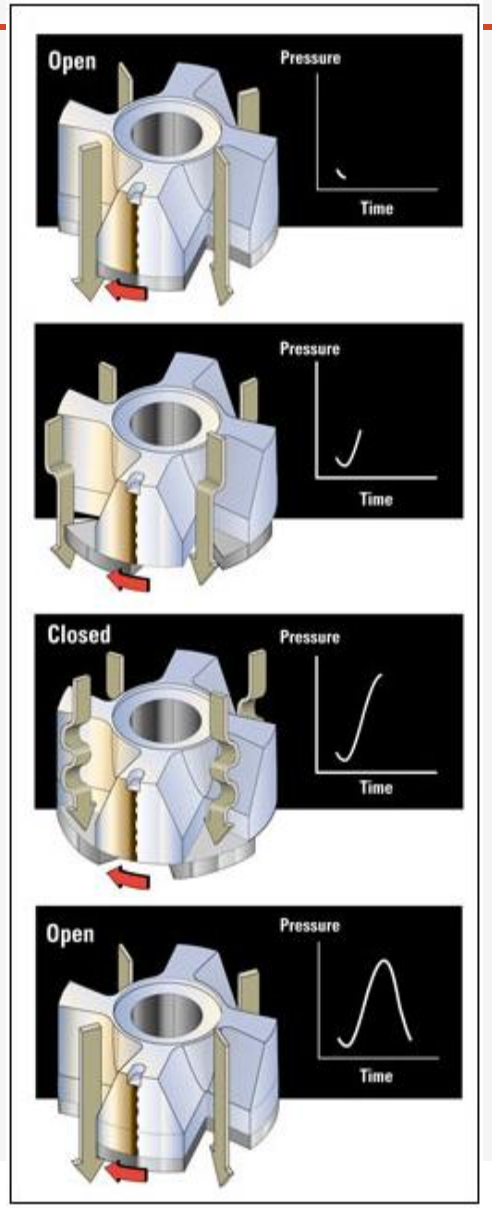
MWD PULSE PROCESS



+ Pulse



- Pulse

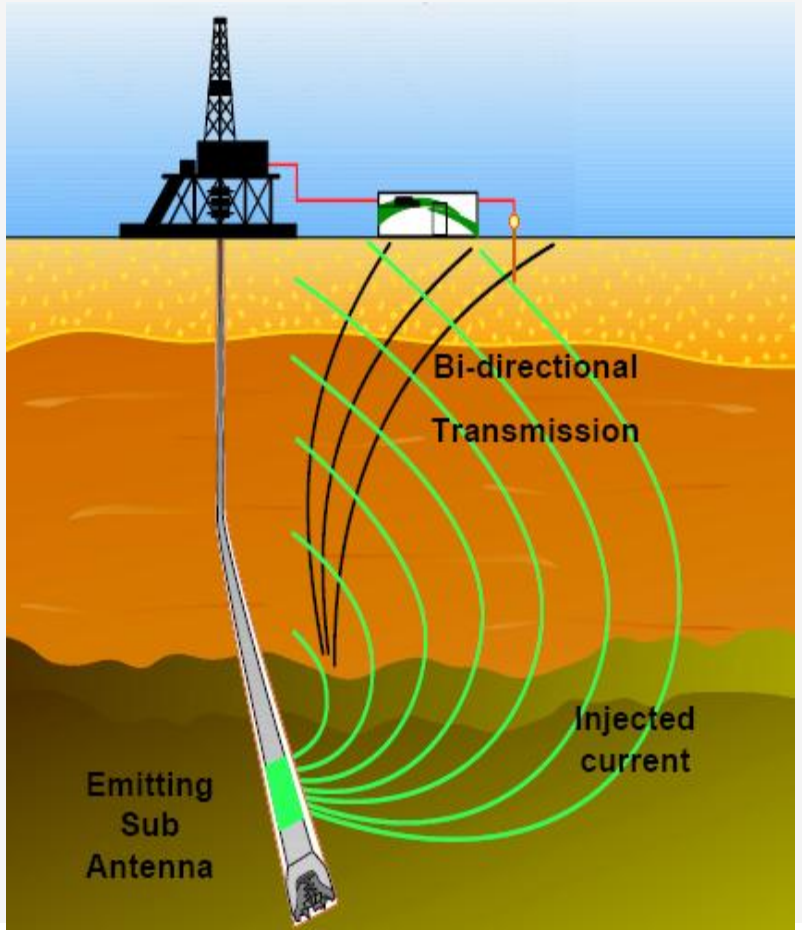


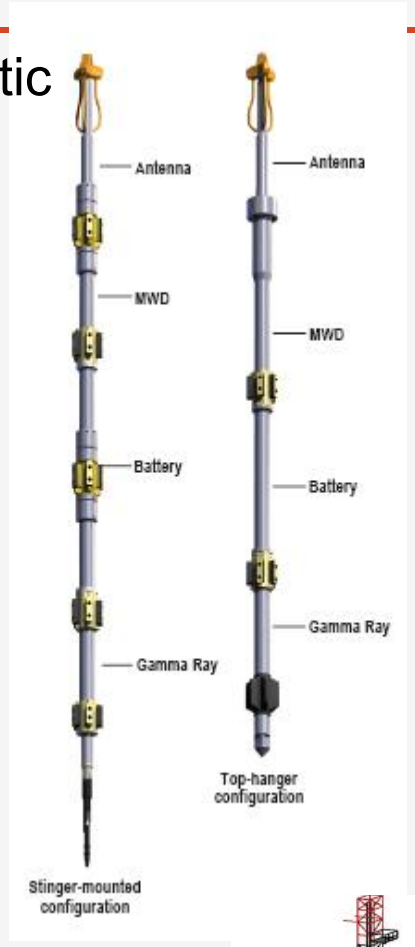
PowerPulse technology utilizes zero gap modulation and data compression techniques to achieve data transmission rates up to 16 bits per second

+ Pulse, Hi-speed

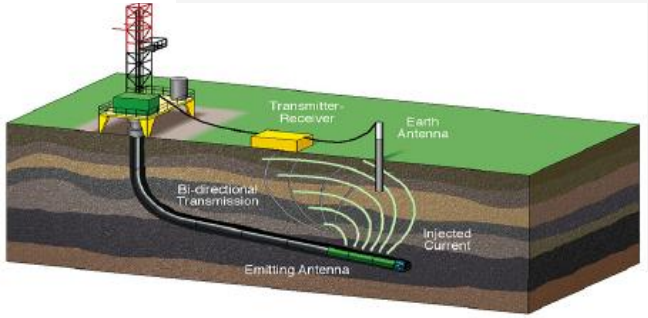
EM (Electromagnetic MWD) – if any Total Mud Losses

MWD dg Gelombang Electromagnetic

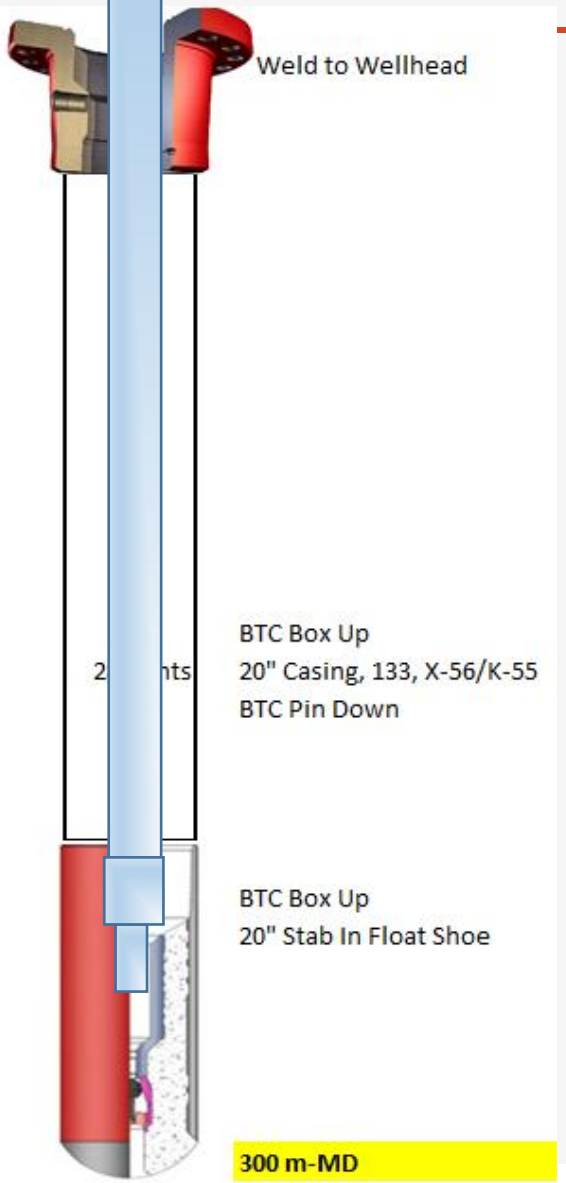








Run 20″ Surface Casing and Cement



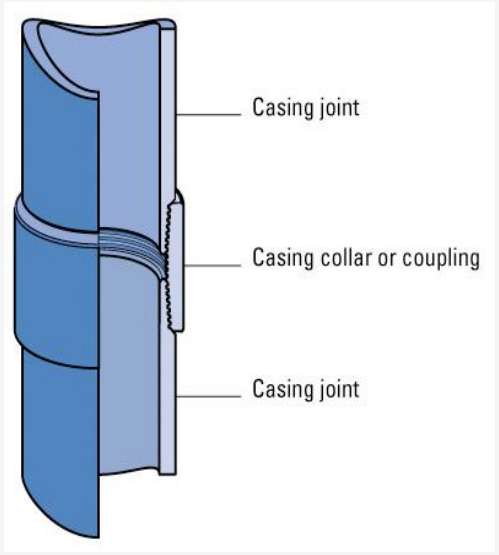
1. Run 20″ Casing:
   20″ Stab In Float Shoe, 20″ Casing, 21-1/4″ Pre-weld Wellhead
2. Run 5″ Inner String
3. Stab in and Circulation prior to pumping cement slurry
4. Pump Cement slurry
5. Pull out 5″ Inner String
6. Wait on Cement to hard
7. Install BOP



BIG CASING SIZE: DON'T USE BUTRESS CONNECTION

Pros	Cons
• Cheap • MU to tri-angle mark → no torque turn monitoring • API	• Not for gas well • Low Temperature and pressure • Limit DLS • Longer time to MU (i.e. 20" casing → 20-30 minutes per joint





Buttress (BTC)

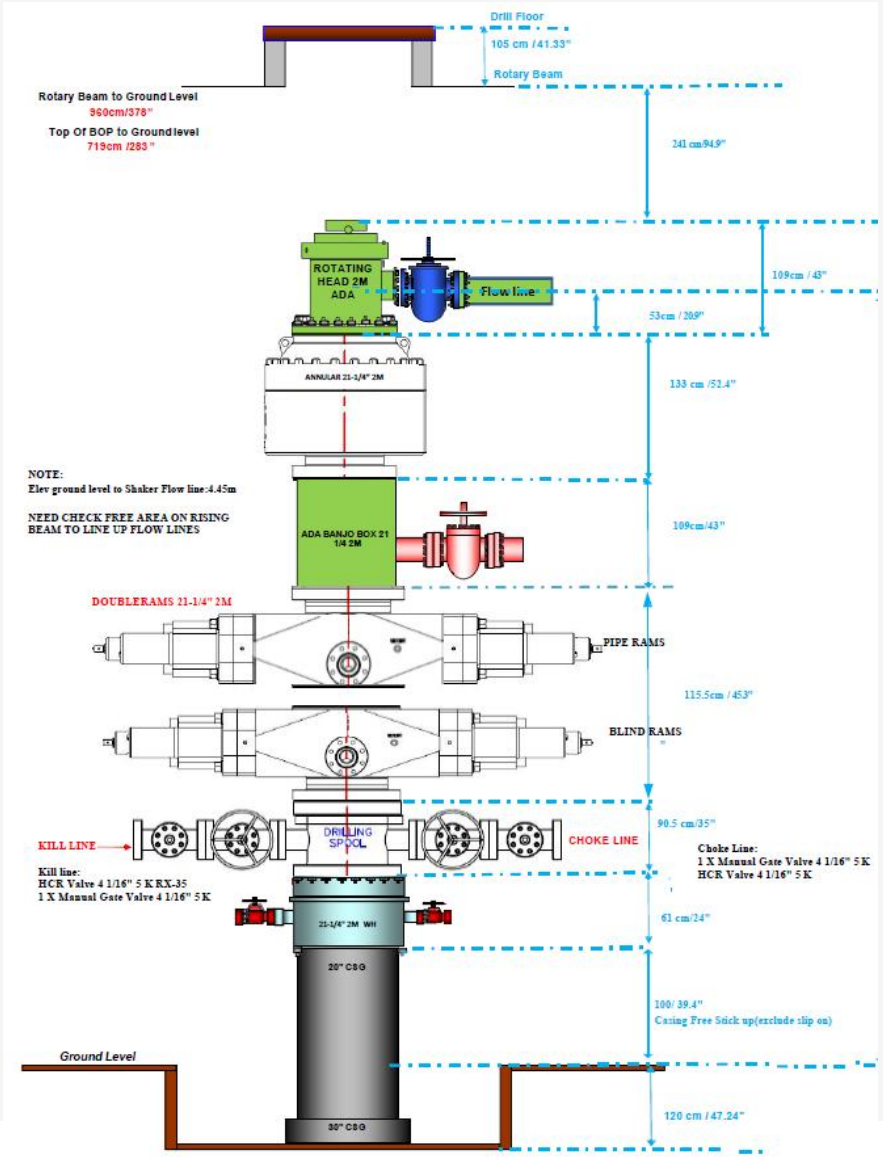




QUICK Connector

- QUICK Connector
- < 5 minutes per joint
- 15 minutes rig time saving.
- Rig Cost = USD 3000 /hrs
- Cost saving per joint: USD 750.
  - Additional Cost for QUICK connection: USD 200
  - Net cost saving: USD 550 /jt
  - 45 joint per well: USD 24750

Drilling Operation - Install 21-1/4" BOP



- Pick up 21-1/4" Wellhead (Section-A)
- Weld on 20" Surface Casing
- Nipple Up 21-1/4" " BOP
- Function and Pressure Test

- Ready to drill 17-1/2" Hole section

Question

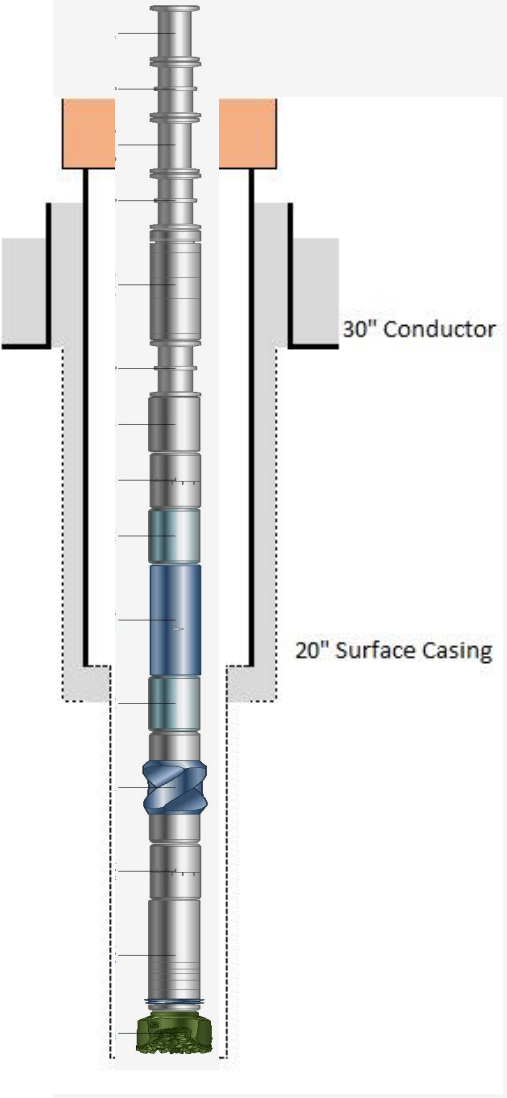


1. Type of MWD ?
2. Type of Drill Collar ?
3. Hole Size or Bit Size to pass 30” Conductor?.

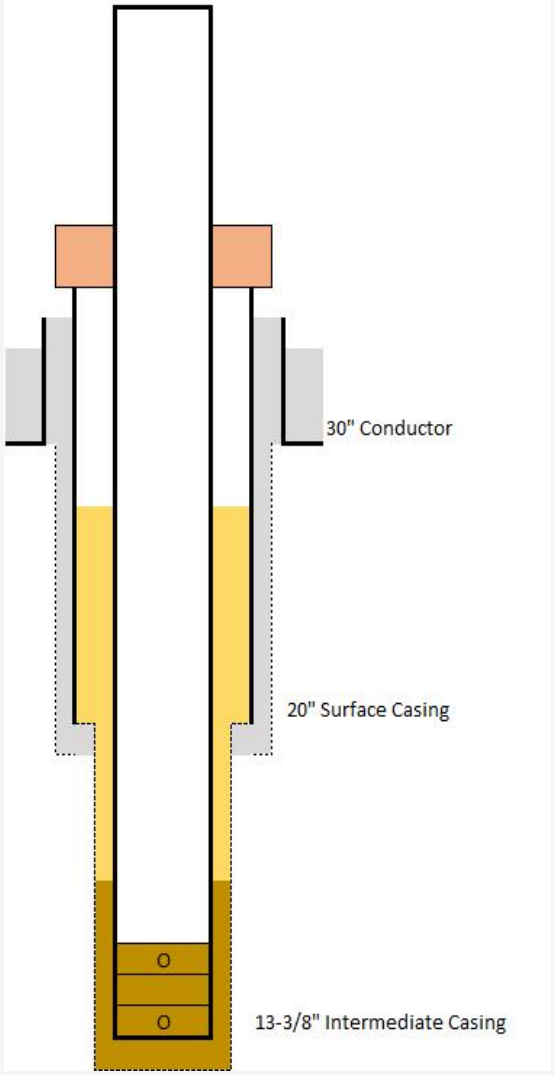
DRILLING OPERATION

17-1/2″ HOLE SECTION

Drilling Operation - 17-1/2" Hole Section



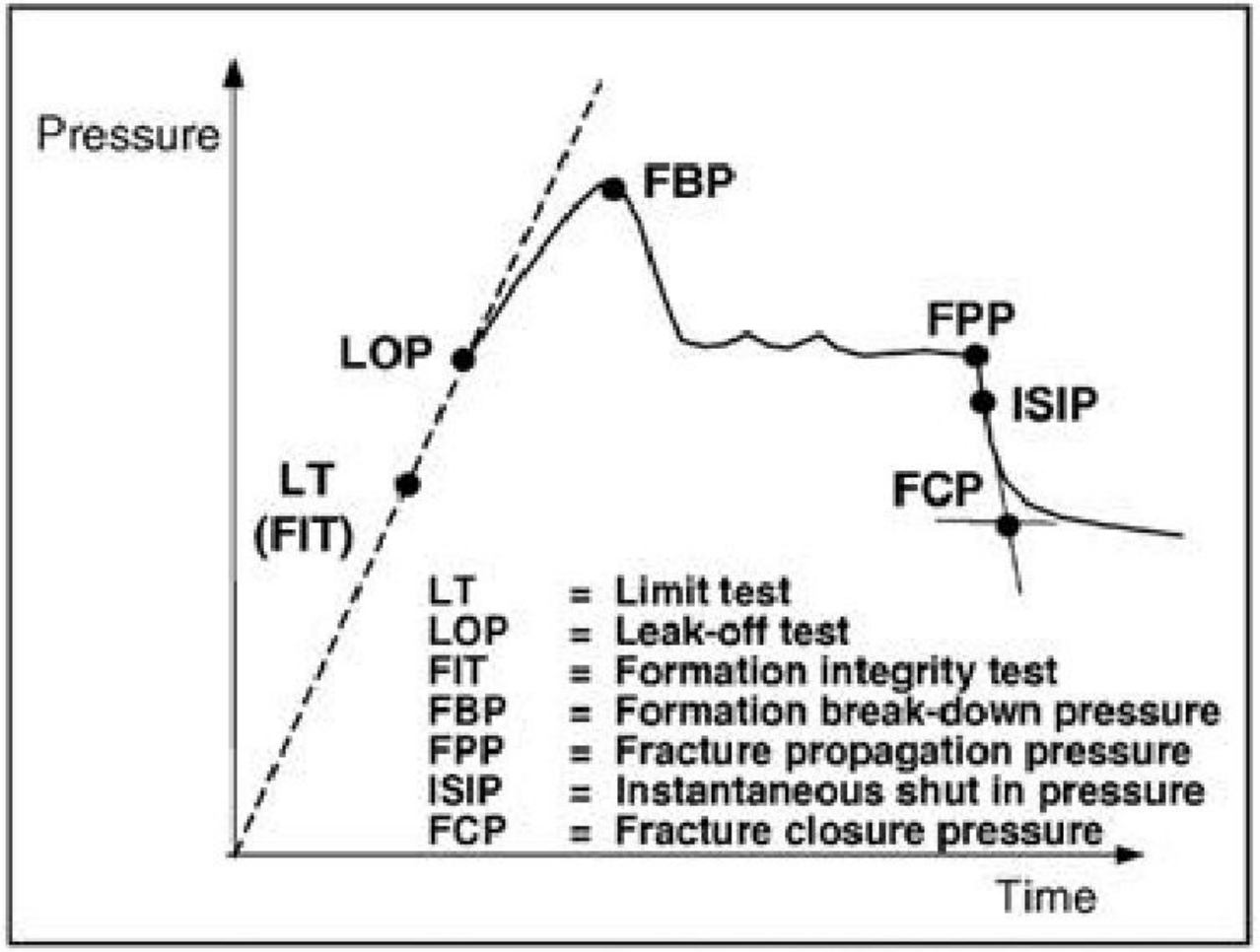
Drill 17-1/2" Hole

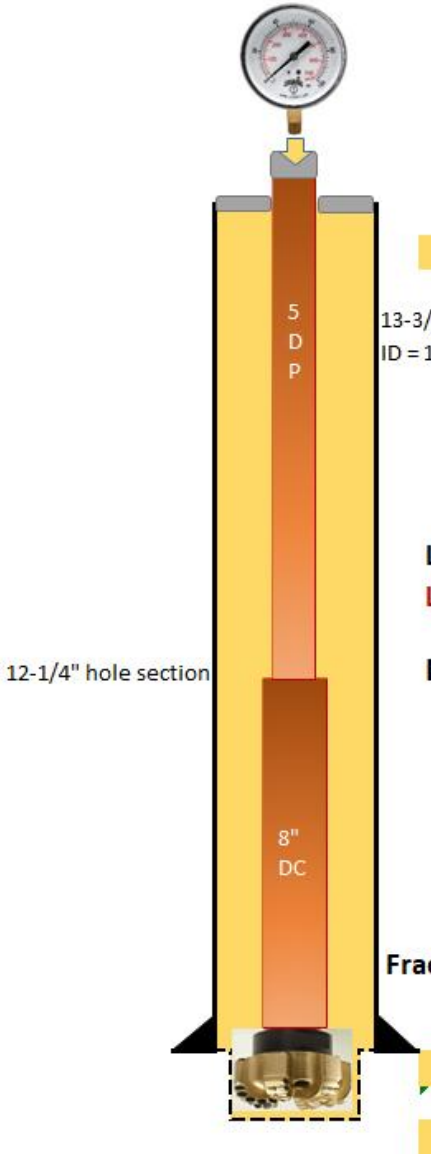


Run 13-3/8" casing and Cemented

- Make Up 17-1/2" Bottom Hole Assembly (BHA)
- Run in hole and tag TOC (Top of cement)
- Drill out cement.
- Pressure test 20" casing
- Continue Drill out shoe track and 3 m new formation.
- Perform LOT (Leak Off Test) or FIT (Formation Integrity Test
- Drill 17-1/2" hole to section TD (Total Depth)
- Circulation hole clean
- Wiper trip
- Pull Out Of Hole (POOH)
- Run 13-3/8" Casing and Cement.

LOT (Leak Off Test) or FIT (Formation Integrity Test)





LOT (Leak Off Test):

LOP (Leak Off Pressure) = 480 psi

LOT Equivalent MW (ppg) = Pressure (Psi) / 0.052 X TVD (ft)

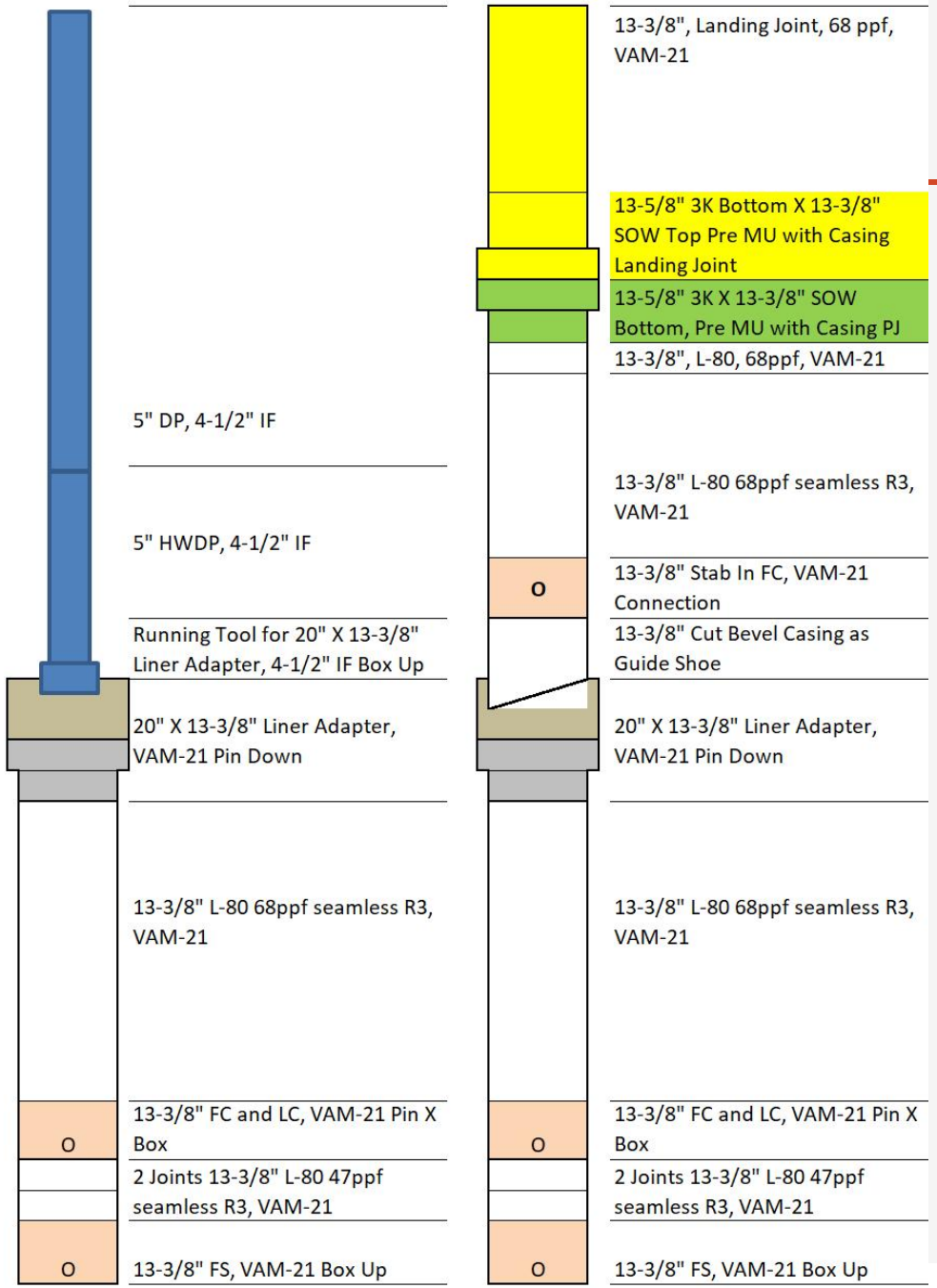
= 480 psi / (0.052 X 3281 ft) ppg

= **2,81** ppg Equivalent MW

MW Static = **12,7** ppg

Fracture Gradient = **15,5** ppg

Run and Cement: 13-3/8" Liner and Tie Back

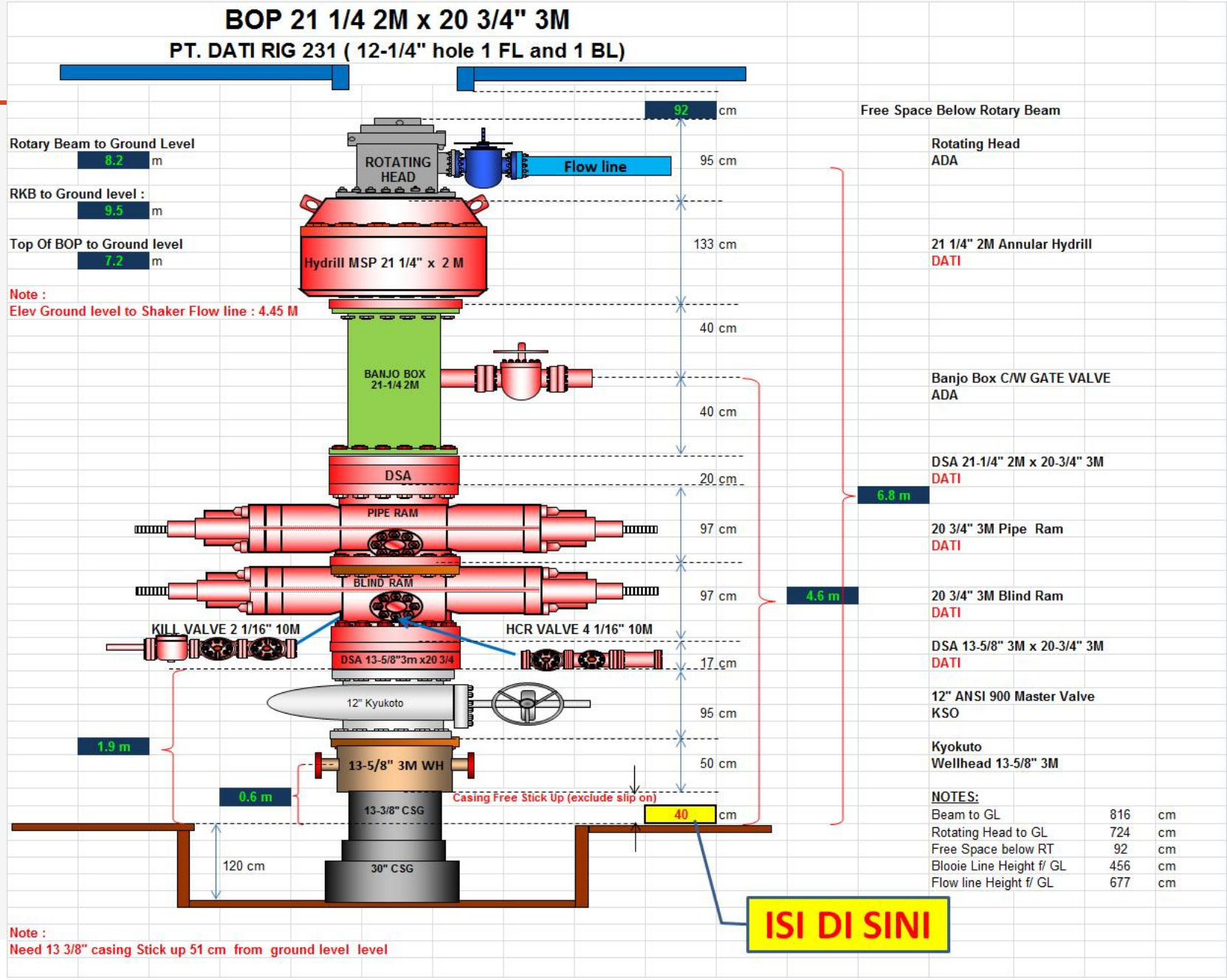


Casing Technical Datasheet

Hole In	Size In	Depth m-MD / TVD	ID In	Specification	Burst Psi	Collapse Psi	Make Up Torque, Ft-Lbs Min	Opt	Max	MU Loss In
36	30	33 / 33	29.000	X56, 156.5 ppf, SAW			NA	NA	NA	None
26	20	450	18.730	K55, Seamless, 133 ppf, Geoconn, R3	3060	1500	27,200	30,200	33,200	4.913
17-1/2	13-3/8	400	12.415	Tie Back Casing L80, 68 ppf, VAM-21, R3	5020	2260	37,700	41,900	46,050	5.555
	13-3/8	1400	12.415	Liner L80, 68 ppf, VAM-21, R3	5020	2260	37,700	41,900	46,050	5.555
	13-3/8		12.347	Liner L80, 72 ppf, VAM-21, R3	5380	2660	39,400	43,800	48,150	5.555
12-1/4	9-5/8	Contingency	8.681	Liner L-80, 47 ppf, DWC/C-IS, R3	6870	4750	40,000	47,700	54,800	4.81
12-1/4	10-3/4	2800	10.050	Perforated Liner L-80, 40.5 ppf, DWC/C-IS, SCC, R3	4560	1730	13,900	16,600	28,000	4.81
9-7/8	8-5/8	Contingency	8.097	Perforated Liner K55, 24 ppf, DWC/C-IS, SCC, R3	2950	1370	7,600	9,000	10,400	4.81
7-7/8	7	Contingency	6.366	Perforated Liner K55, 23 ppf, DWC/C-IS, SCC, R3	4360	3270	10,100	11,500	12,900	4.5

BOP CONFIGURATION
Geothermal

- Use One size BOP, 21-1/4” 2 M for drilling all hole section.
- Less RU time and cost saving

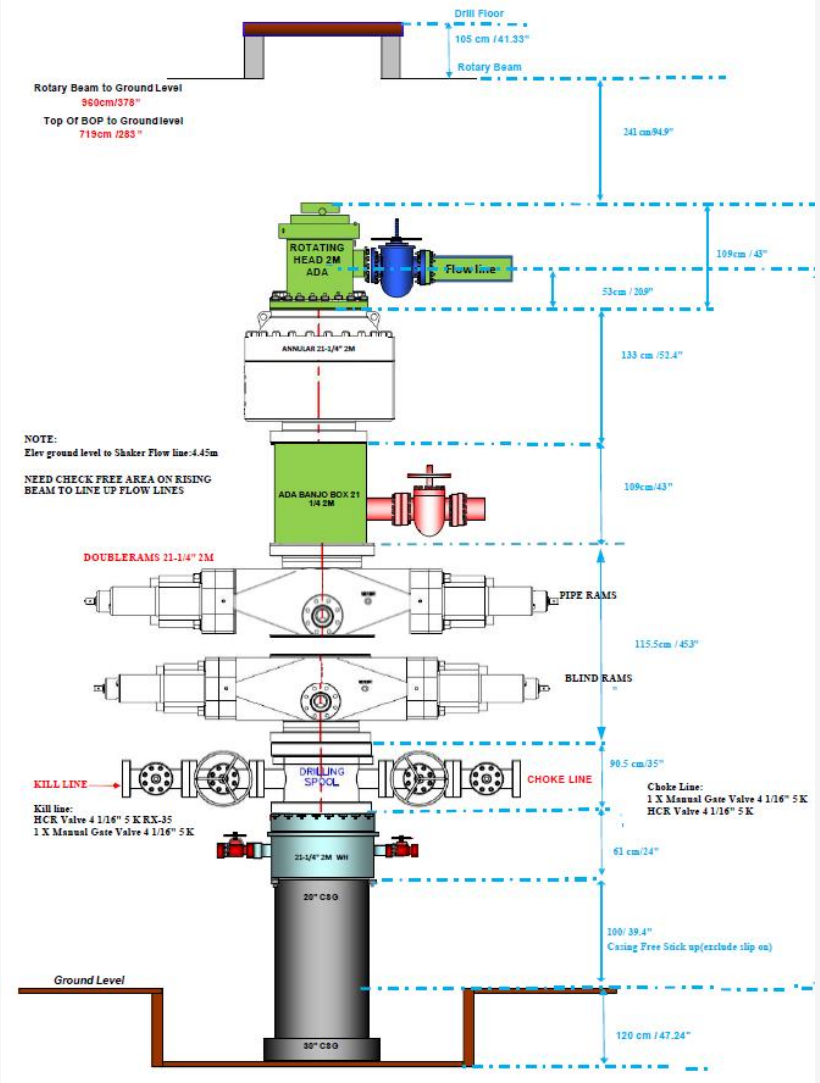


DRILLING OPERATION

12-1/4” HOLE SECTION

Drilling Operation - 12-1/4" Hole Section

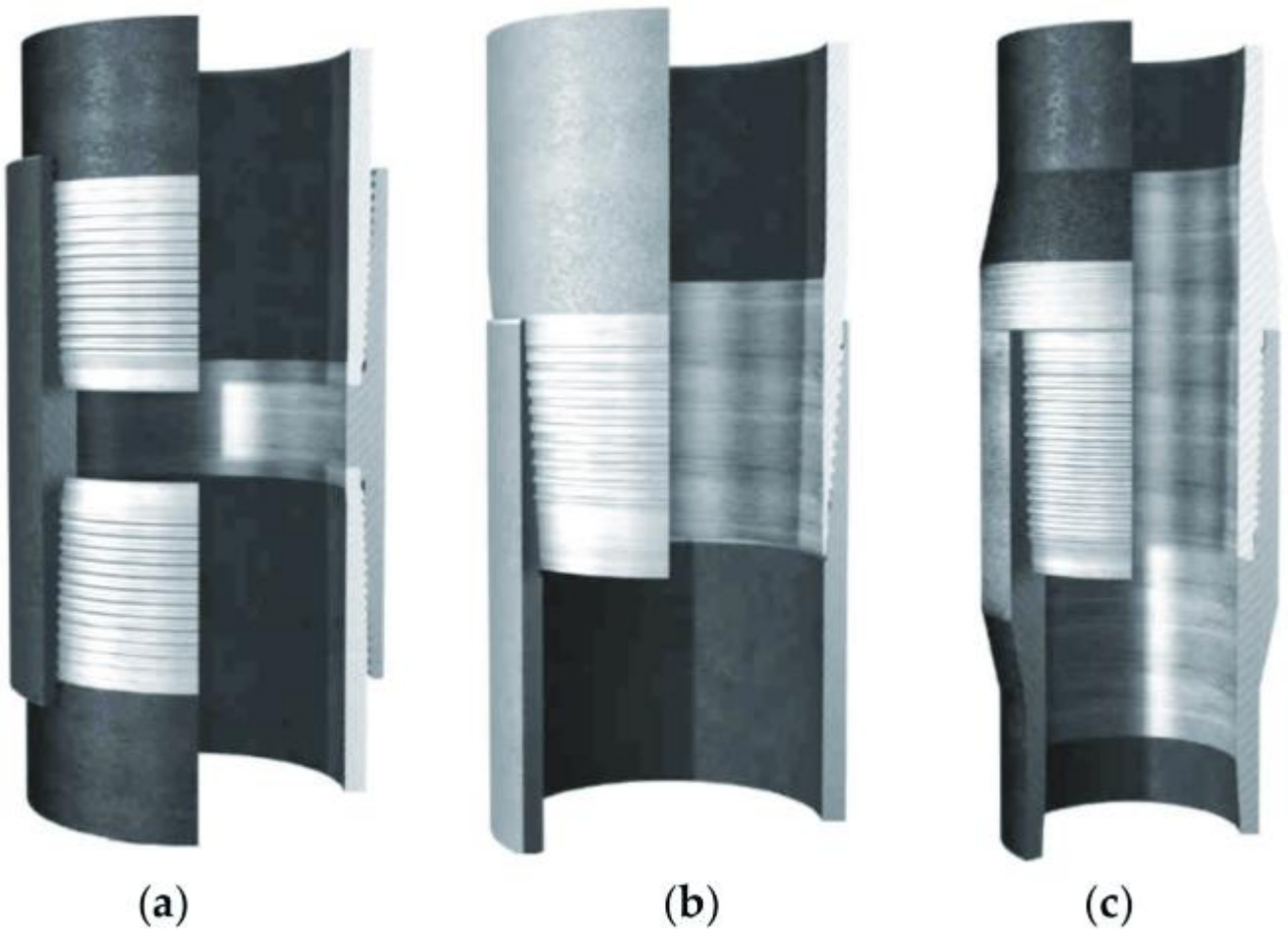
- Install 13-5/8" Wellhead
- Nipple Up 21-1/4" BOP
- Pressure Test BOP

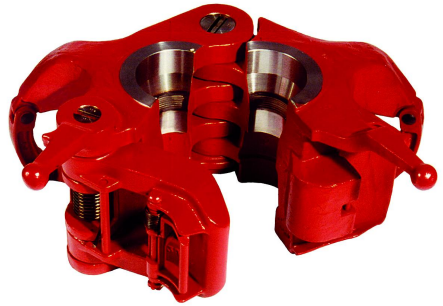


Drilling Operation - 12-1/4" Hole Section

- Make Up 12-1/4" BHA
- Run in hole and tag TOC
- Drill out half shoe track.
- Pressure test 13-3/8" casing
- Continue Drill out shoe track and 3 m new formation.
- Perform LOT or FIT
- Drill 12-1/4" hole to section TD
- Circulation hole clean
- Wiper trip
- POOH
- Run 10-3/4" Perforated Liner.

Casing Connection Type → Casing Handling





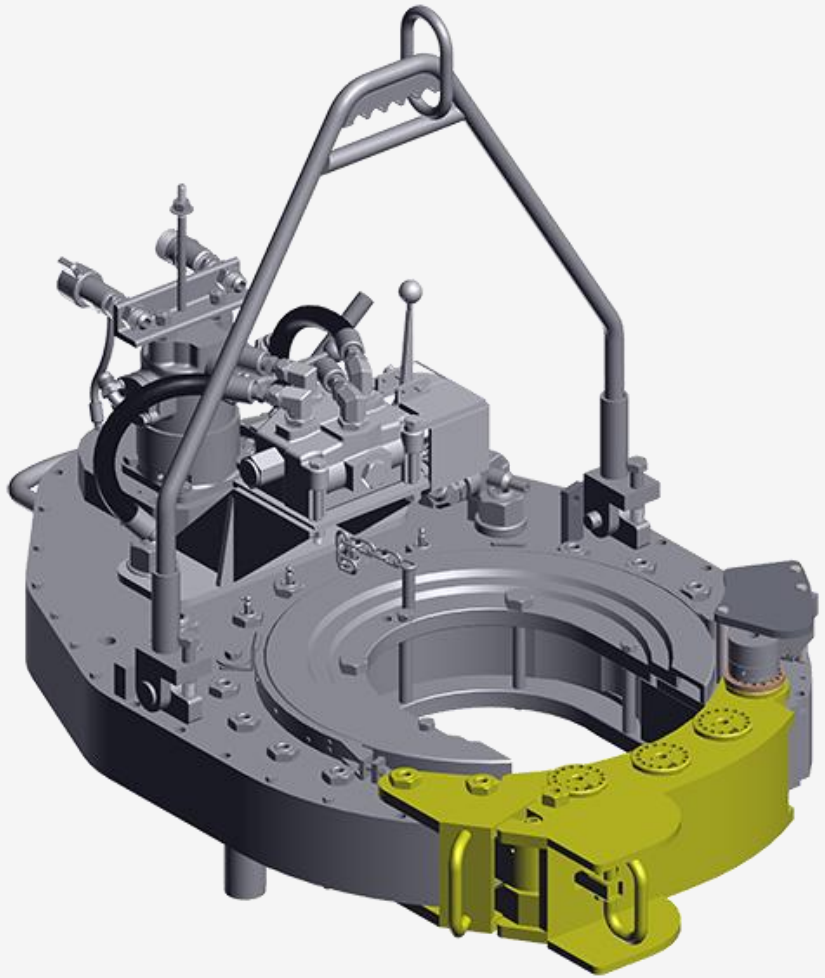
a → use Side Door Elevator

(a) Threaded and coupled;
(b) Integral flush;
(c) Integral upset connection

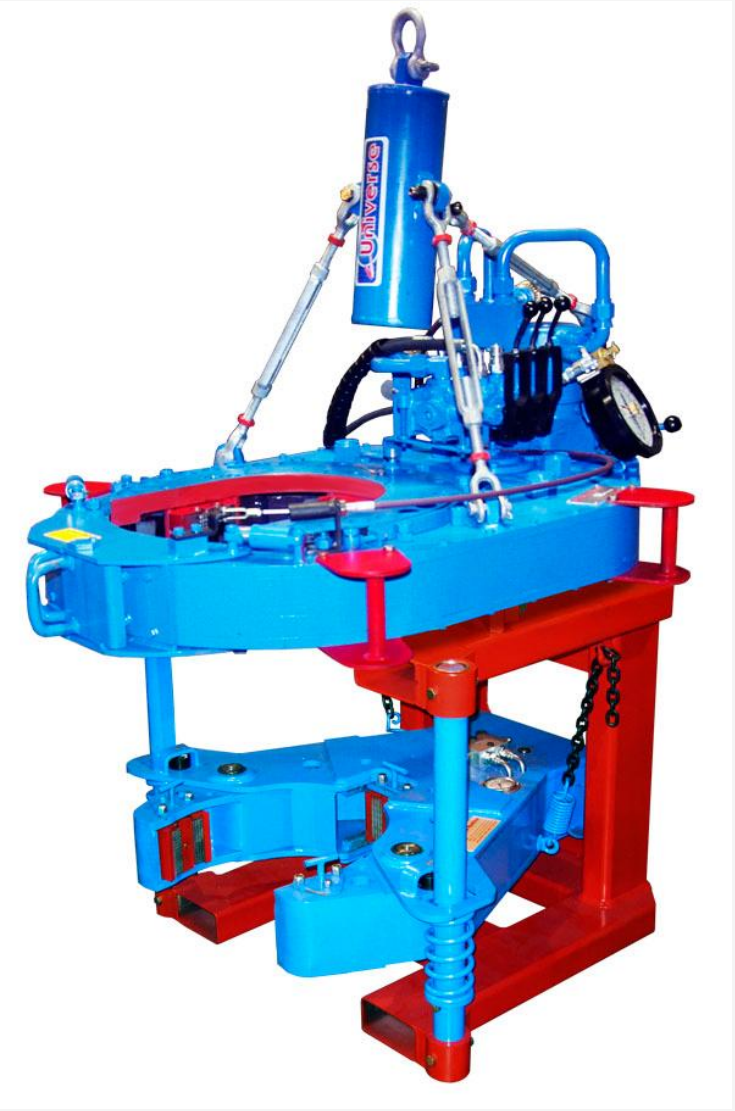


b and c → use Slip Type Elevator

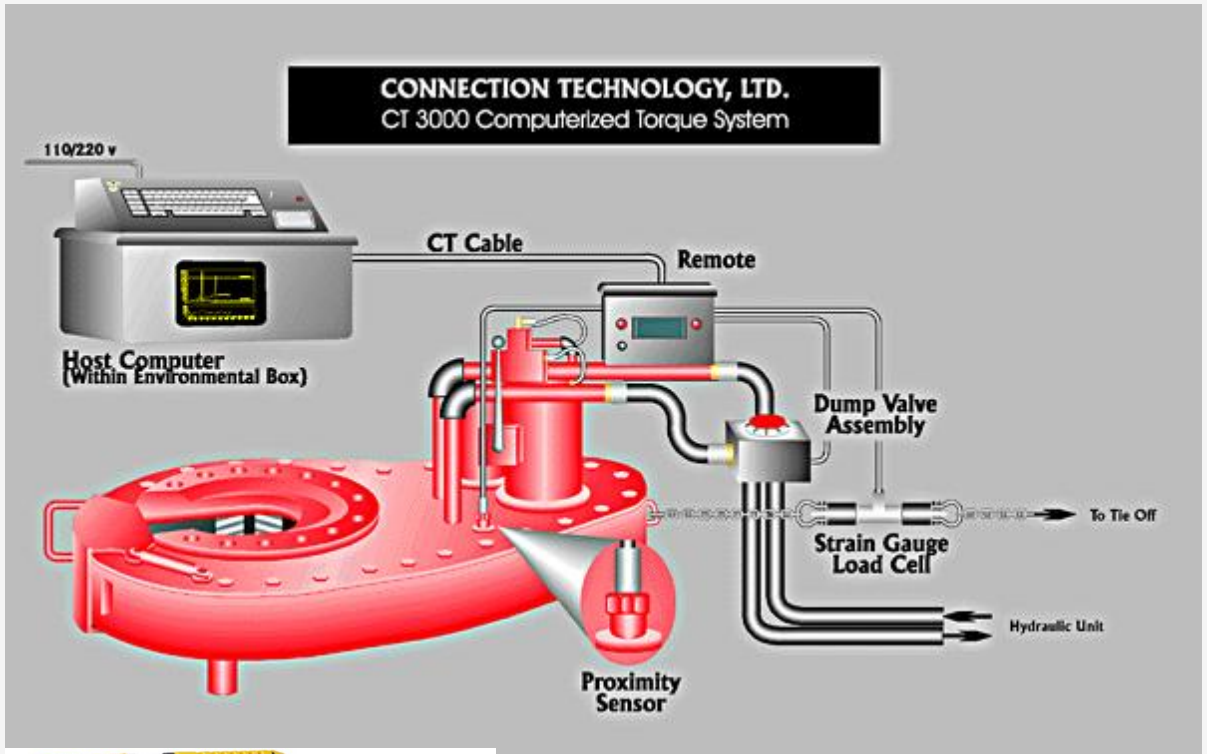
Casing Handling Equipment



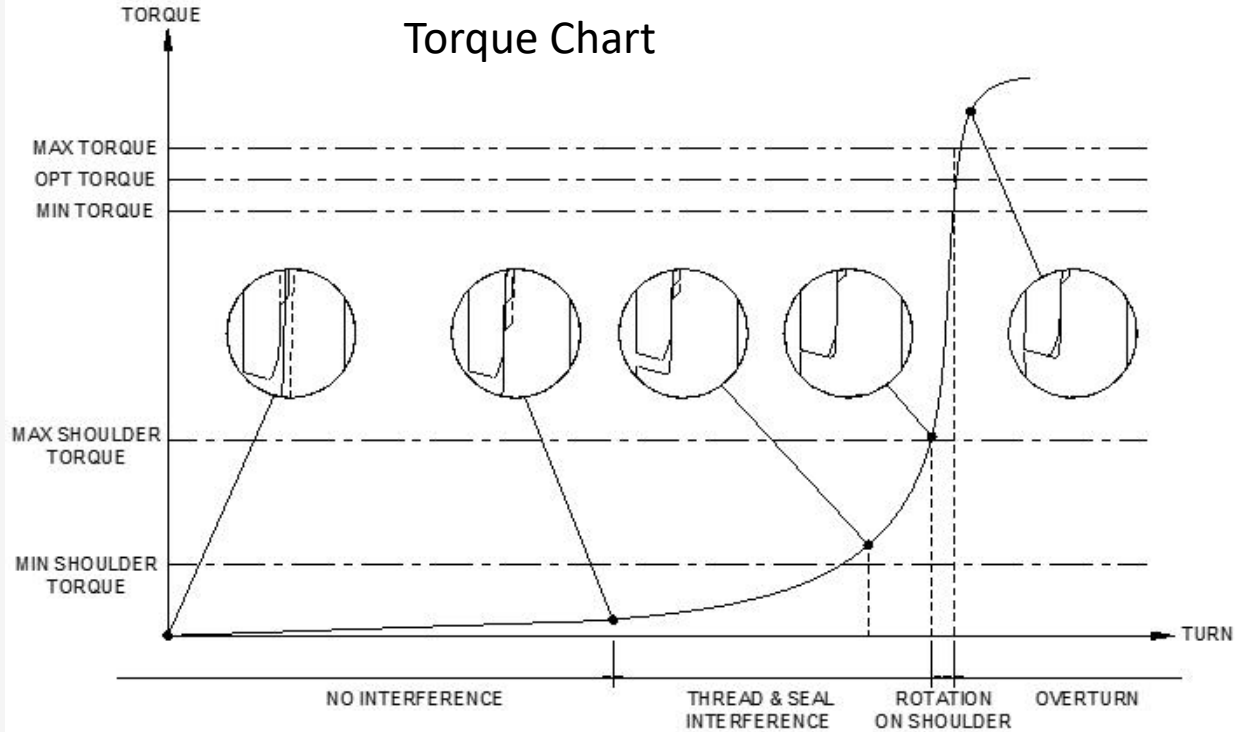
Hydraulic Power Casing Tong



Casing Handling Equipment

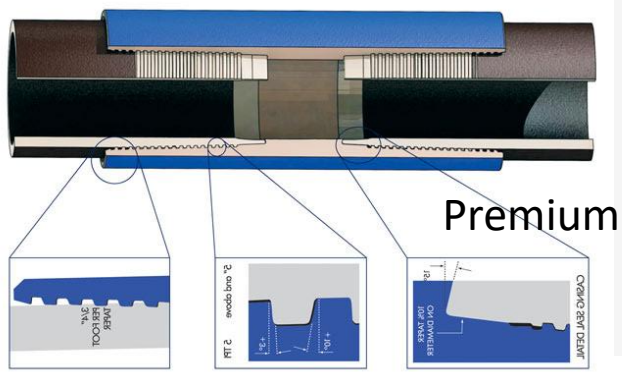


Torque Chart





Torque Turn Equipment



Premium Thread

Casing Handling Equipment – CDS (Casing Drive System)



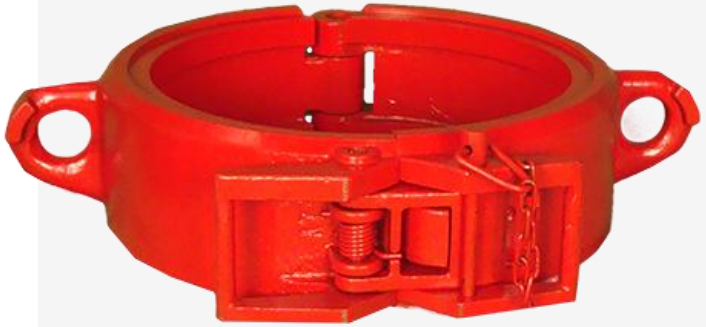
Top Drive System



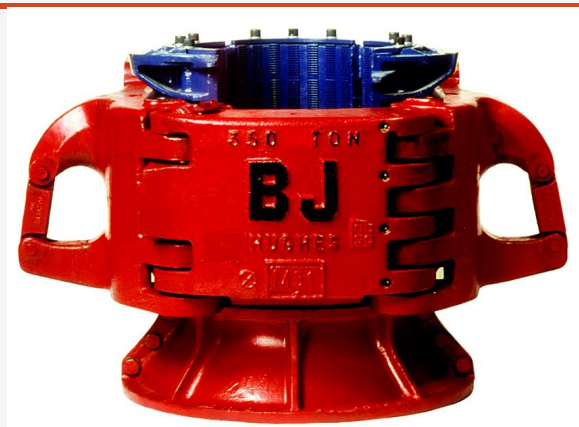
350 Ton External Tool | 500 ton Internal Tool | 750 ton Internal Tool

Casing Drive System Assembly

Casing Handling Equipment



Single Joint Elevator



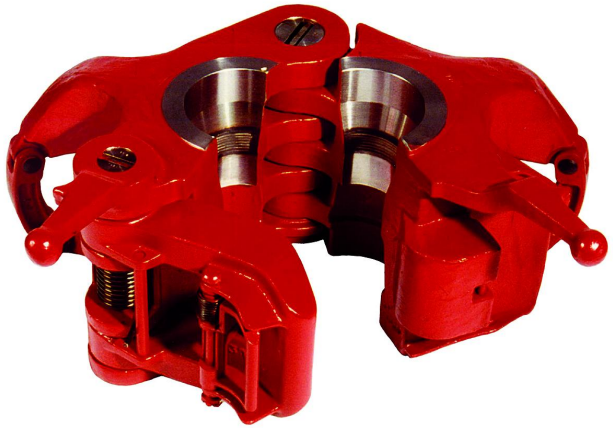
Spider Elevator



Casing Slip



Clamp On



Side Door Elevator



Spider Slip



Safety Clamp



Casing Drift



Stabbing Guide

Discussion during Drilling New Formation

a. Drilling Parameters;
    - WOB, RPM, Torque
    - Flowrate
b. Hole Cleaning issue
c. Wiper trip
d. BHA trip:
    - Change drill bit
    - BHA change
    - Downhole failure
    - Pipe leakage
    - Hole instability
e. Total Depth
f. E-log
g. Drilling Problems

Casing Handling Equipment



1. List of Casing Handling Equipment ?
2. Casing Elevator type for handling Integral upset connection ?
3. CDS ?

DRILLING OPERATION – 12-1/4" Hole Section

CEMENTING EQUIPMENT

Cementing Equipment

- 1set of Twin Triplex Pump Cementing Unit
- 1 set of Twin 50 bbls Batch Mixer
- 4 ea of Dry bulk storage vertical pneumatic silos with total capacity of 1000 cuft each
- 1 ea of Cement surge tank
- 3 ea of Water storage with Total capacity of 1000 bbl.
- 2 ea of Air compressor with capacity 350 scfm
- 1 ea of Diesel-powered Centrifugal Pump
- 1 ea of Cement dust catcher
- 1 lot of Complete set of inter-connecting low-pressure hoses for cementing system components. (Waterline, cement bulk line, etc)
- 100 m of High-Pressure lines (1502 WECO)
- Complete set of Air hose, electrical cable, and two cementing equipment to rig
- Adequate rain shelter
- Capable to pump acidic fluid and other
- 1 ea of 20" & 13-3/8" 10-3/4", 9-5/8", 8-5/8" and 7" casing, minimum 1 ea per casing size. Casing Circulating Swedge, BTC connection. And Premium X-over
- 2 ea of 4-1/2" IF Drillpipe Circulating Swedge x 2" 1502 WECO
- 2 ea of Water Bushing for Drillpipe 4-1/2" IF Box to all size of Casing

CEMENTING EQUIPMENT

Cementing Pumping Unit





Cementing Cilo



Batch Mixer



WECO 1502 CHIKSAN



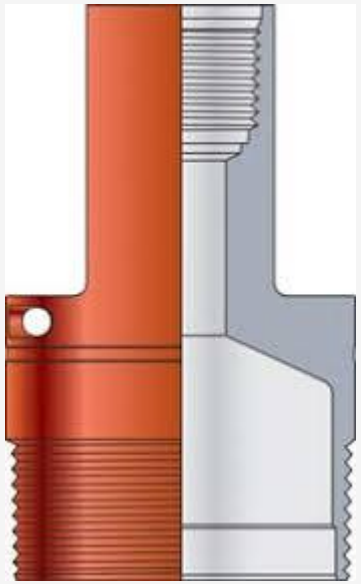


CEMENTING EQUIPMENT

Circulation Swage



Water Bushing



Cementing Head



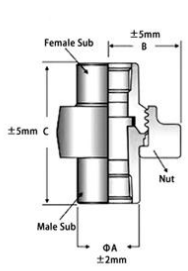




FIG.1502 CWP 15000 PSI





Cementing Equipment

- 1 ea of Single Plug Cementing head
- 1ea cementing head for inner string cementing method
- 1 ea of Plug Catcher w/ 4-1/2" IF Box x Pin c/w dart for 5" DP and shear pin
- 2 sets of Complete Stab In Stinger with 4-1/2 " IF Box and DP centralizers
- 5" Drill Pipe Cementing wiper dart
- 5" Drill pipe Foam Cementing Balls
- Minimum 4 ea of Radio Communication (optional)
- 1 unit of Pressurized Mud Balance

BULK PLANT EQUIPMENT (OPTIONAL)

Cementing Equipment

Laboratory in Jakarta

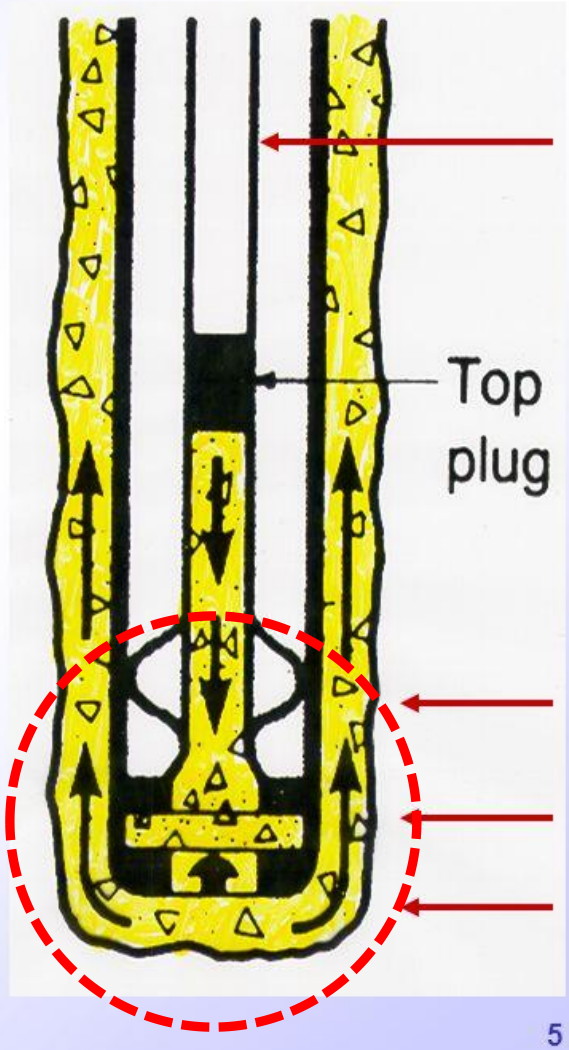
a. 1 unit of Electronic Lab Balance
b. 1 unit of Speed Controlled Waring Blender
c. 1 unit of Atmospheric Consistometer
d. 1 unit of Fann Viscometer
e. 1 unit of HP/HT Filter Press
f. 2 units of HP/HT Consistometer Cells
g. 2 units of HP/HT Curing Chambers
h. 2 units of Ultrasonic Cement Analyzer (UCA)
i. 1 unit of Pressurized Mud Balance
j. 1 unit of Hydraulic Press for Crush Tests
k. 1 unit of Variable Speed Lab Stirrer
l. 1 unit of Water Bath
m. 1 unit of B.P. Type Settlement Testing Apparatus

Primary Cementing Process – Inner String

Large-Hole Cementing

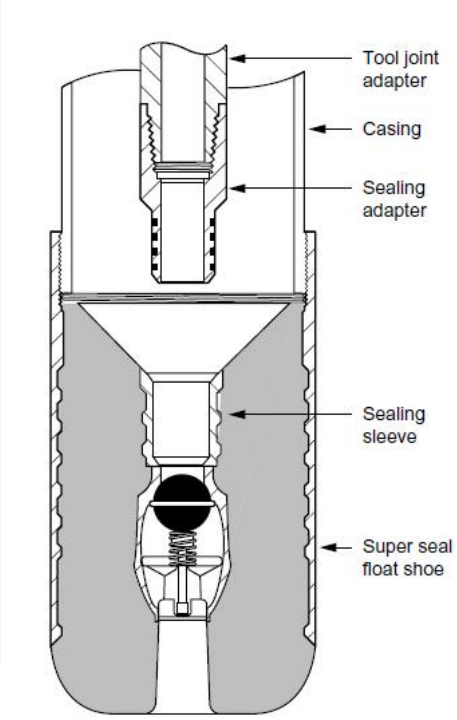
Inner String Cementing

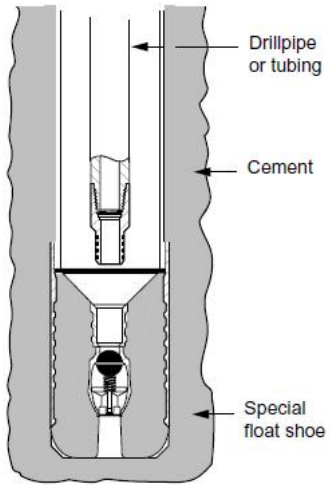
- Down the inside of the DP
- Use top wiper plug
- Stab-in adapter
- Much shorter displ. time











Primary Cementing Process – Single Stage Cementing

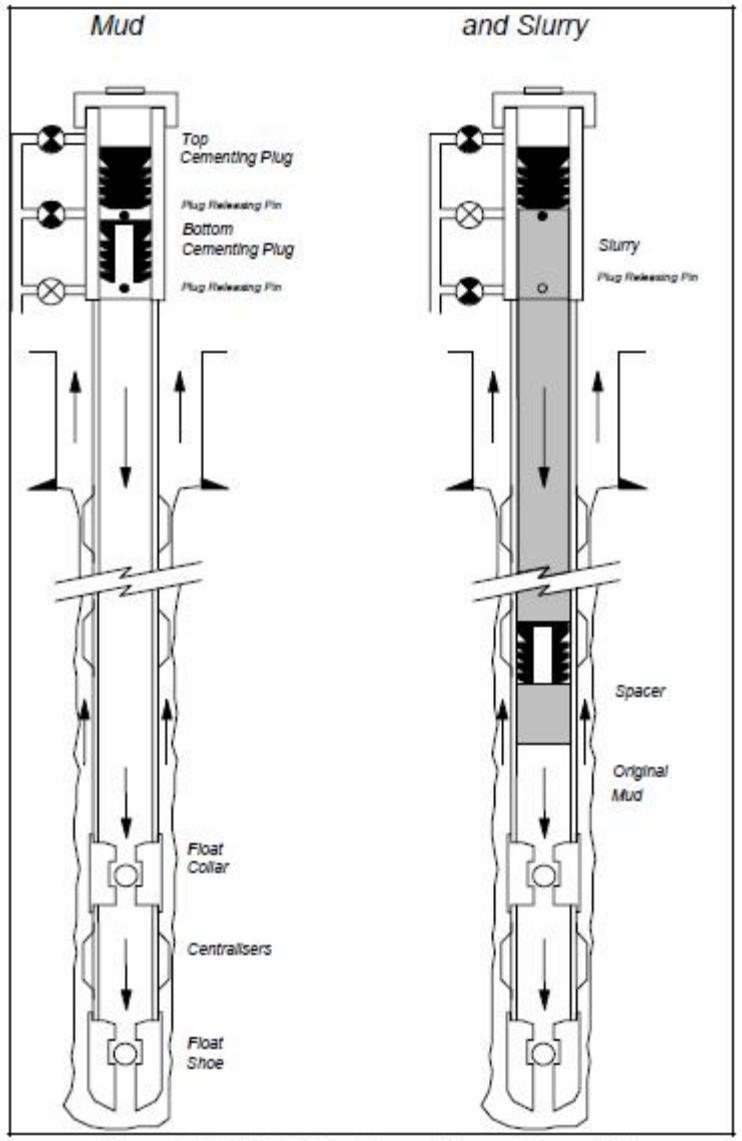


Figure 16-3- Typical One-Stage Primary Cement Job

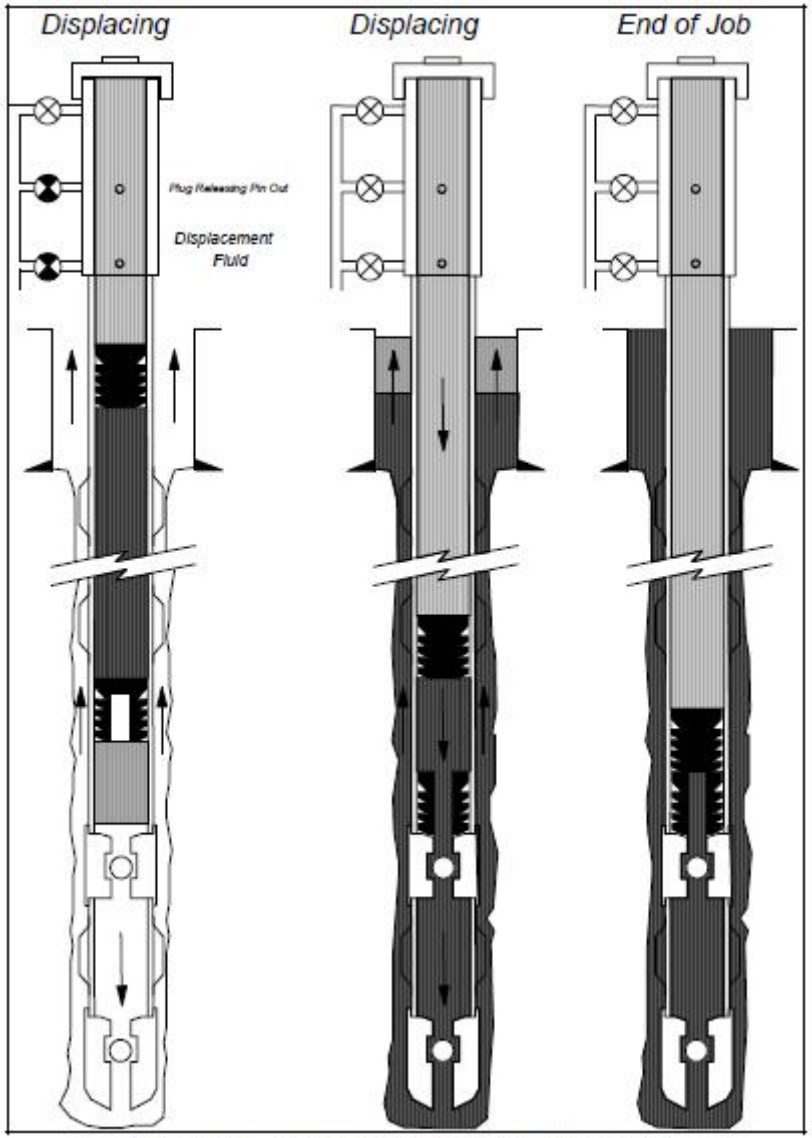
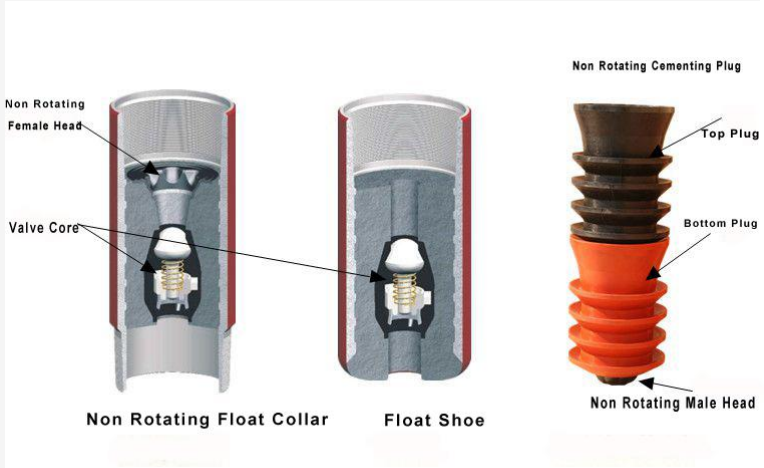


Figure 16-4- Typical One-Stage Cement Job (Cont)





Dual Stage Cementing

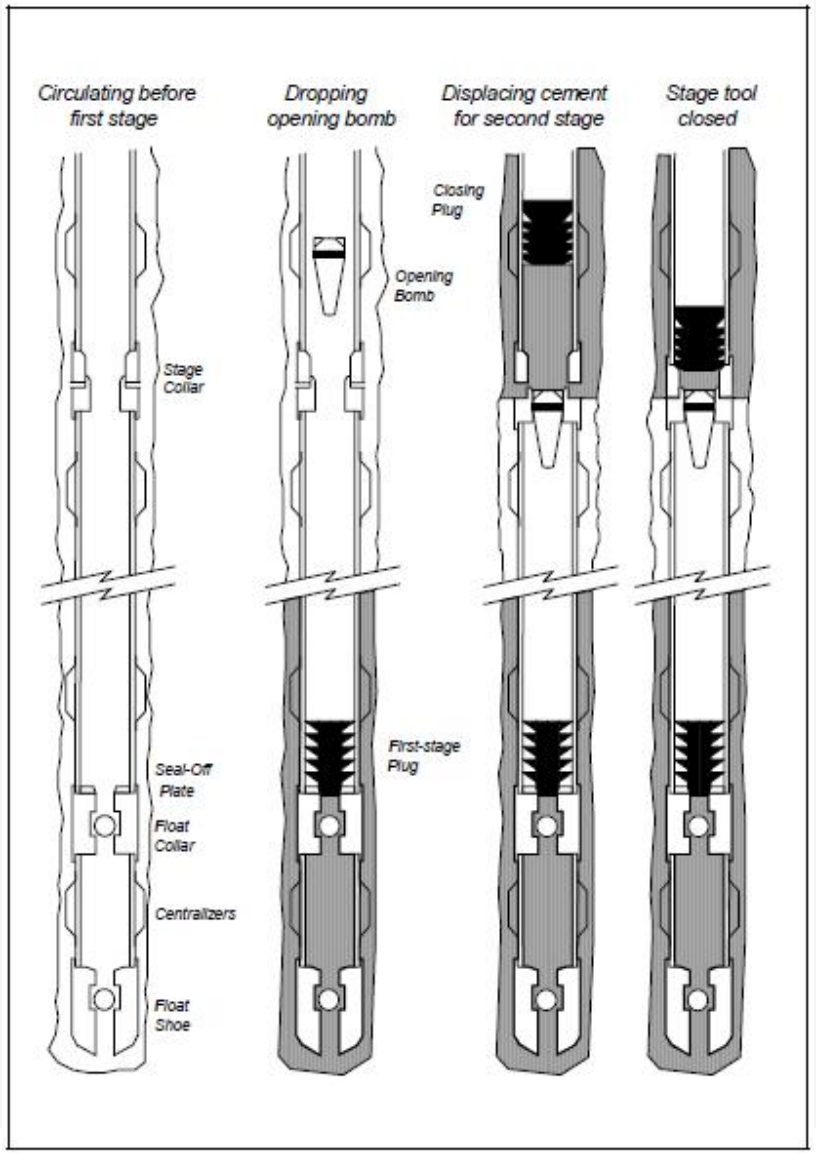


Figure 16-5- Two-Stage Cement Job (single plug system)

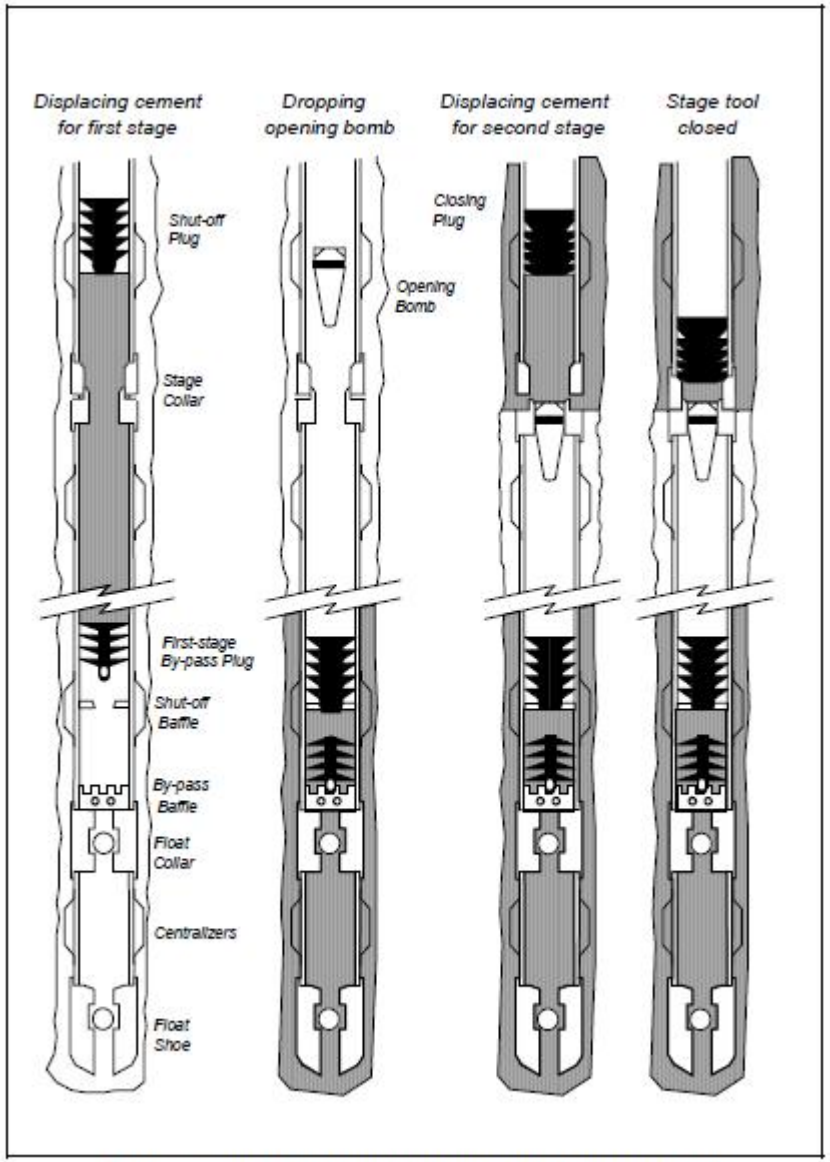
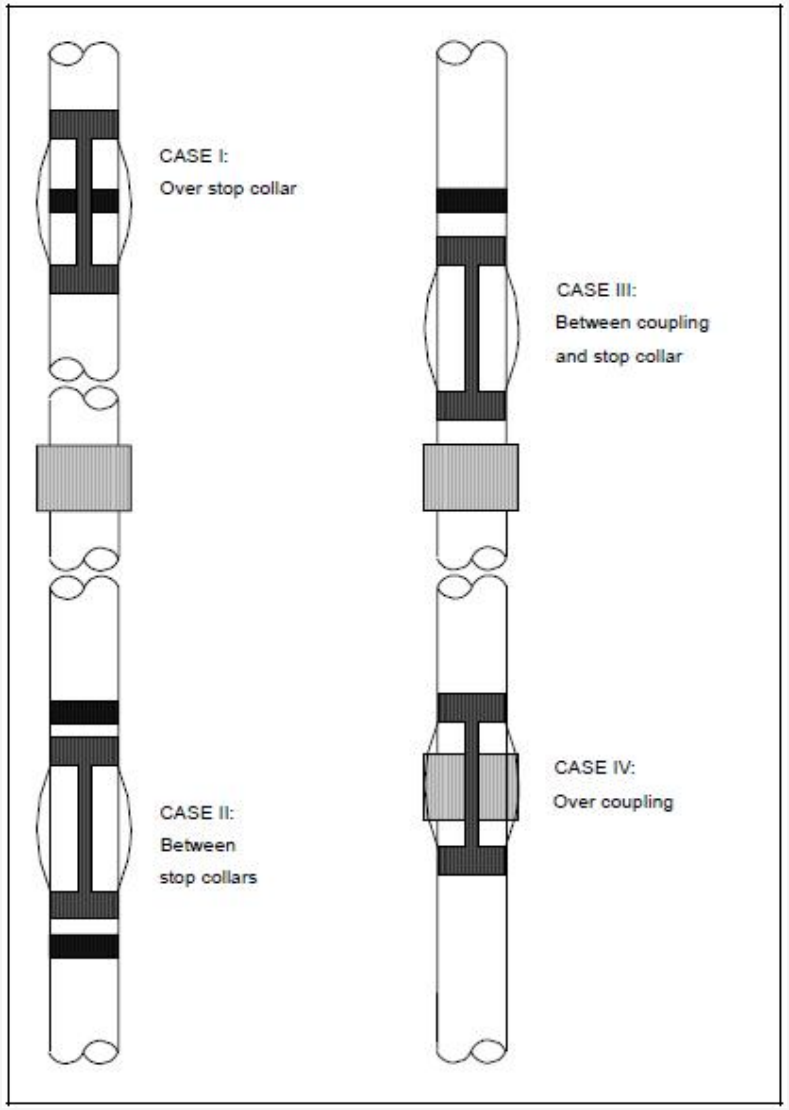
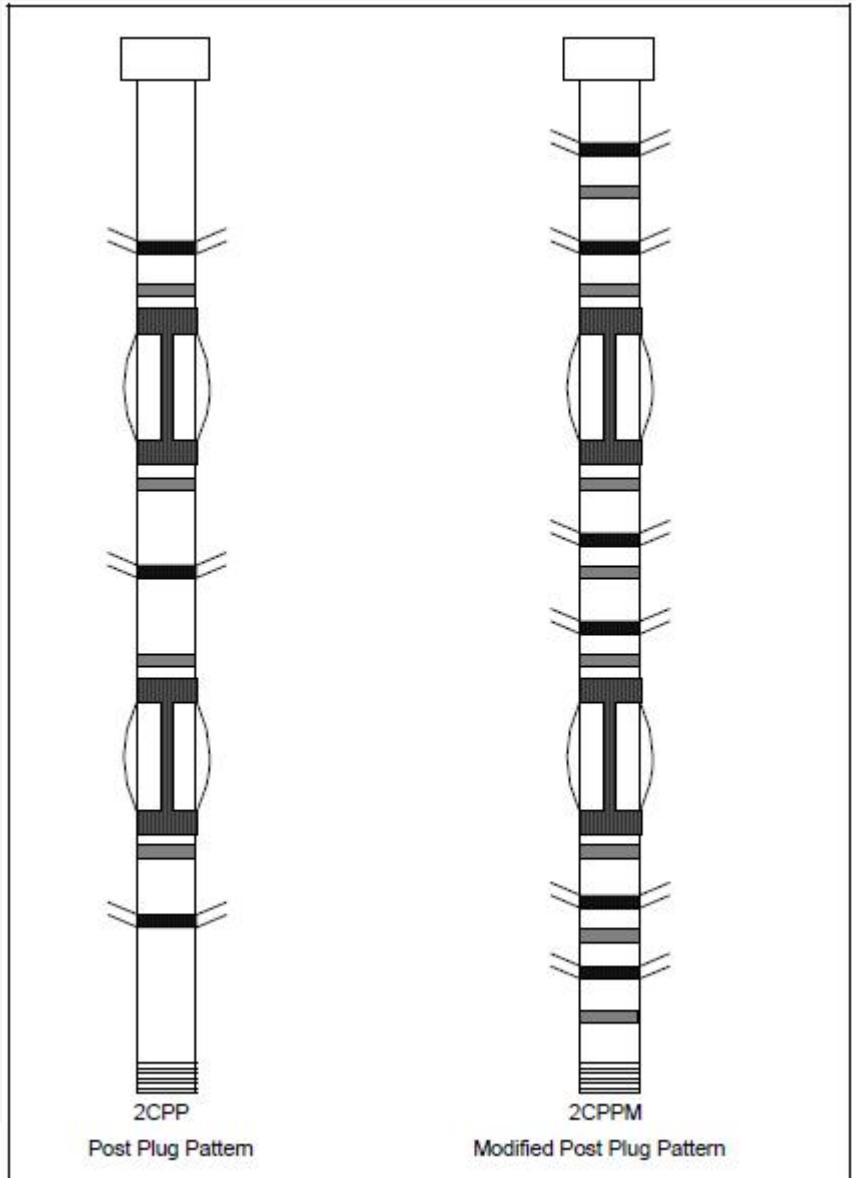


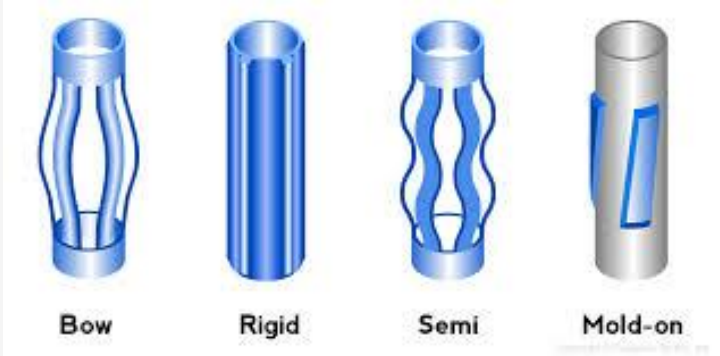
Figure 16-6- Two Stage Cement Job (double plug system)



Cementing Accessories













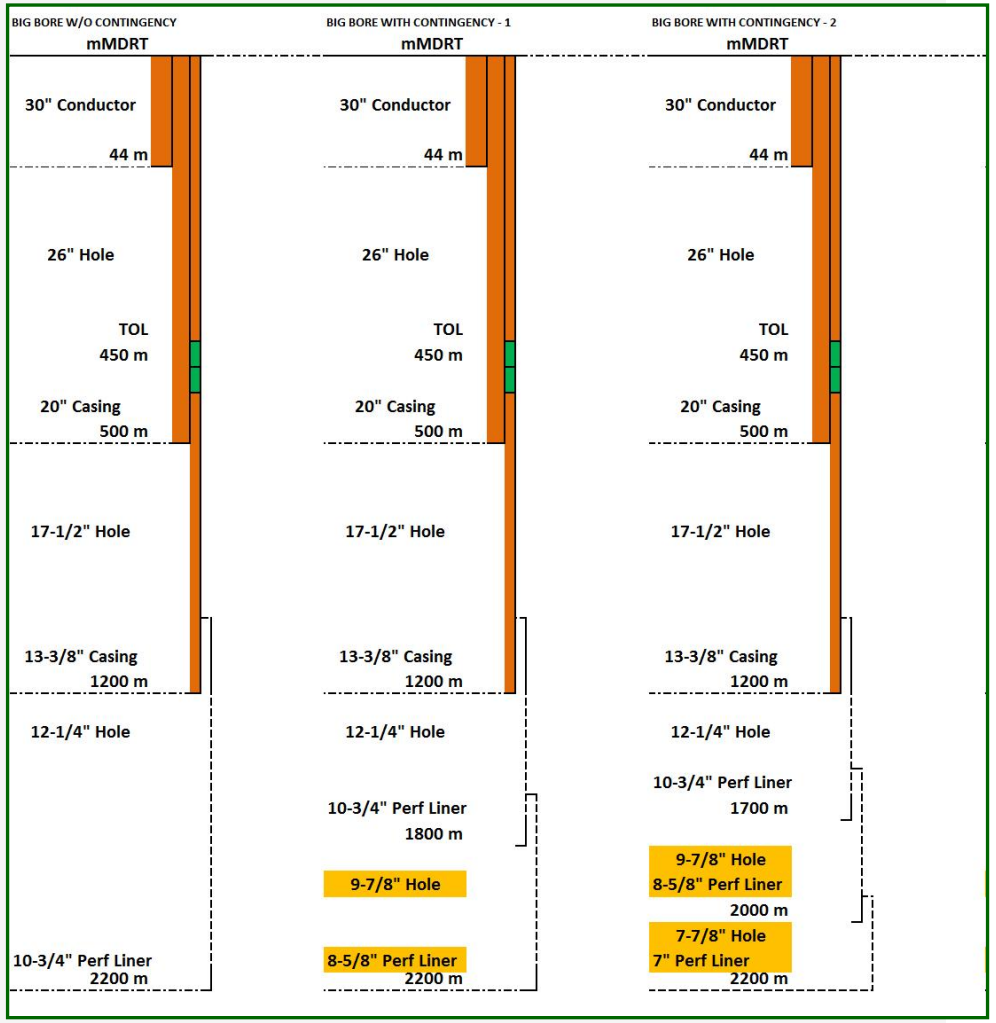
Drilling Operation - Contingency Hole

Drill 9-7/8" Hole Section

- Make Up 9-7/8" BHA
- Run in hole
- Drill out 10-3/4" Guide Shoe.
- Drill 9-7/8" hole to TD
- Circulation hole clean
- Wiper trip
- POOH
- Run 8-5/8" Perforated Liner

Drill 7-7/8" Hole Section

- Make Up 7-7/8" BHA
- Run in hole
- Drill out 8-5/8" Guide Shoe.
- Drill 7-7/8" hole to TD
- Circulation hole clean
- Wiper trip
- POOH
- Run 7" Perforated Liner



Question



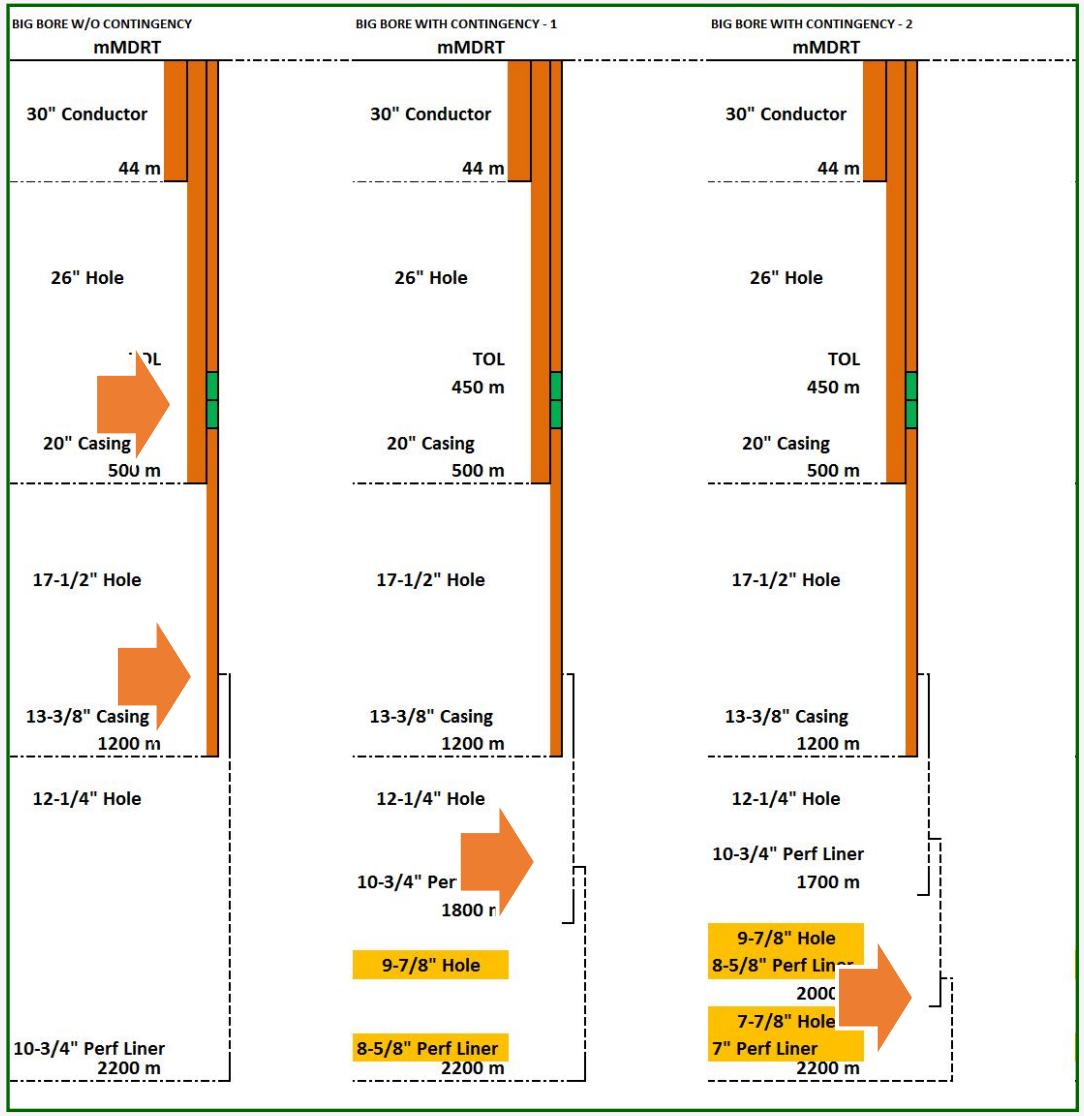
1. 3 Method of Primary Cementing ?
2. List Cementing Equipment ?.

DRILLING OPERATION – Production Hole

LINER HANGER

- Casing → set pipe to surface
- Liner → set pipe just overlapped with previous casing. Casing will be hanged on LINER HANGER

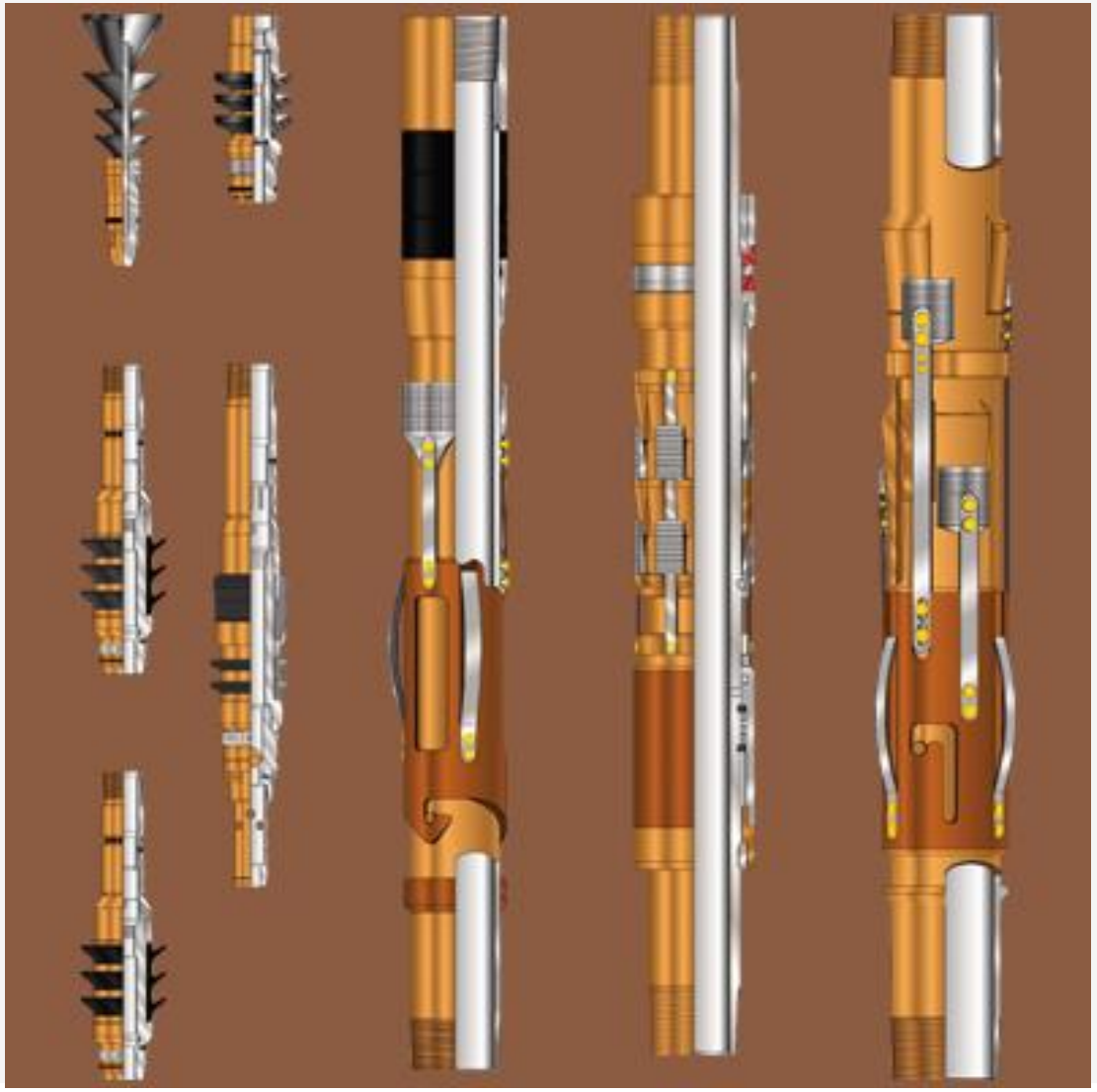
Liner Hanger / Liner Adapter for Geothermal



Liner Adapter Size:

- 13-3/8″
- 10-3/4″
- 8-5/8″
- 7″

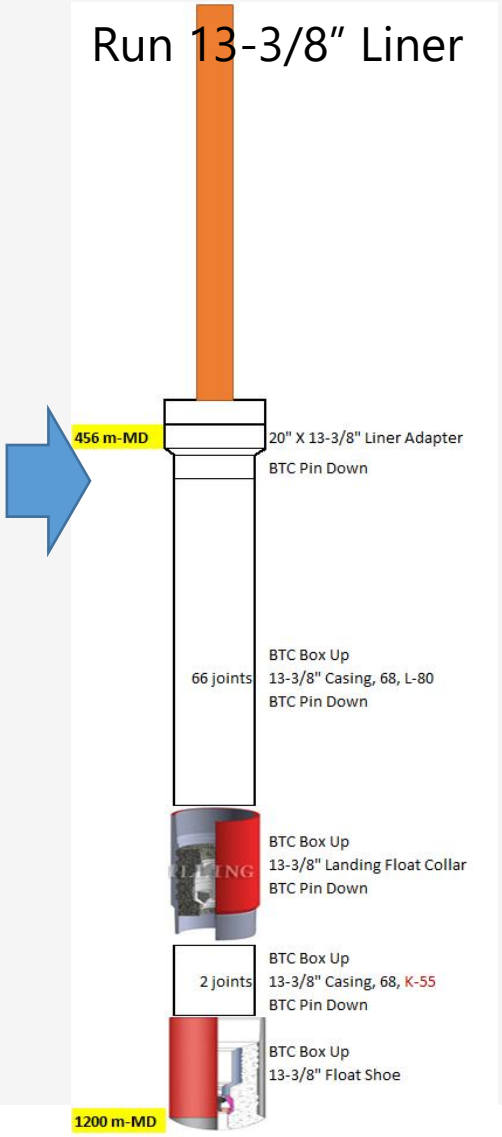
LINER HANGER – Oil & Gas



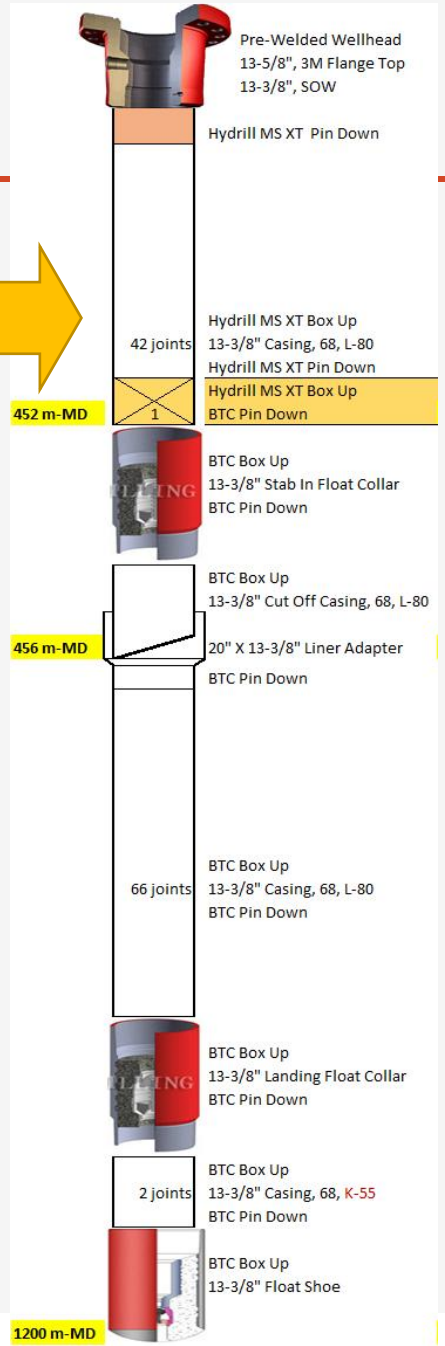
- Liner Hanger
- Liner Hanger Packer
  - Component:
    - Hanger
    - PBR (Polish Bore Receptacle)
    - Tie Back
    - Running Tool
  - Type:
    - Non Rotating
    - Rotating

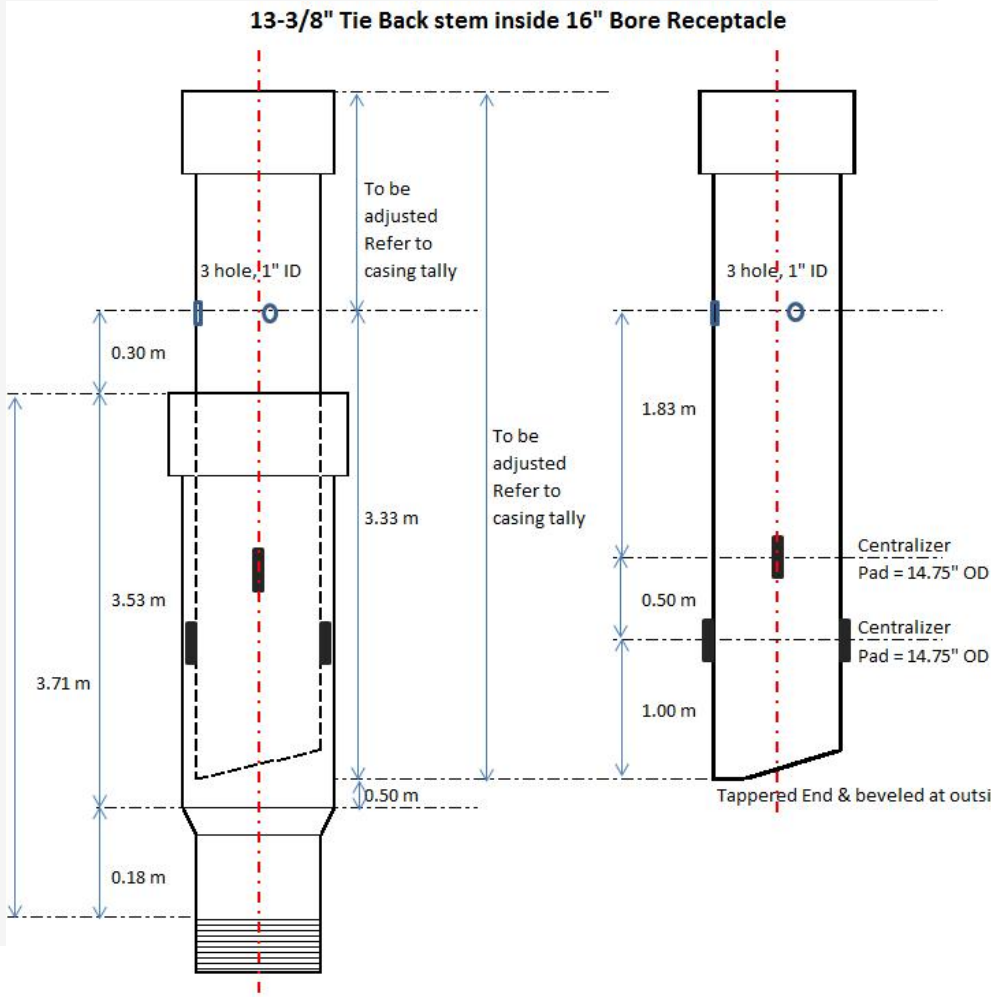
13-3/8″ Liner Adapter and Tie Back (Geothermal)

Run 13-3/8″ Liner

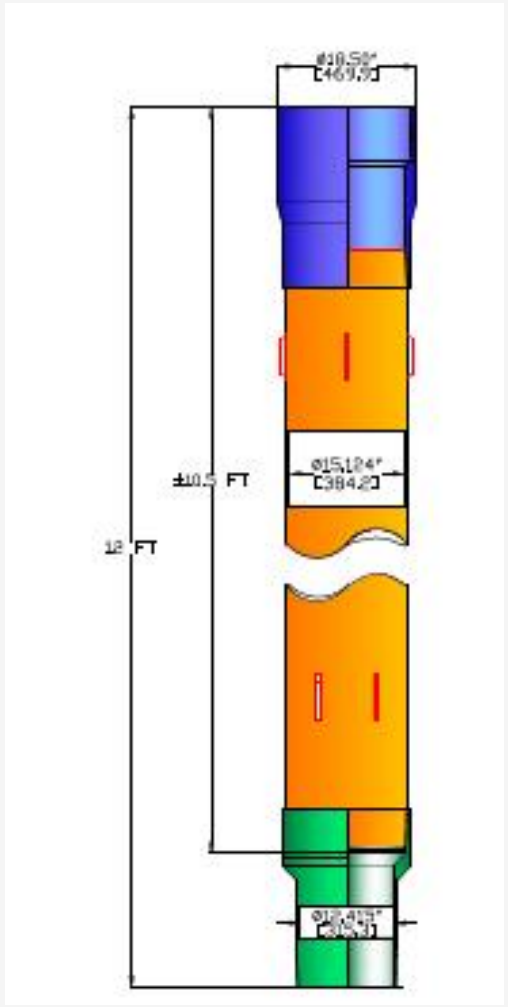


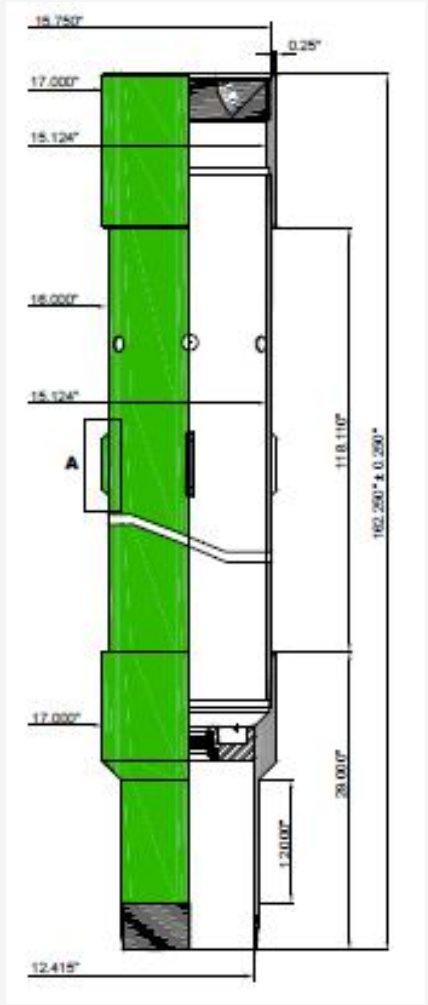
Run 13-3/8″ Tie Back



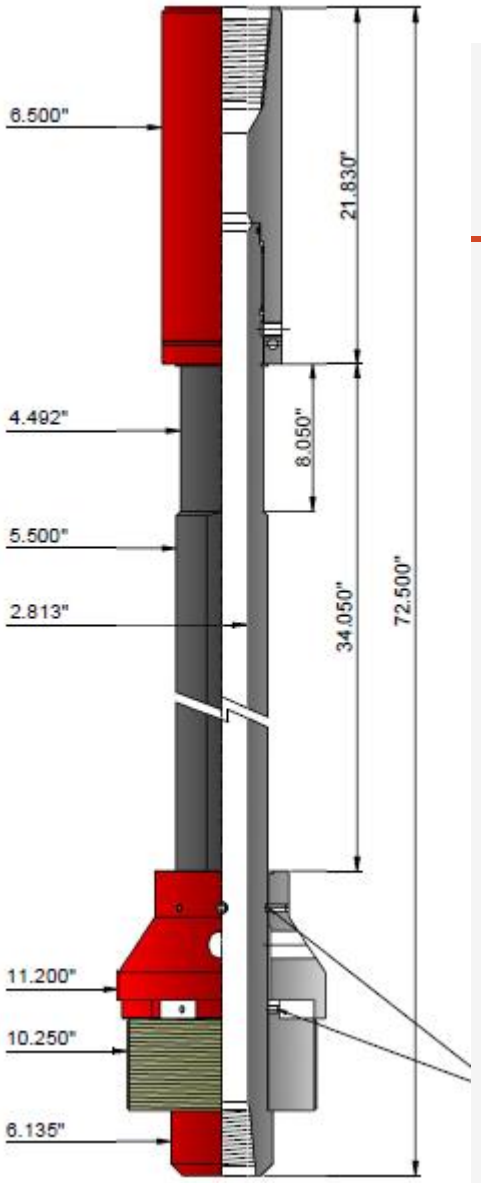


LINER ADAPTER - Geothermal

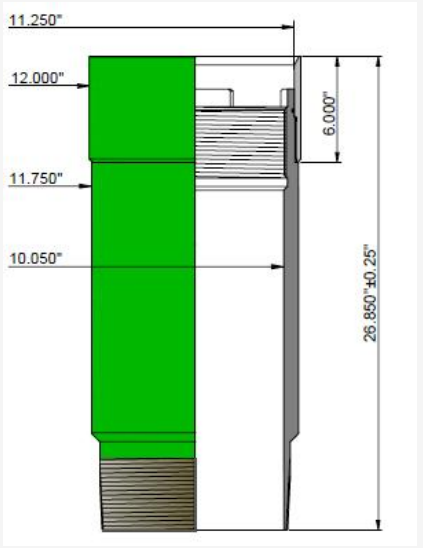




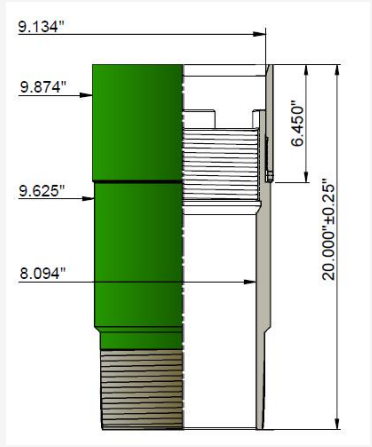
20” X 13-3/8”



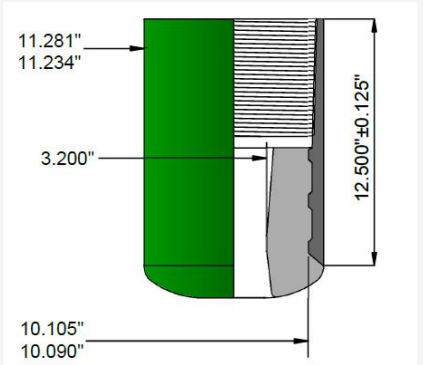
Running Tool



13-3/8” X 10-3/4”

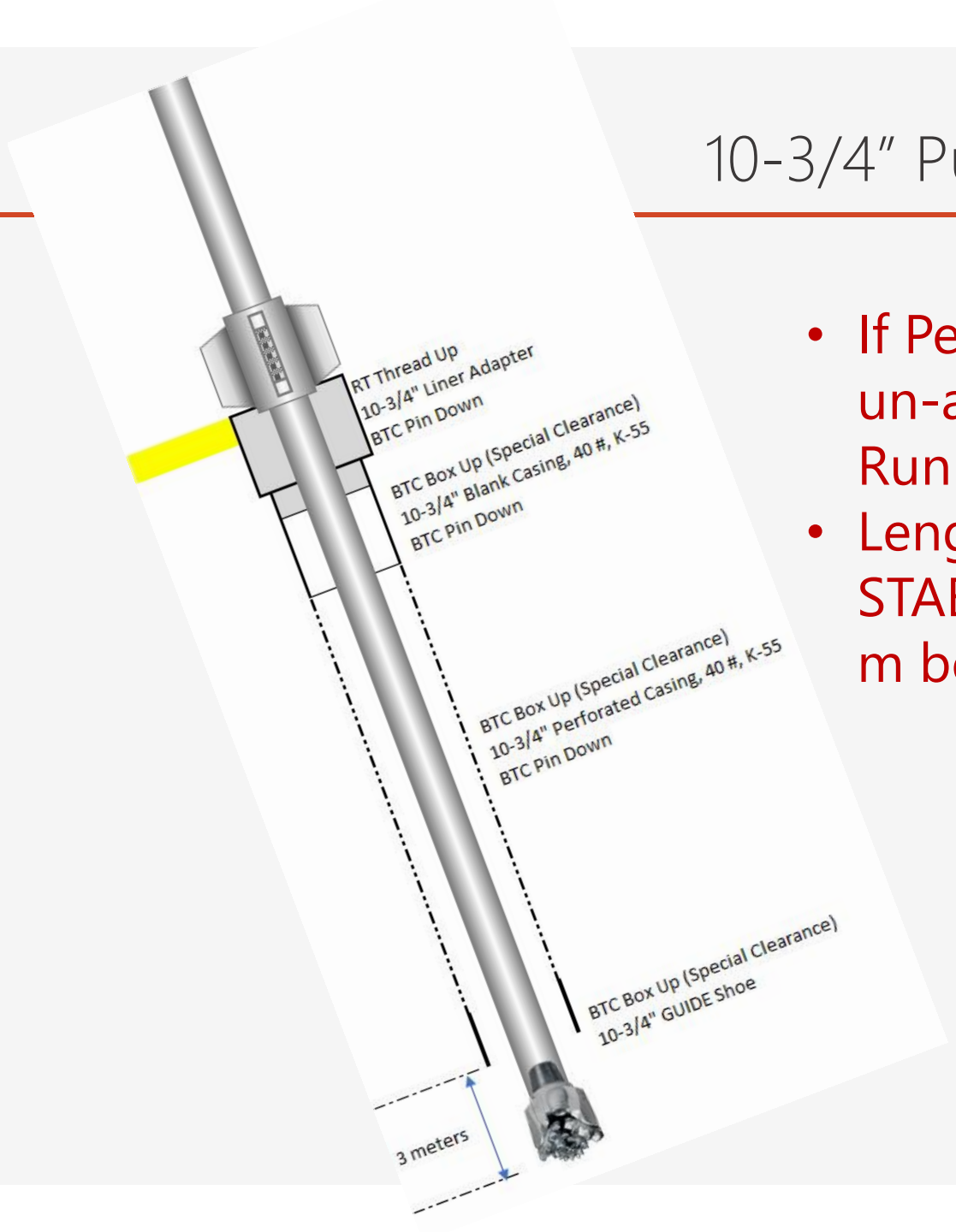


10-3/4” X 8-5/8”



10-3/4” Guide Shoe

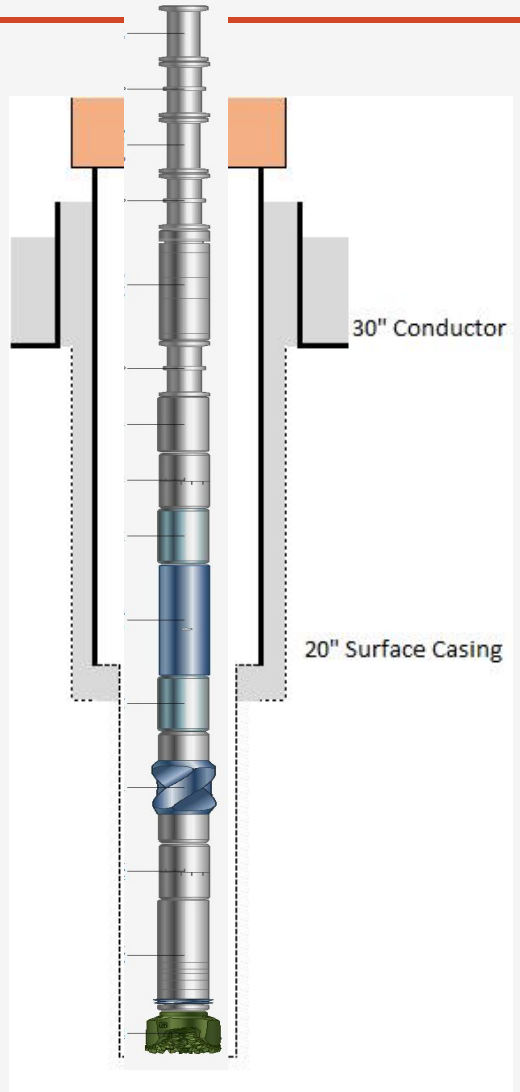
10-3/4" Push BHA



- If Perforated liner tagged weight and un-able to run to bottom → released Running Tool and run PUSH BHA.
- Length of BHA below pushed STABLIZER → space out to have bit 1-2 m below the shoe.

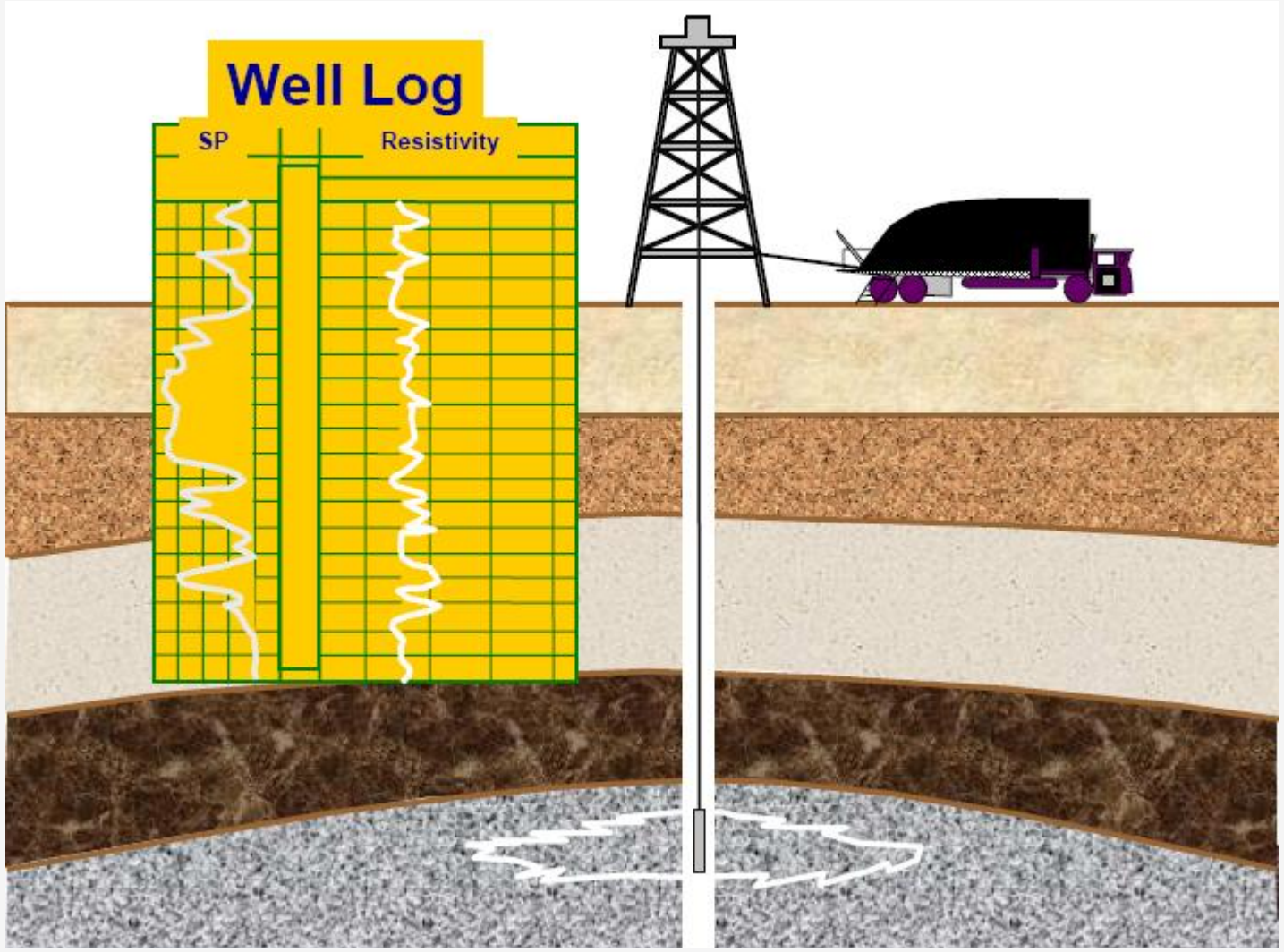
Check List Prior to Running BHA

- BHA tally
- Dimension of each BHA Component, including inner part.
- Bit Nozzles,
- Bit Hydraulic
- Weight below Jars → Available WOB
- Jars running hours
- Mud Motor running Hours
- Surge / Swab Pressure – Running speed



- At shoe: check off bottom torque
- At bottom:
  - check off bottom torque
  - Check SPR.

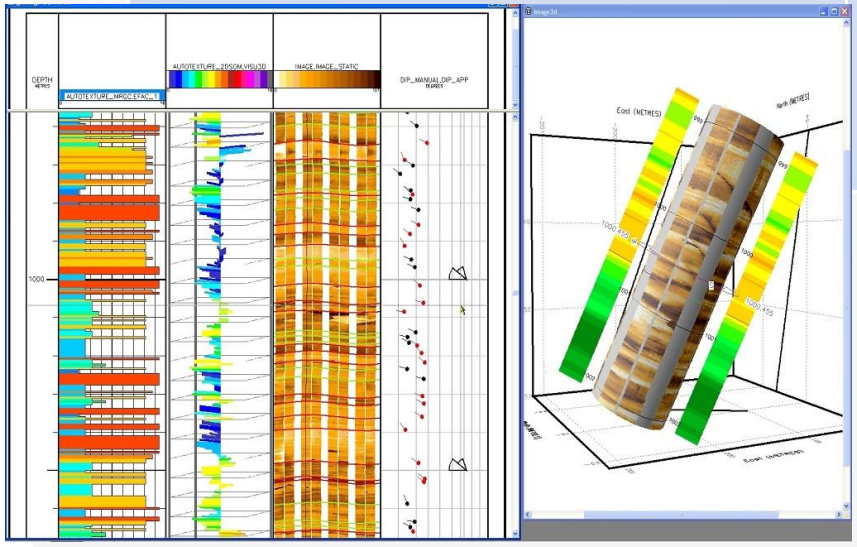
ELECTRIC LOGGING OPERATION

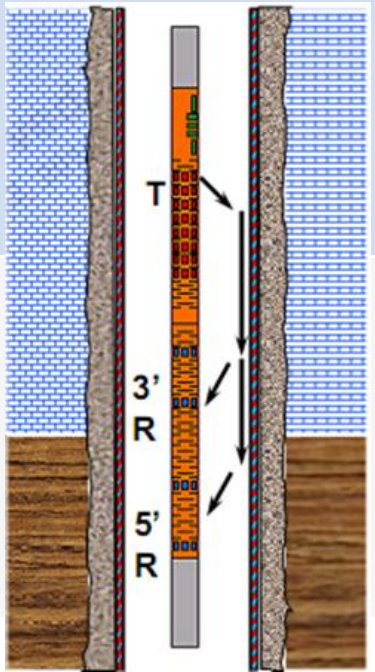


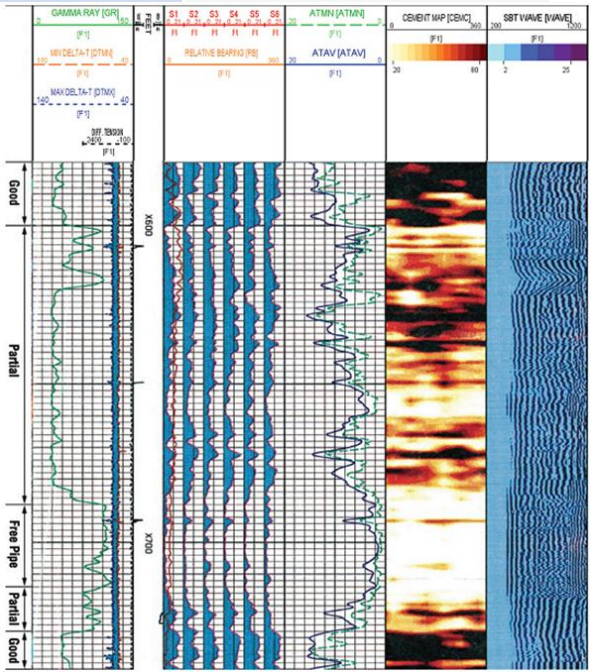
- E-Log Tool configuration.
- MDT:
  - Discuss time for tool in static condition
- Check fishing tool for E-Log
- Wiper trip BHA during running E-Log
- E-Log time duration → may require double crew
- Tool calibration
- Discuss mud properties effected to E-Log reading

E-Log (Wireline Log)

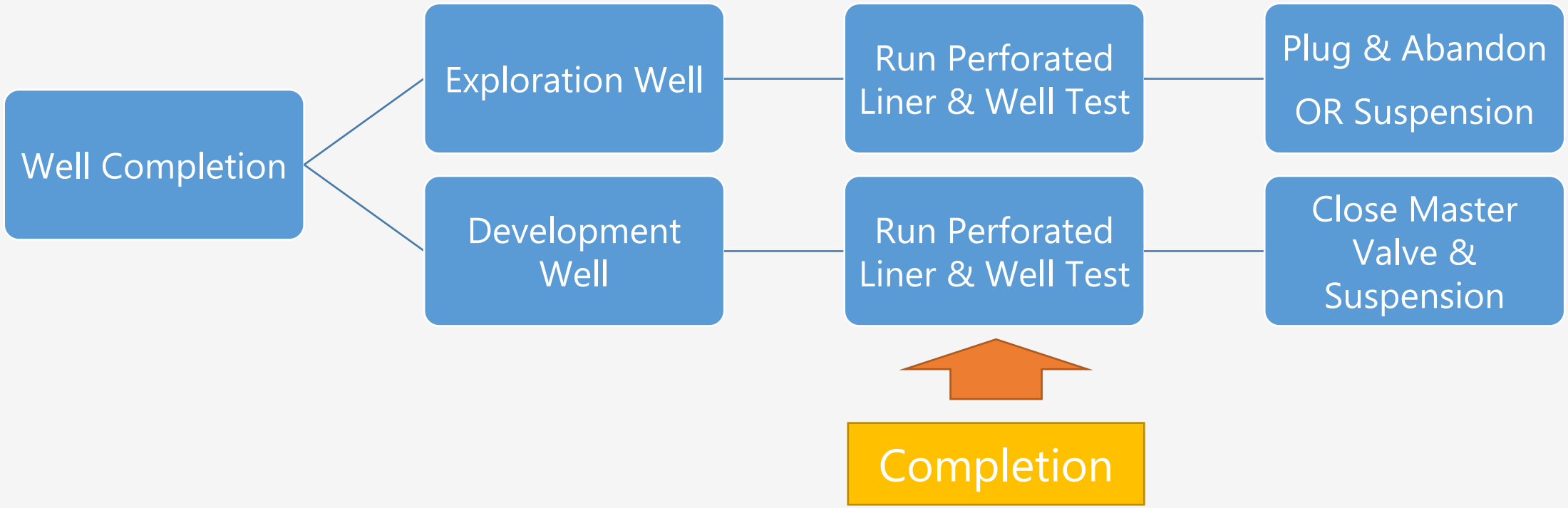
Open Hole Log	Case Hole Log
• FMI • etc	• GR • CBL-VDL • USIT • Set Cement Retainer / Bridge Plug • Pipe Recovery • Casing Perforation • VSP







Well Completion



CASING PRESSURE TEST

CASING PRESSURE TEST

1. Casing and liners shall be pressure tested prior to drilling out the shoe. The Well Operations program shall state the pressure test values.

2. Note that for work-over or well intervention, the pressure test value of any of the casings must not exceed the pressure test value to which the casing was exposed during the original drilling of the well.

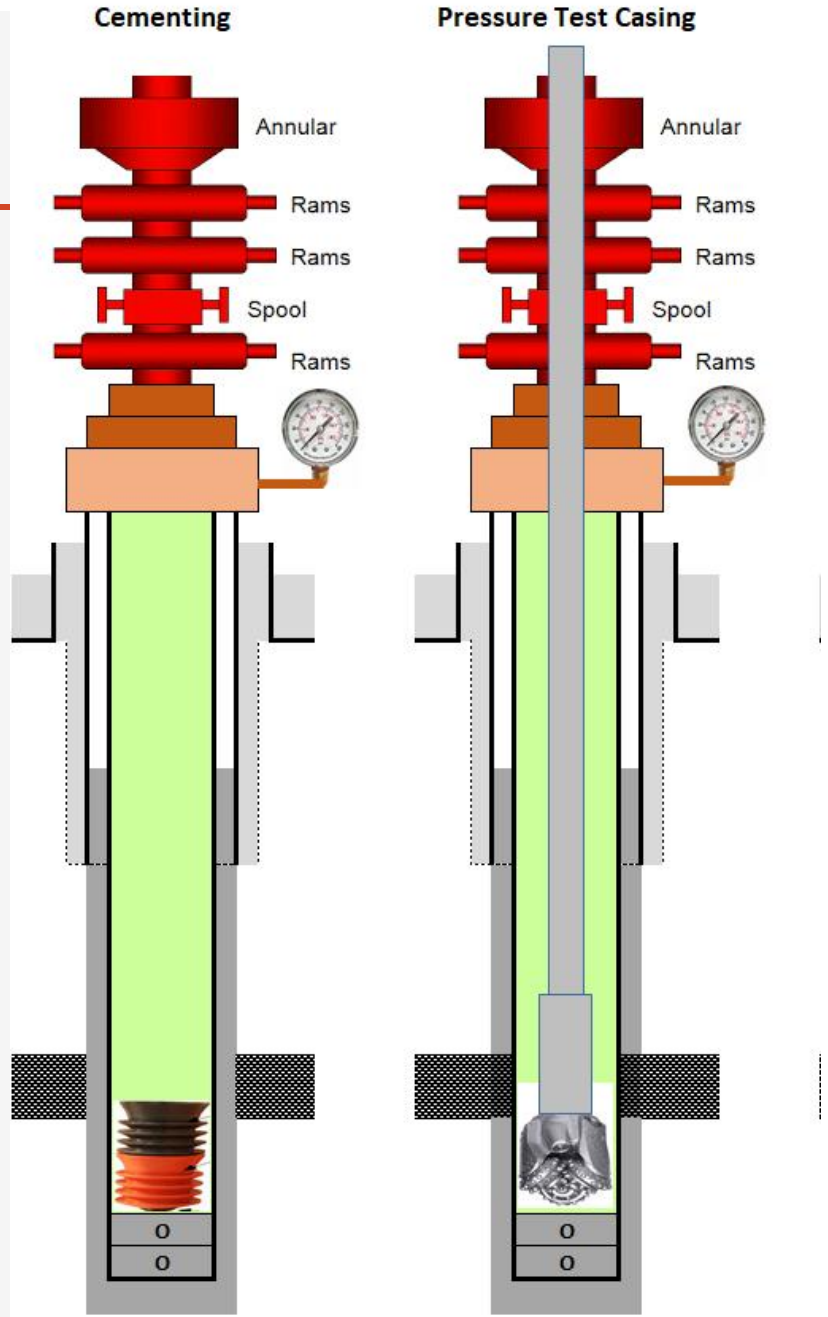
3. Production casing and liners will be pressure tested and inflow tested on critical wells prior to well testing. Any displacement to a lighter fluid which can result in an underbalanced condition shall be performed in such a way that any potential for an influx can be detected and controlled.

4. The Well Operations Supervisor must ensure that the pressure testing results is reported on the IADC and Daily Drilling Report. A test report will be filled in and test charts attached.

Casing Pressure Test

- Wet Pressure Test:
  - 5 minutes if tested after bump plug
- Dry Pressure Test:
  - 30 minutes

- Pressure tested immediately after bumping the plugs OR
- At least 24 hours have elapsed from the time the cement slurry was in place OR
- As per pressure chart shown at UCA test if unable to bump the plug during cementing job.
- All pressure tests shall be recorded (using barton chart or digital recorder)

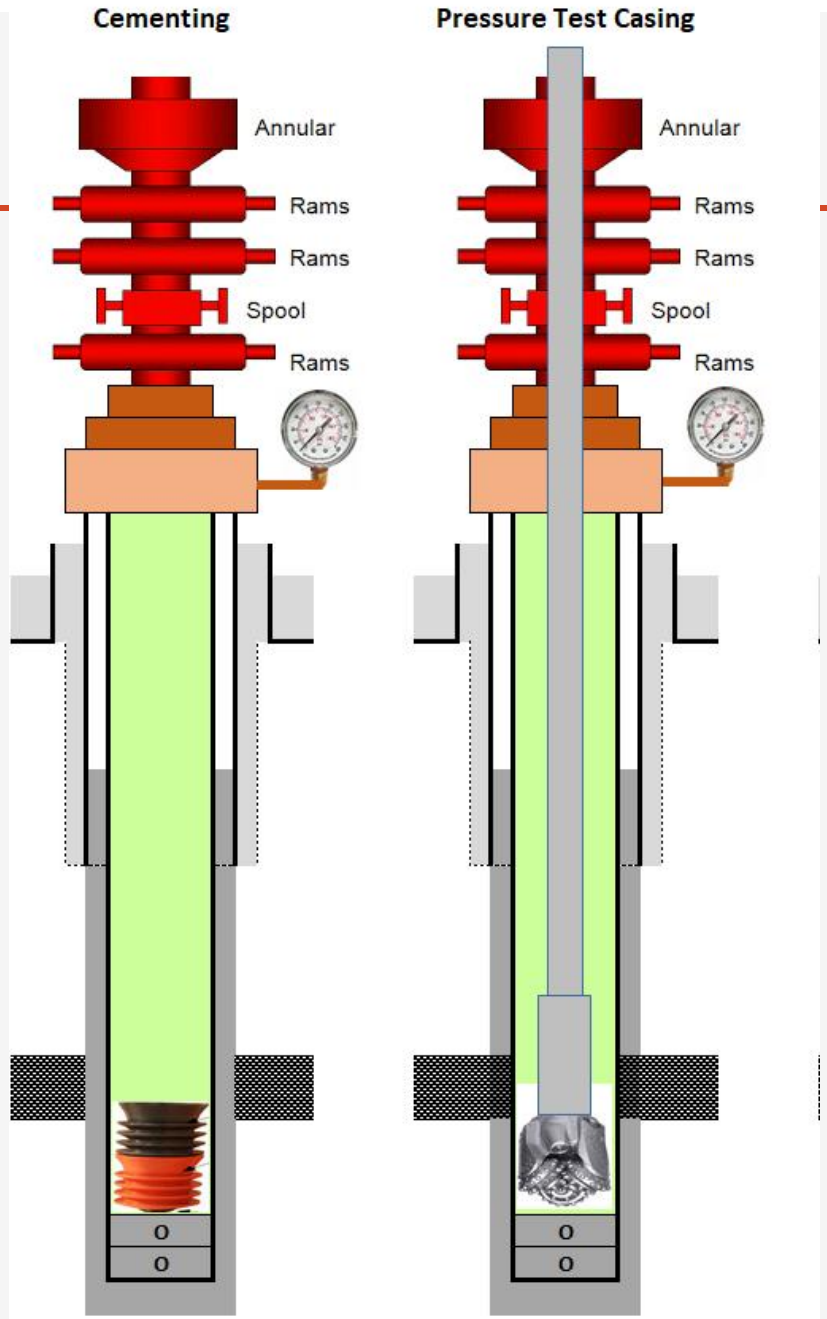


Casing Pressure Test

- Dry Pressure Test

Casing Pressure Tests shall be conducted after nippling up and pressure testing the BOP stack, by either:

- Closing the blind rams and pumping down the choke/kill line OR
- Closing the pipe rams/annular around the drill string and pressure testing prior to drilling out the shoe track.



CASING PRESSURE TEST

Maximum surface test pressure for new casing shall not exceed the lowest value of:

a. Eighty per cent (80%) of casing API internal yield rating of the weakest casing or connection in the string,
b. Wellhead working pressure,
c. BOP working pressure.

- Pressure = 0.052 x MW x ft-TVD

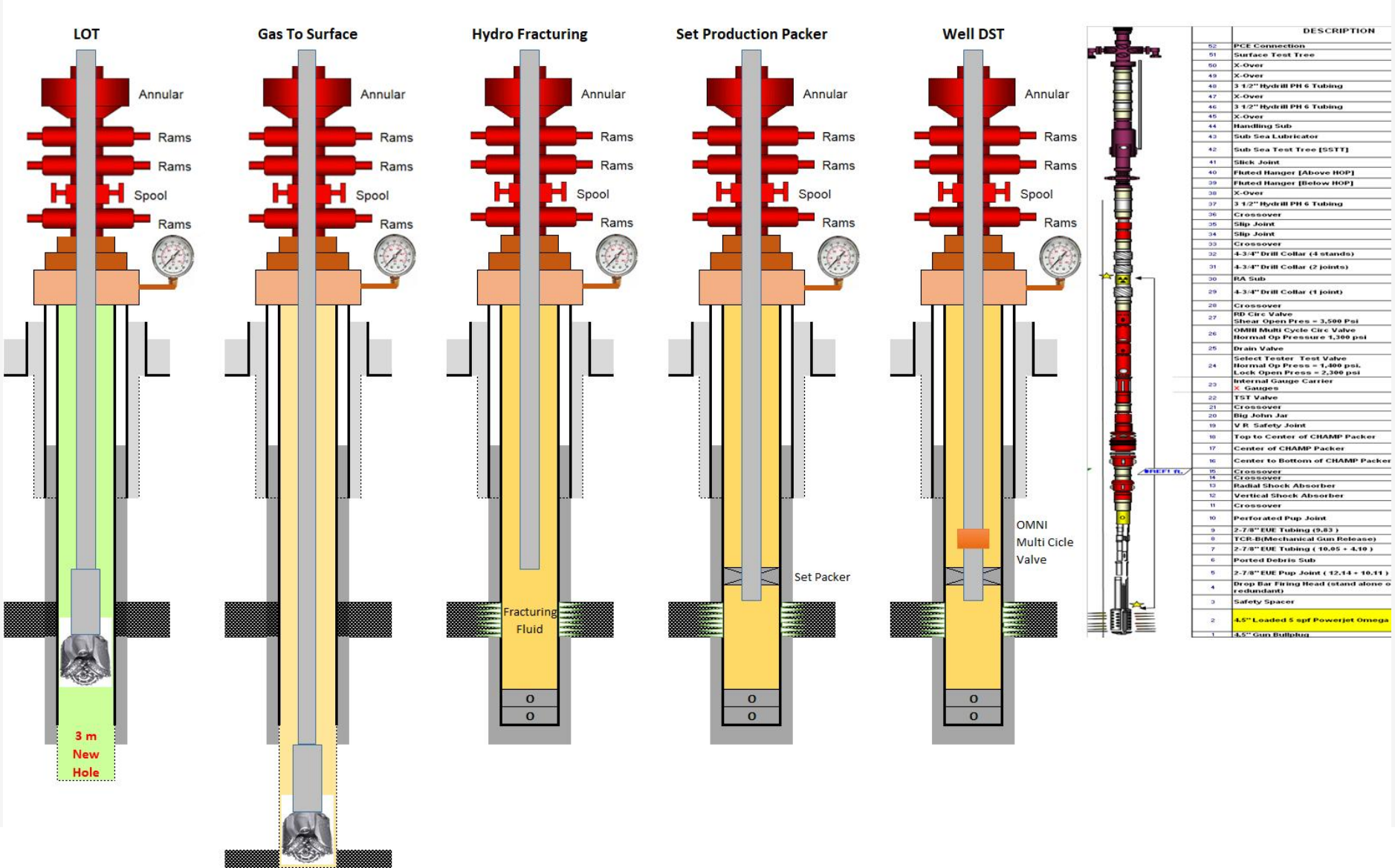
CASING PRESSURE TEST

Due consideration shall be given to:

a. The density of fluid columns inside and outside the casing,
b. The design factors,
c. The impact of **pressure testing on tensile loads,**
d. The impact of pressure testing on cement sheet,
e. Casing wear and/or corrosion.

- Axial Load = Pressure x Area

Casing Pressure Test – How much?



CASING PRESSURE TEST

- Minimum test pressure for **surface and intermediate casing** shall be calculated as follows: **surface pressure with the well full of gas limited by the expected formation fracture gradient at shoe plus 500 PSI**.

- Minimum test pressure for **production casing and liners** shall be equivalent to the shut-in tubing pressure on top of the annulus completion fluids.

- For drilling, **liner overlaps** shall be pressure tested to a minimum of 7.0MPa **(1000psi) over formation leak-off pressure at the shoe of the previous casing**.

- For testing the **liner overlaps** will be pressure tested to the pressure corresponding to **tubing leak over DST annulus fluid**.

CASING PRESSURE TEST

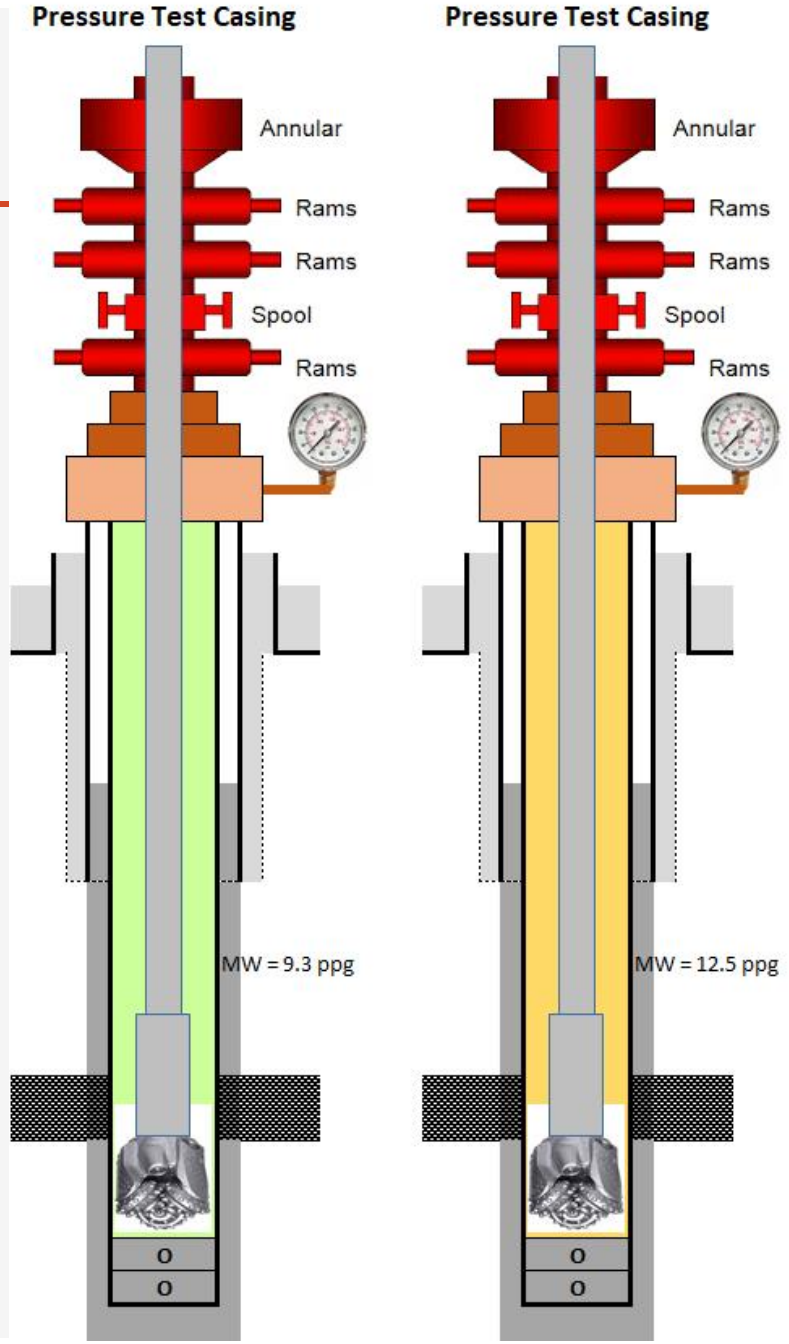
- The integrity requirements of any additional working loads, **such as annular pressure operated tools, shall also be taken into consideration**.

- An **inflow test** shall be done when the hydrostatic head at the top of the liner is expected to drop below the formation pressure isolated by the liner.

- **Tubing strings** shall be pressure tested to the maximum anticipated **wellhead pressure, plus a safety margin of 500psi for gas wells and 1000psi for oil wells**.

Pressure Test Casing Discussion

- If Pressure Leak: ?
  - Check any leakage on surface lines
  - Check if any air trap
  - ?

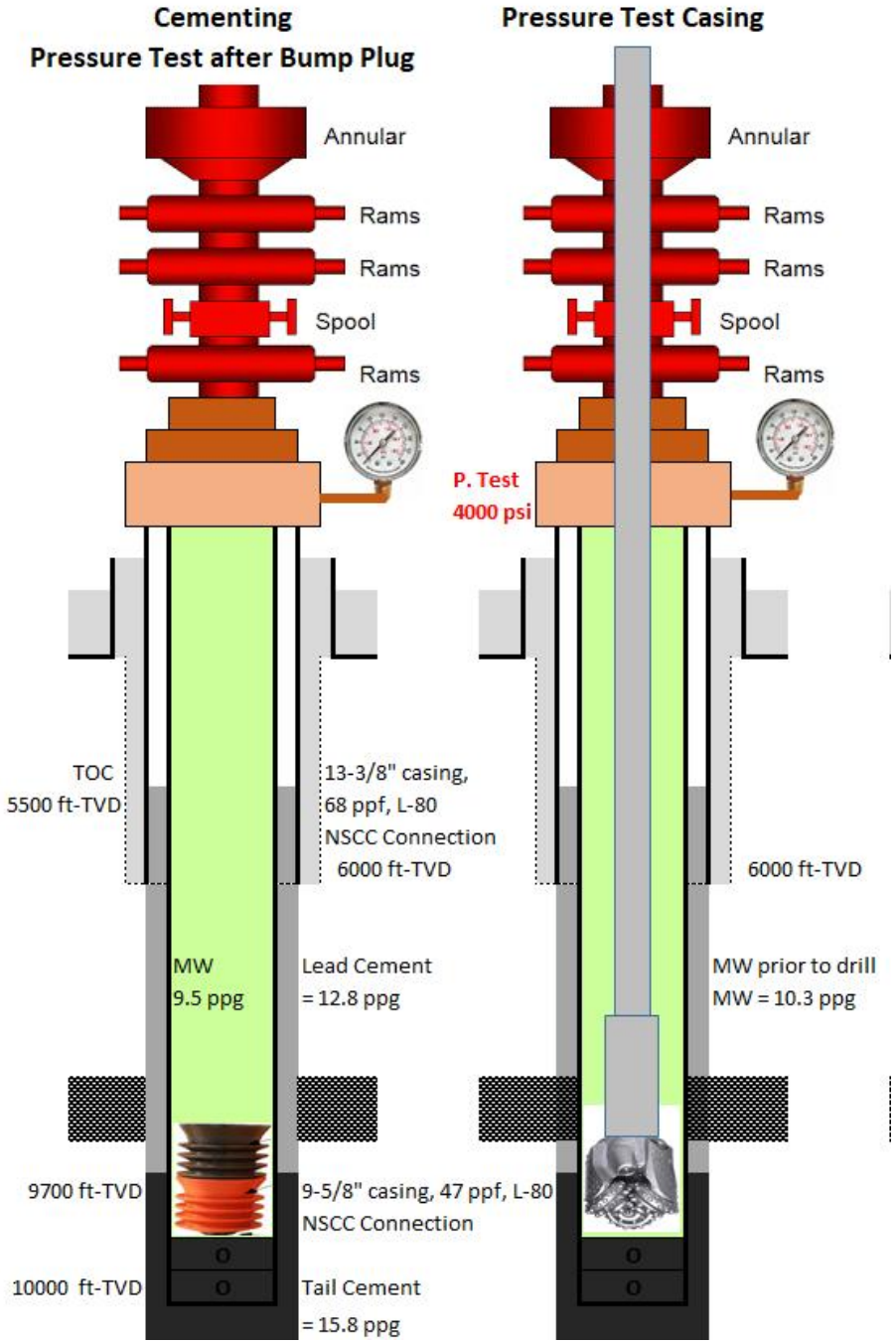


Pressure Test Casing Discussion

- **Check if casing burst ? If not, how many % of casing burst pressure.**
- **Check tensile load due to casing pressure test.**

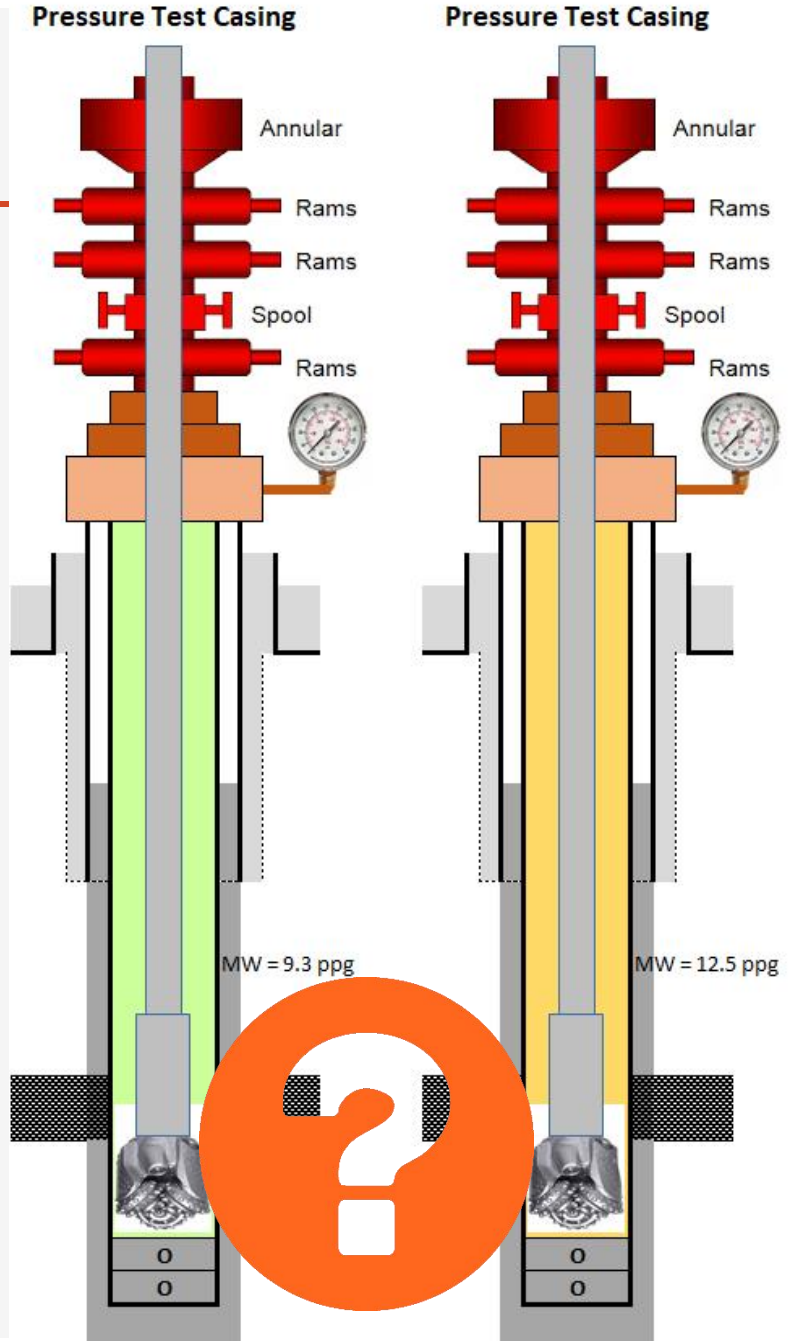
Pressure Test cases:

1. Pressure Test casing after cement bump plug
2. Pressure Test prior to drill out shoe track



Pressure Test Casing Discussion

- Same casing depth, but different Mud Weight.
- Do we apply same surface pressure ?



WELL STIMULATION

HYDRAULIC FRACTURING



PLUG AND ABANDON

Plug & Abandon Operation

Objectives:

1. Isolation of Hydrocarbon zones
2. Protection of Freshwater aquifers
3. Prevention of the migration of formation fluids within the wellbore to the mudline or surface
4. Clearance of sites so as to avoid conflict with other uses of the area around the drilling site.

References:

- Company Policy
- SNI 13-6910-2002

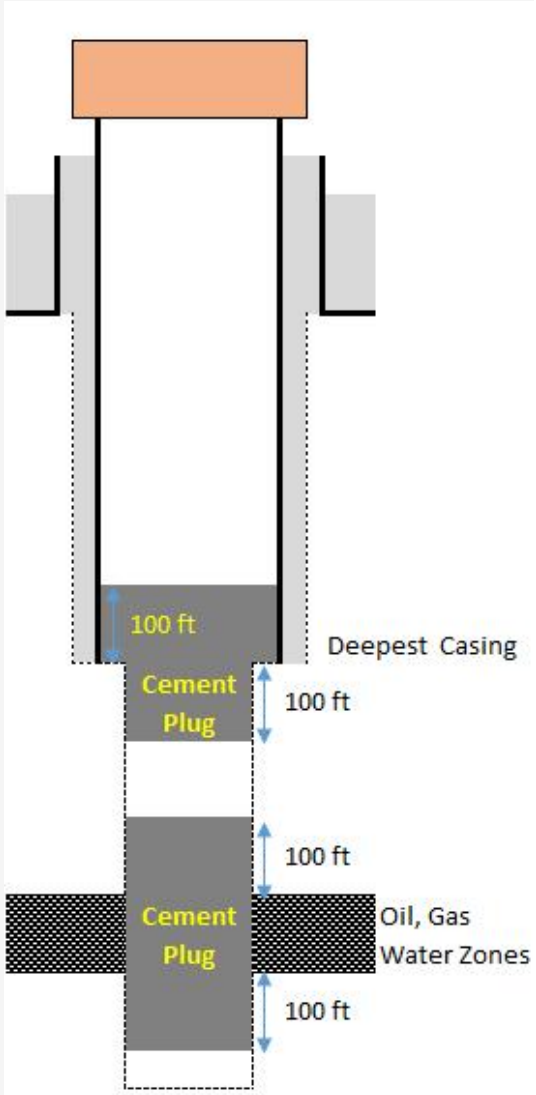
Plug & Abandon Operation - SNI 13-6910-2002:

Approval.

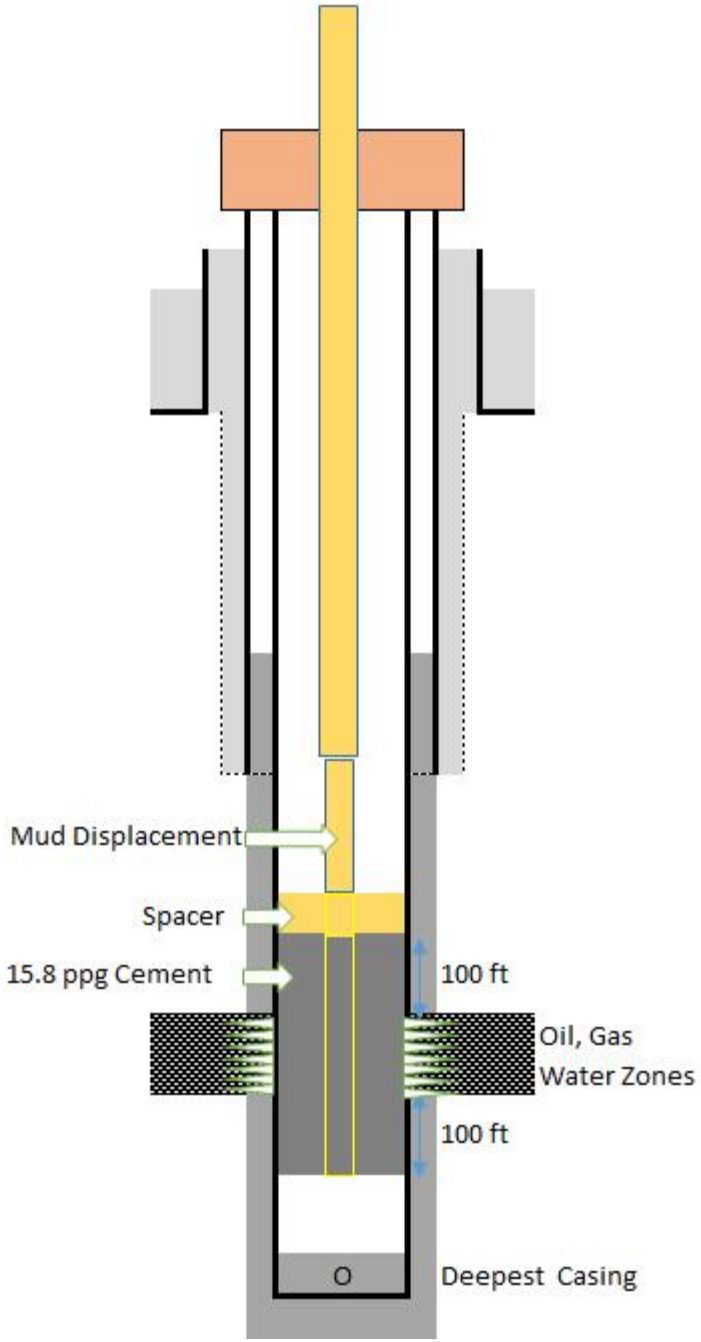
1. Notice of Intent to abandon well
   - MOM with Exploration team (Company + SKK Migas)
   - Drilling to prepare P&A program and Approval Form
2. Subsequent report of abandonment (max 30 days from completion of the works)

Plug & Abandon Operation - SNI 13-6910-2002:

1. Isolation of Zones in Open Hole
   - Cement plugs shall be set to extend minimum 100 ft below and above of oil, gas or fresh water zones.
2. Isolation of Open Hole
   - At the deepest casing by cement plug minimum 100 ft below and above casing shoe



Balance Plug Cement



Plug & Abandon Operation - SNI 13-6910-2002:

2. Isolation of Open Hole – Alternate-1
    - Set Cement Retainer at 50-100 ft above casing shoe
    - Cement plug: minimum 100 ft below casing shoe and 50 ft above retainer

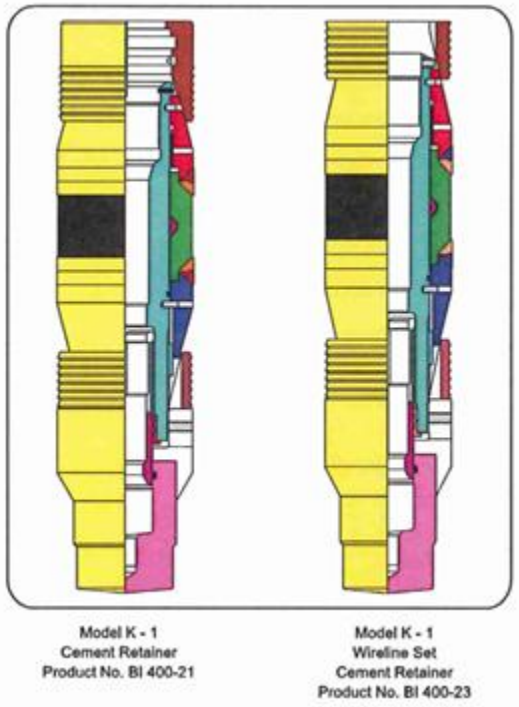
2. Isolation of Open Hole – Alternate-2 (if loss circulation)
    - Set Permanent type Bridge Plug at 150 ft above casing shoe.
    - Cement plug: minimum 50 ft above Bridge Plug. Not required to be tested.

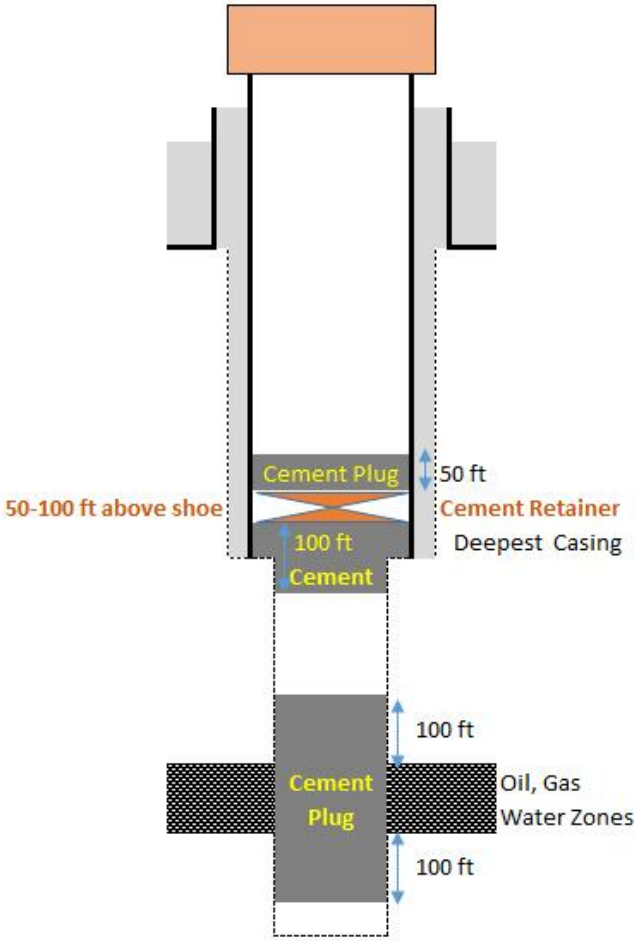
Test Cement Retainer Or Bridge Plug:
    - Tagged with 15 Klbs weight OR
    - Pressure test with 1000 psi, 15 mins, pressure drop less than 10%.

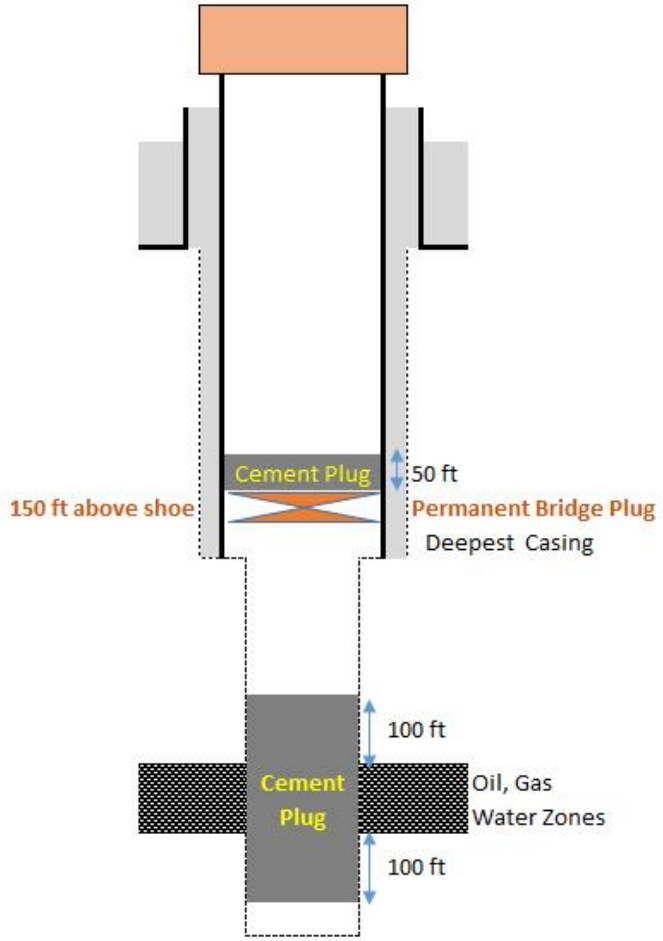
Plug & Abandon Operation - SNI 13-6910-2002:

2. Isolation of Open Hole:

Alternate-1 Alternate-2 (if loss circulation)





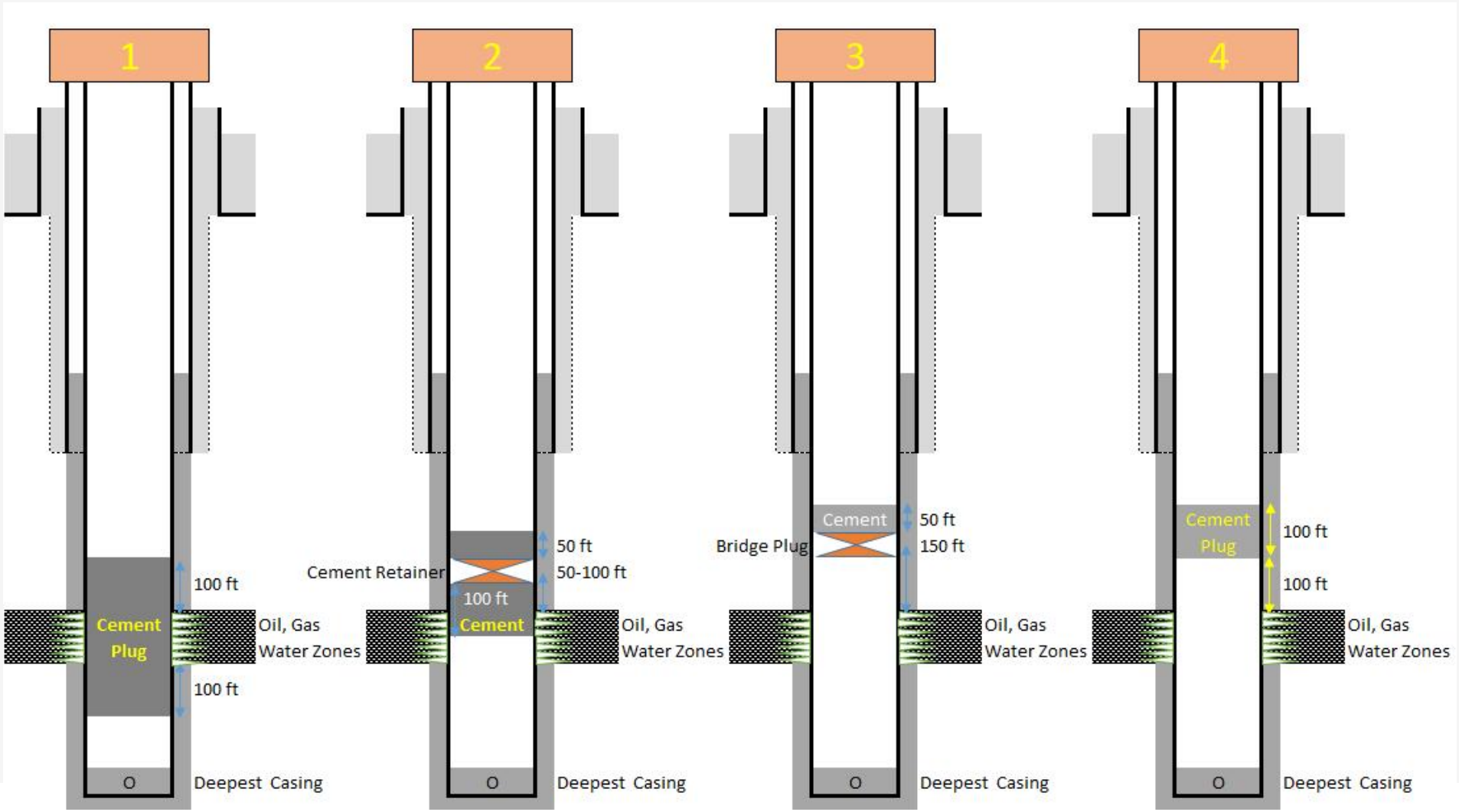


Setting Cement Retainer:

- **With Drill Pipe**
- **With Electric Line - Explosive**

Plug & Abandon Operation - SNI 13-6910-2002:

3. Isolating Perforated Interval



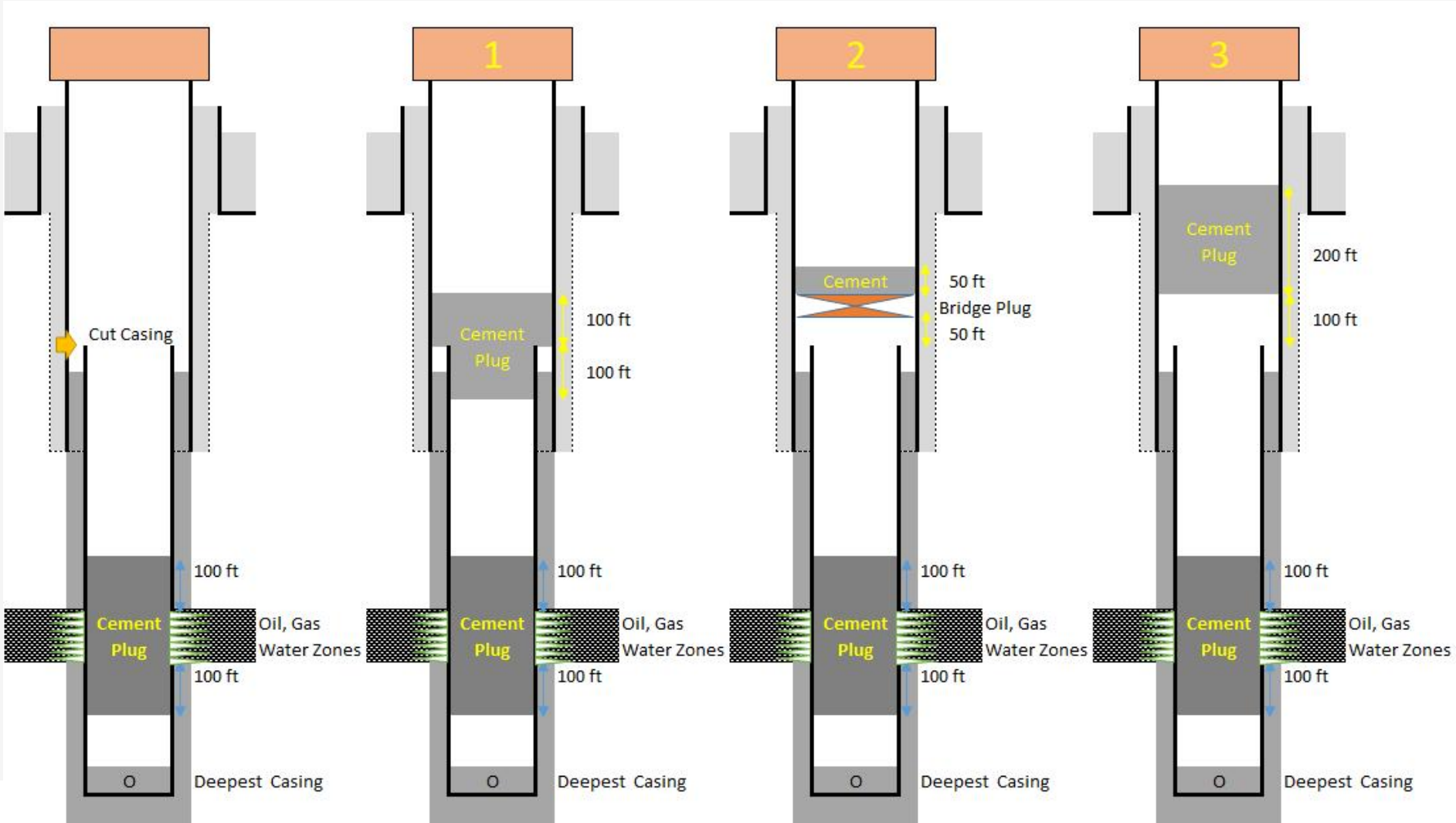
Cut Casing





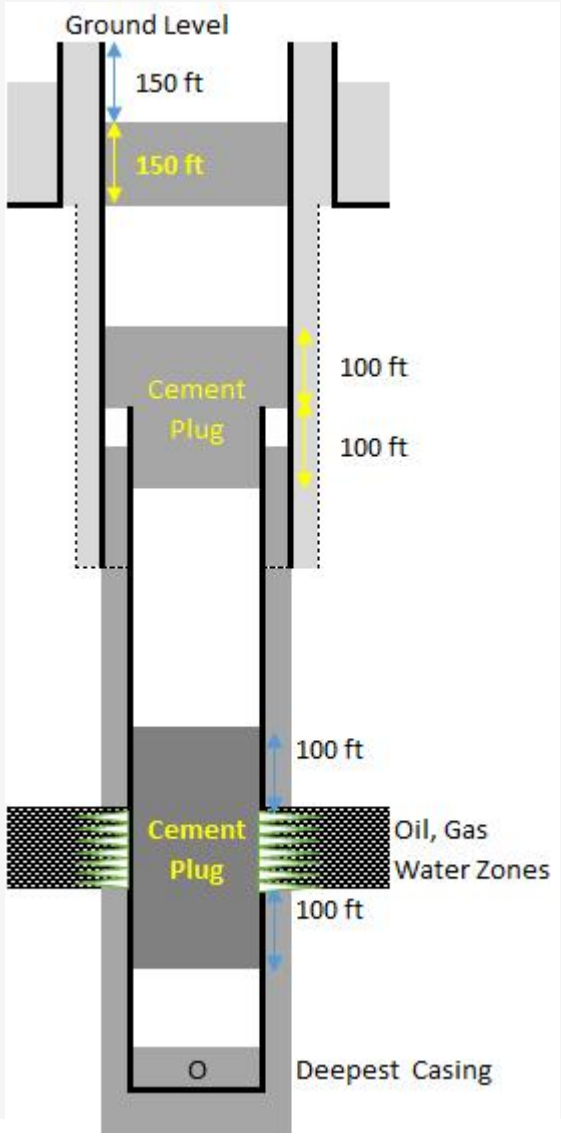
Plug & Abandon Operation - SNI 13-6910-2002:

4. Plugging of Casing Stubs

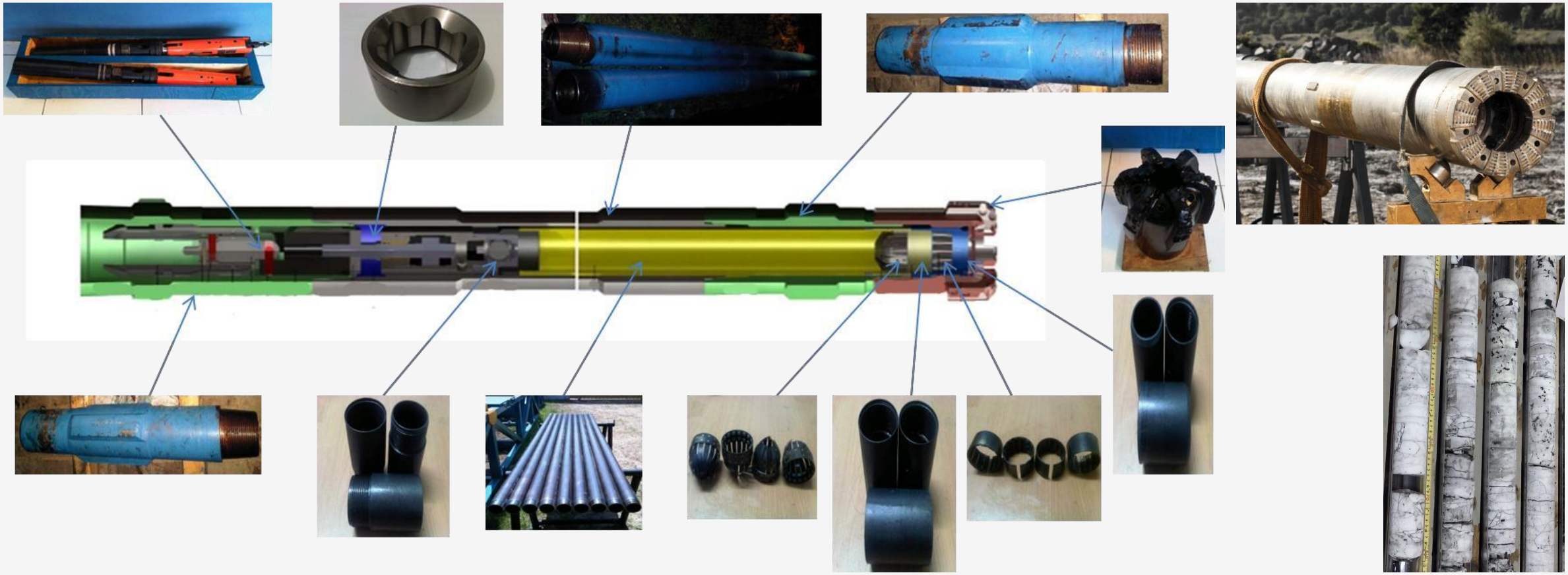


Plug & Abandon Operation - SNI 13-6910-2002:

5. Surface Plug



Coring



DRILLING PROBLEM







Typical Drilling Problem

Rig Move, Rig Up	Drilling	Completion, Well Test	Rig Down, Rig Move
• Heavy Equipment accident • Not sufficient site • Poor access road • Bridge height • Bridge tonnage capacity • In Proper lifting gear • Leveling of site • Soil compression strength	• Hole cleaning • Stuck Pipe • Washed pipe • Drill String Parted • High Torque • Gas Kick • Blowout • Hole stability • Casing tag weight • Tight hole • Casing stuck • Premature set cement • Poor cement bond • Mud Losses • Spillage	• Tubing Leak • Production Packer leak • Downhole tool malfunction • Well could not flow	• Not enough space to lowering rig mast • Rig Company request to extend rig demobilization time • Drop object • Early termination • Local crew termination

DRILLING PROBLEMS - Geothermal

No	Hole Section	Well Name							
1	**36" hole section**								
	- Loss Circulation		o						
	- Others:								
2	**26" hole section**								
	- Stuck Pipe (free)								
	- Stuck Pipe (LIH)								
	- Steam Kick								
	- Running Casing Problem (sticky formation)			c					
	- Loss Circulation during drilling				o				
	- Loss Circulation during running casing and cementing		o	o					
	- Others :								
3	**17-1/2" hole section**								
	- Tight hole								
	- Stuck Pipe (free)	@							
	- Stuck Pipe (LIH)					x			
	- Loss Circulation					o	o	o	o
	- Casing On Bottom								
4	**12-1/4" hole section**								
	- Tight hole							T	T
	- Stuck Pipe (Free)		@	@	@	@	@	@	@
	- Stuck Pipe (LIH)		x					x	
	- Loss Circulation	o	o	o	o	o	o	o	o
	- Logging Problem (couldn't go to bottom)	V							
	- Steam Kick	+	+	+	+	+	+	+	+
5	**9-7/8" hole section**								
	- Tight hole						T		T
	- Stuck Pipe (Free)								
	Stuck Pipe (LIH)								
	- Loss Circulation								o
	- Steam Kick								+
6	**8-1/2" hole section**								
	- Tight hole								T
	- Stuck Pipe (Free)								
	- Stuck Pipe (LIH)							x	
	- Logging Problem (couldn't go to bottom)						V		
	- Casing Stop							c	
	- Loss Circulation						o		
	- Steam Kick						+		

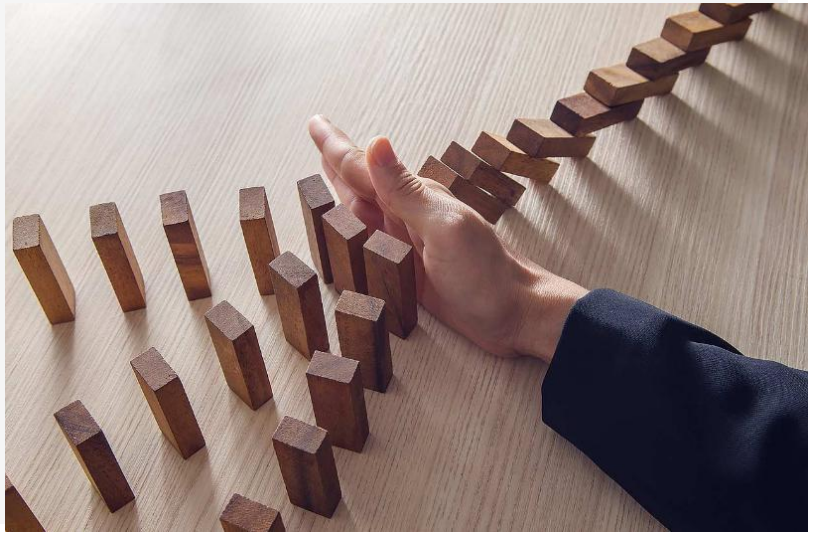
100% Mud losses

90% Stuck Pipe

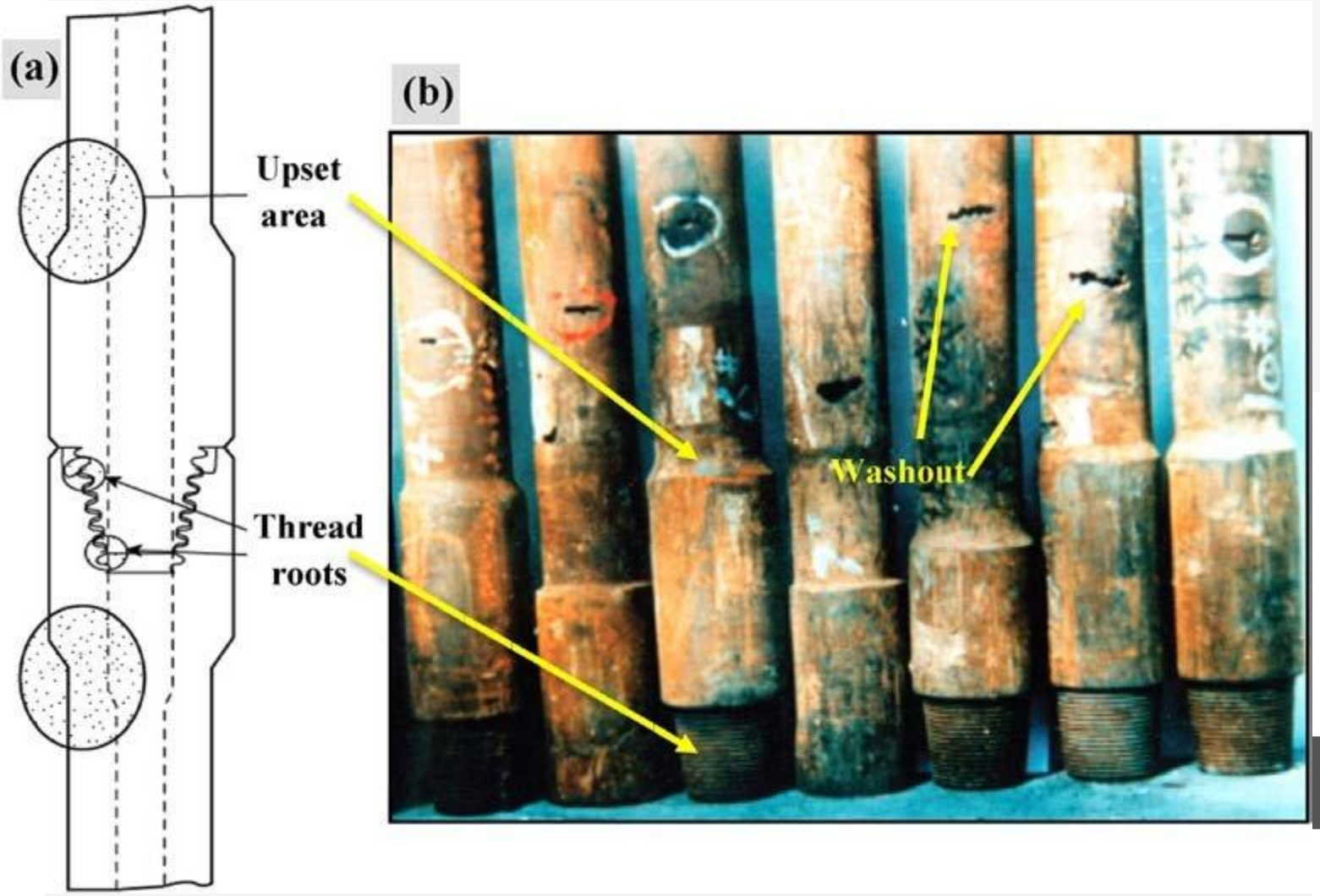
25% LIH

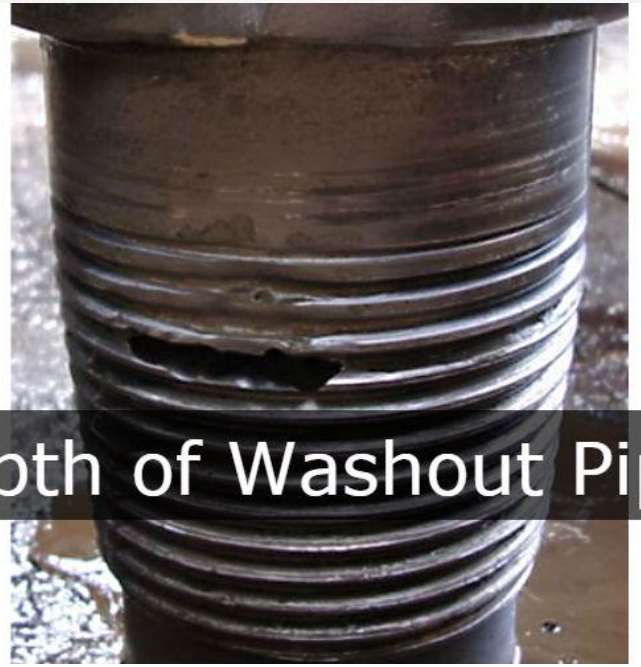
25% Hole Problem

12% Casing, kick etc



DRILL PIPE WASHED OUT





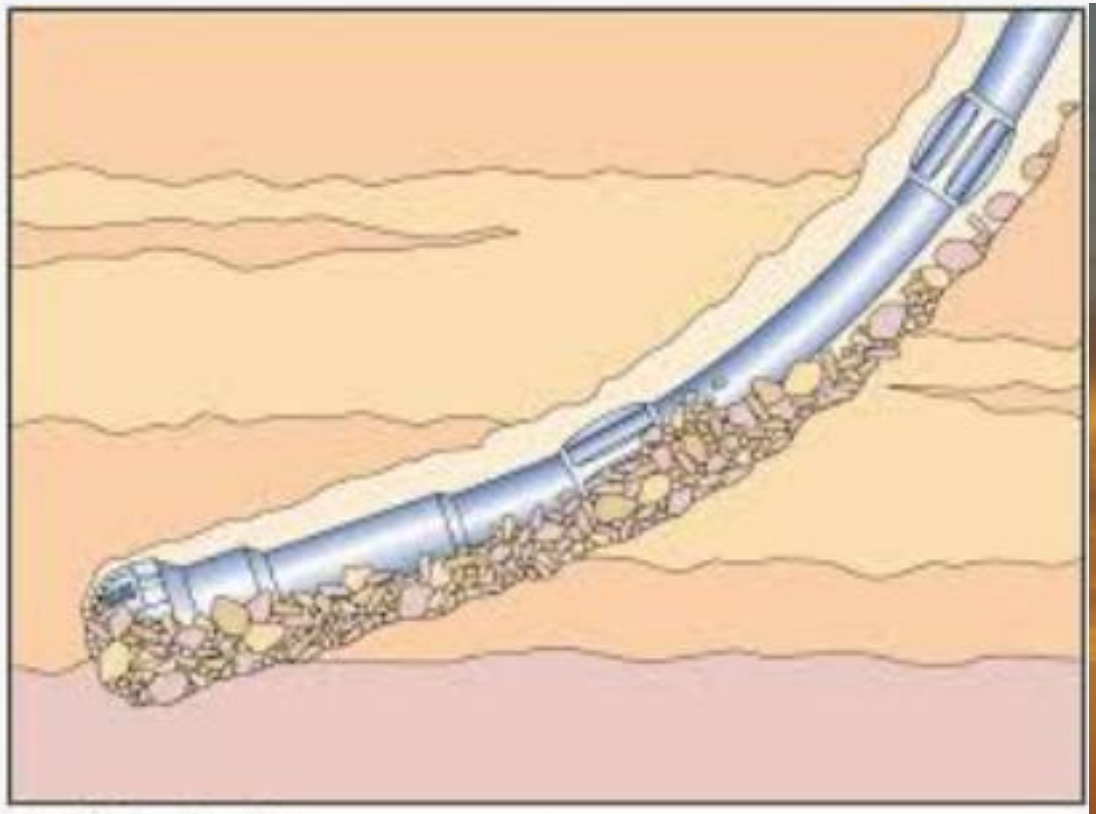
Depth of Washout Pipe

DRILL STRING PARTED

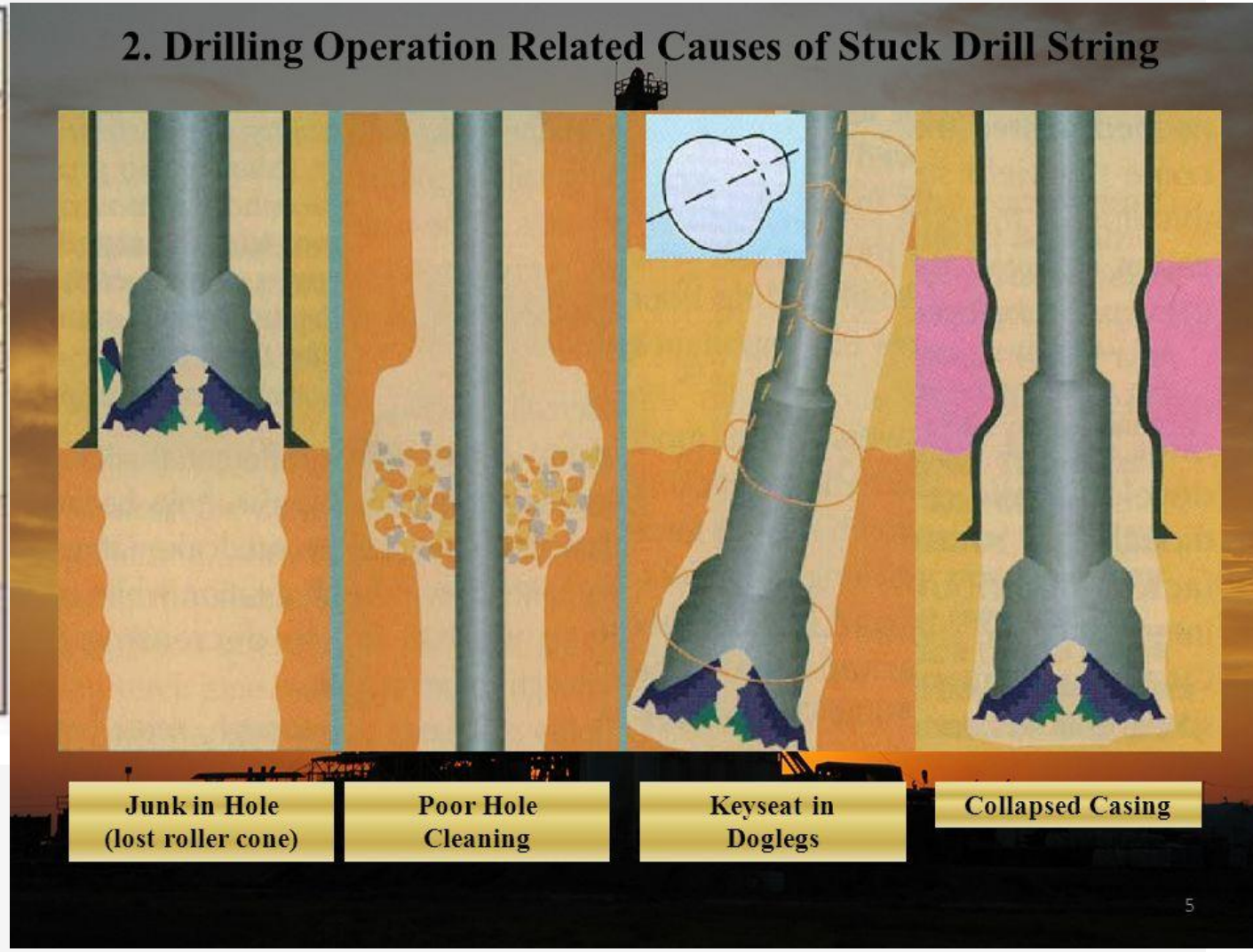




HOLE CLEANING PROBLEM AND STUCK PIPE



Poor Hole Cleaning

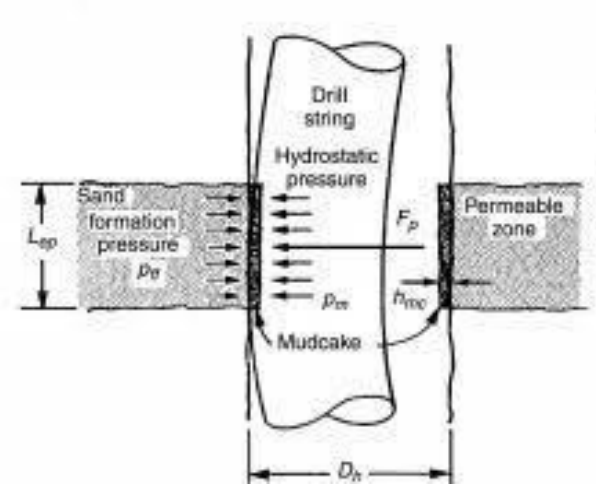


STUCK PIPE

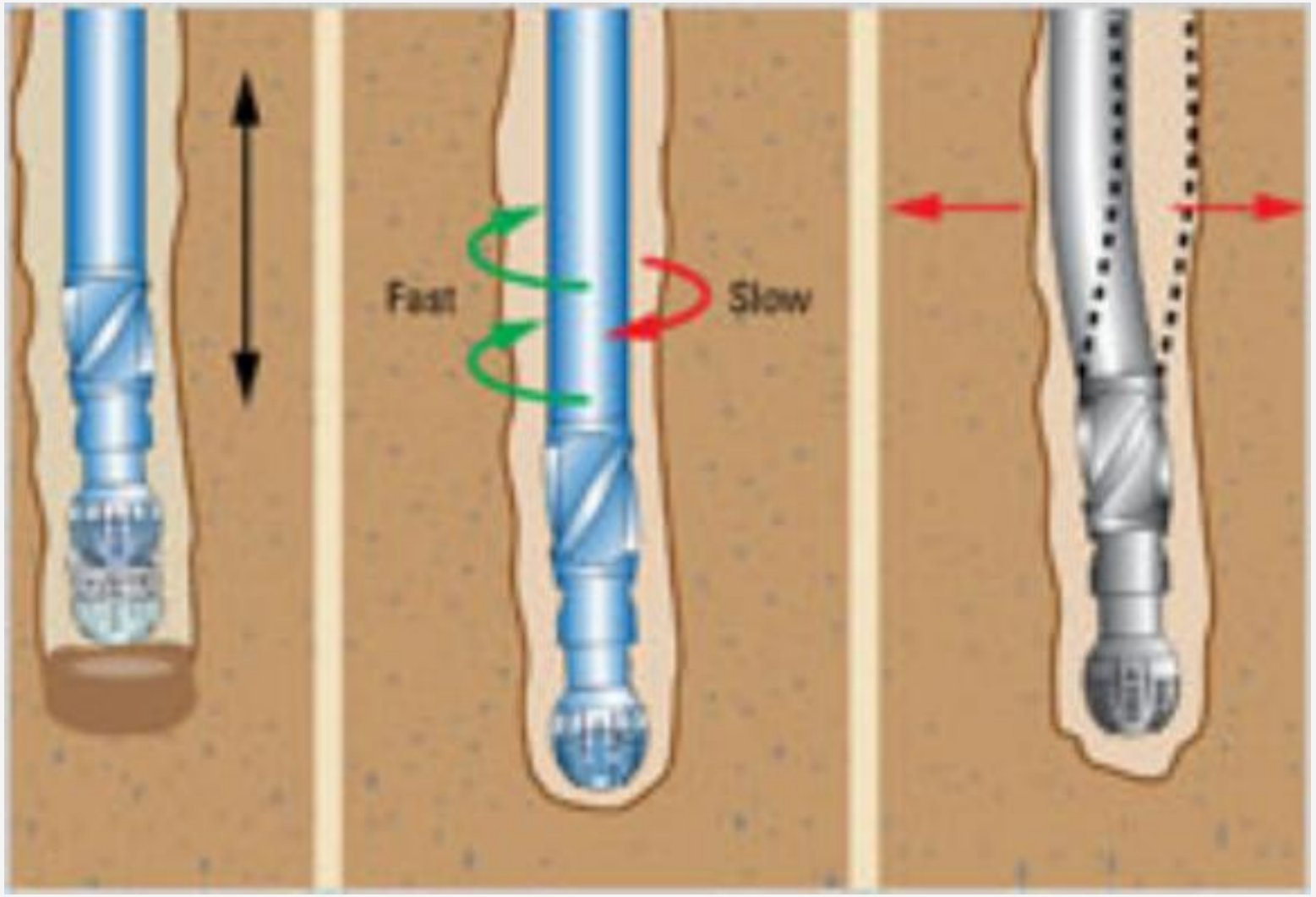






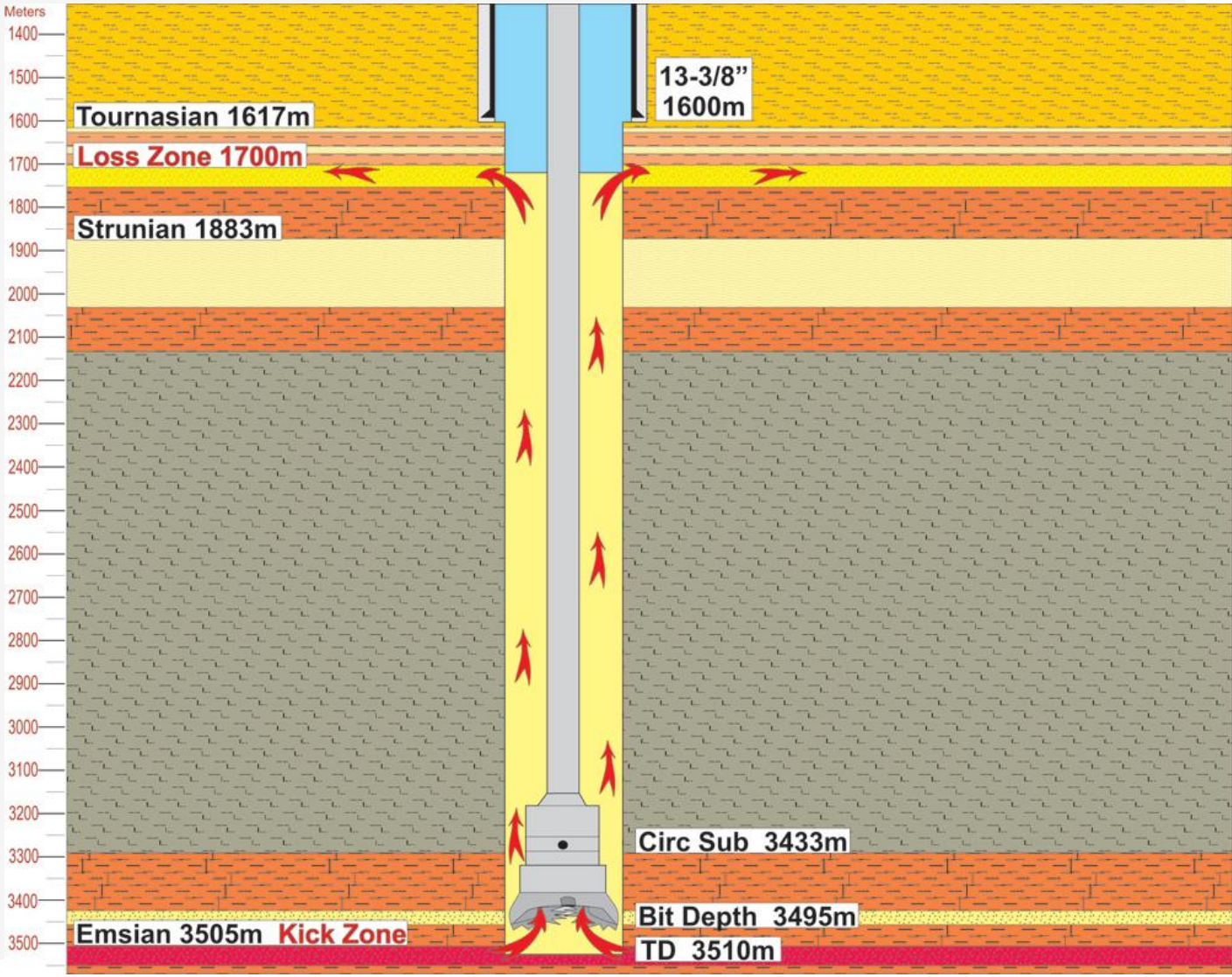

Drill string
Hydrostatic pressure
Sand formation pressure
p_f
L_{ep}
F_p
Permeable zone
p_{cr}
h_{mc}
Mudcake
D_h

DRILLING VIBRATION



MUD LOSSES AND WELL KICK





Blow Out







COMMUNITY ISSUE



















Landslide















RIG MOVING PROBLEM











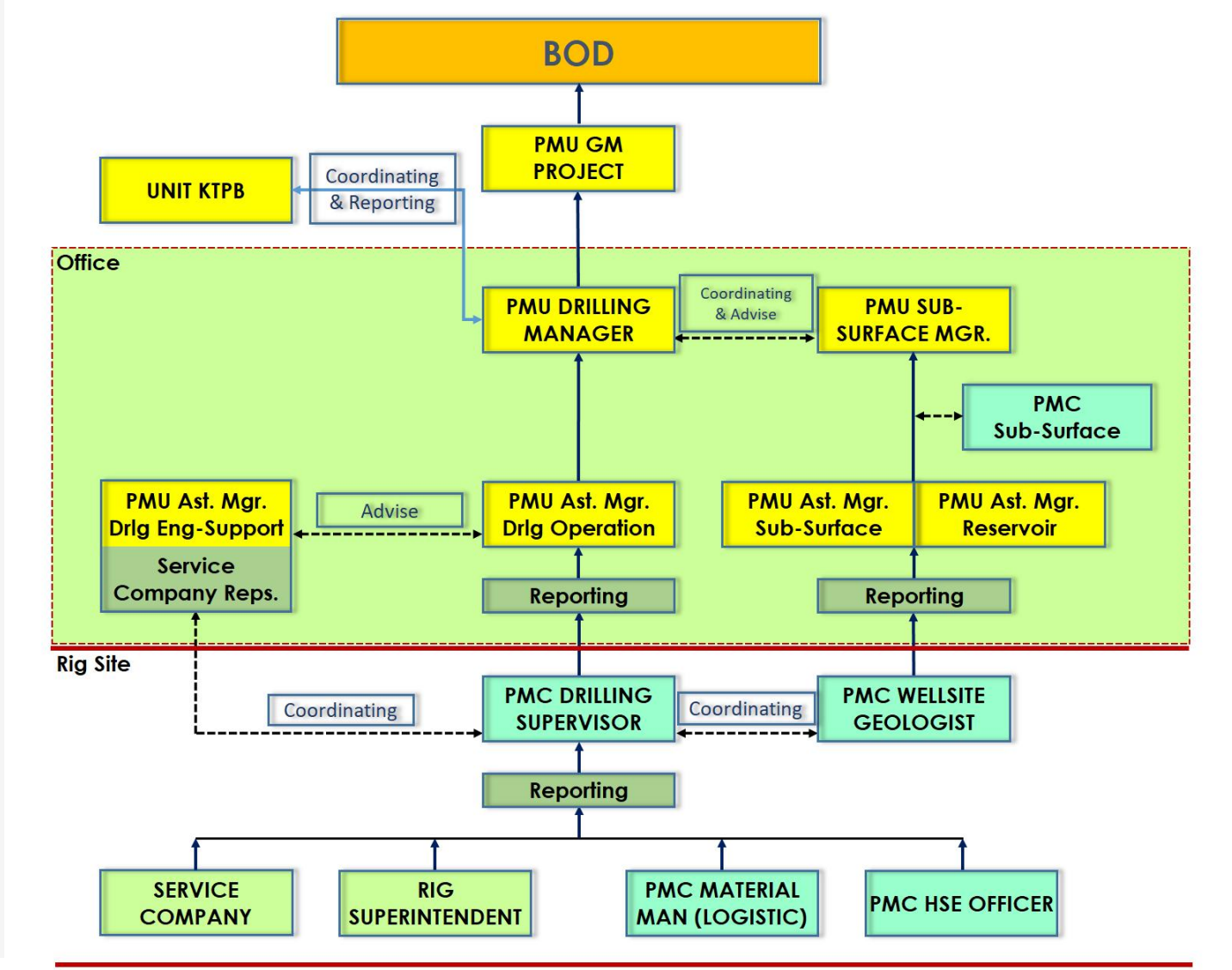


WDT (Well Decision Team)

Critical Decision

Strategic Decision

Routine Drilling Operation



Nice Working Environment



Strong Team Work




Thank You!
ophir
ophir